CAUTION

IF A NON-VARIABLE HEAT GUN IS USED WHEN AMBIENT TEMPERATURE IS LESS THAN 65 F (18.3 C) DEGREES, IT CAN CAUSE DRY-OUT TO NOT BE COMPLETED. THIS CAN CAUSE DAMAGE.

(a) Use a heat gun, COM-15157, to heat the PS3 and P3B pressure ports for 4 minutes before you continue with the water removal.

NOTE: Refer to the table below to set the heat gun temperature (Table 201).

Table 201/73-21-11-993-802-G00 Heat Gun Temperature

Ambient Temperature	Maximum Heat Gun Setting
65 to 50°F (18.3 to 10.0°C)	250°F (121.1°C)
49 to 32°F (9.4 to 0.0°C)	300°F (148.9°C)
31 to 14°F (-0.5 to -10.0°C)	400°F (204.4°C)
Less than 13°F (-10.6°C)	500°F (260.0°C)

SUBTASK 73-21-11-020-001-G00

WARNING

USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

CAUTION

KEEP THE PS3 SENSE LINE AND P3B SENSE LINE AND OPEN PSS UNIT PORTS AWAY FROM CONTAMINATION. IF NOT, OUTSIDE PARTICLES AND SMALL OBJECTS CAN CAUSE DAMAGE TO THE EQUIPMENT.

(2) Disconnect the air tubes from the PSS [1] as follows:

(a) Cut and remove the safety wire [5] from the coupling nut (P3B) [6] and coupling nut (PS3) [7].

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

(b) Disconnect the coupling nut (P3B) [6] and coupling nut (PS3) [7] from the PSS [1].

(c) Install the protective caps on the air tube openings.

SUBTASK 73-21-11-160-004-G00

CAUTION

REMOVE AS MUCH STANDING WATER AS POSSIBLE BEFORE YOU START THE VACUUM PUMP. IF YOU DO NOT REMOVE THE STANDING WATER IT WILL REDUCE THE CAPABILITY OF THE VACUUM PUMP AND CAN CAUSE DAMAGE TO THE VACUUM PUMP.

(3) Do a PS3 and P3B transducer visual water removal:

(a) Insert a clean disposable medicine dropper, STD-1067, (or equivalent) into the opening of the PS3 transducer port on the PSS.

(b) Remove the visible standing water.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that you remove as much of the standing water as possible.

(c) Examine the medicine dropper, STD-1067, for dirt or contamination.

1) If it has dirt or contamination, discard the medicine dropper, STD-1067.

NOTE: You can use a clean medicine dropper, STD-1067, again.

(d) Insert a clean disposable medicine dropper, STD-1067, (or equivalent) into the opening of the P3B transducer port on the PSS.

(e) Remove the visible standing water.

1) Make sure that you remove as much of the standing water as possible.

(f) Examine the medicine dropper, STD-1067, for dirt or contamination.

1) If it has dirt or contamination, discard the medicine dropper, STD-1067.

NOTE: You can use a clean medicine dropper, STD-1067, again.

SUBTASK 73-21-11-160-005-G00

CAUTION ALWAYS FOLLOW THE MANUFACTURER'S INSTRUCTIONS WHEN YOU OPERATE THE EQUIPMENT. IF YOU DO NOT FOLLOW THE MANUFACTURER'S INSTRUCTIONS, DAMAGE TO THE EQUIPMENT CAN OCCUR.

CAUTION MAKE SURE THAT YOU CHANGE THE OIL IN THE VACUUM PUMP AFTER 50 OPERATIONS. IF YOU DO NOT, IT CAN DECREASE THE VACUUM PUMP PERFORMANCE.

(4) Use the EEC dryout kit, SPL-14895, to do a vacuum dry-out procedure as follows (Figure 202):

(a) Connect one end of teflon-6 SST braided teflon hose assembly to the P3B pressure port on the PSS unit.

(b) Connect one end of other teflon-6 SST braided teflon hose assembly to the PS3 pressure port on the PSS unit.

(c) Connect the two loose ends of the teflon-6 SST braided teflon hose assemblies to the two horizontal ports on the J6T union tee SS fitting.

(d) Connect the vacuum hose to the T-end of the J6T union tee SS fitting.

(e) Tighten all fittings to 270 in-lb (30.5 N·m) - 300 in-lb (33.9 N·m).

(f) If the ambient temperature is less than 65°F (18.3°C), do these steps:

CAUTION IF A NON-VARIABLE HEAT GUN IS USED WHEN AMBIENT TEMPERATURE IS LESS THAN 65 F (18.3 C) DEGREES, IT CAN CAUSE DRY-OUT TO NOT BE COMPLETED. THIS CAN CAUSE DAMAGE.

1) Use a heat gun, COM-15157 to heat the PS3 and P3B pressure ports for 4 minutes before you start the vacuum dry-out procedure.

NOTE: Refer to this table to set the heat gun temperature (Table 201).

2) Apply heat from the heat gun, switching from both the PS3 and P3B pressure ports until the procedure is completed

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CAUTION DO NOT LET CONTAMINATION GO INTO THE VACUUM PUMP RELIEF PORT. IF THE PRESSURE INDICATION IS NOT BELOW THE SPECIFIED THRESHOLD, THE PROCEDURE COULD BE UNSATISFACTORY.

(g) Do the vacuum procedure as follows:

1) Turn the vacuum relief valve and blank-off valve handles to the open position (vertical position).

2) Start the vacuum pump and let the pump motor get to operating speed.

CAUTION DO NOT CLOSE THE BLANK-OFF VALVE WHILE YOU REMOVE MOISTURE FROM THE PSS UNIT. IF THE BLANK-OFF VALVE IS CLOSED, MOISTURE WILL NOT BE REMOVED. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

3) Turn the valve handle of the vacuum relief valve to the horizontal position to close it.

4) Make sure that the vacuum pump is pulling adequate vacuum on the PS3 and P3B manifolds.

a) If the airport elevation is less than 5000 ft (1524 m), make sure that the vacuum measurement is in the green-colored region on the vacuum pressure indicator.

b) If the airport elevation is more than 5000 ft (1524 m), find the correct vacuum pressure measurement (Figure 203).

NOTE: The vacuum achieved needs to be corrected based on the pressure altitude at which the pump is used.

c) If the vacuum measurement is not correct, do these steps:

<1> Do a check for leaks in the vacuum hose and connections.

<2> Make sure that the pump valves are in the correct position.

5) Operate the vacuum pump for a minimum of 20 minutes.

6) After 20 minutes of operation, open the vacuum relief valve to release the pressure.

7) Turn off the vacuum pump.

8) Remove the vacuum hose end adapter from the PS3 and P3B pressure ports.

(h) Blow out the PS3 sense line and P3B sense line as follows:

1) Remove the protective caps from the air tubes (PS3 and P3B).

2) Use approximately 100 psig (689 kPa) (not to exceed 200 psig (1379 kPa)) to dry the air tubes.

a) Use the regulated air source, STD-3940, to blow air into the air tubes for a minimum of 3 minutes.

NOTE: For the best results, compressed nitrogen is recommended

b) Continue until you see no moisture coming out of the sense line weep hole, located at the 6:00 o'clock position, in the aft location of the fan compartment.

3) Do a visual inspection of the weep holes on the air tubes to make sure that they are clear of all blockage.

(i) Connect the air tubes to the PSS [1] as follows:

1) Make sure that the mating surfaces are clean and clear of unwanted materials.

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2) Lubricate the threads of the coupling nut (P3B) [6] and coupling nut (PS3) [7] with the oil, D00599 [CP2442].

3) Connect the coupling nut (P3B) [6] and coupling nut (PS3) [7] to the PSS [1].

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

4) Tighten the coupling nut (P3B) [6] and coupling nut (PS3) [7] as follows:

a) Tighten the coupling nut (P3B) [6] and coupling nut (PS3) [7] to 270 in-lb (30.5 N·m) - 300 in-lb (33.9 N·m).

b) Make sure that the coupling nut (P3B) [6] and coupling nut (PS3) [7] are fully seated.

c) If the coupling nut (P3B) [6] and coupling nut (PS3) [7] are not fully seated, do the steps above again to tighten the coupling nut (P3B) [6] and coupling nut (PS3) [7].

5) Install the safety wire [5] (safety wire, G02345 [CP8001]) between the sense line coupling nut (PS3) [7] and the PS3 pressure port nipple of PSS [1].

6) Install the safety wire [5] (safety wire, G02345 [CP8001]) between the sense line coupling nut (P3B) [6] and the P3B pressure port nipple of PSS [1].

(j) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-11-860-012-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-11-860-013-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-11-860-014-G00

(3) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

J. Pressure Sub-System Test

SUBTASK 73-21-11-720-001-G00

(1) Do one of the following steps:

(a) Do this task: Test No. 4 - Idle Leak Test, TASK 71-00-00-790-804-G00.

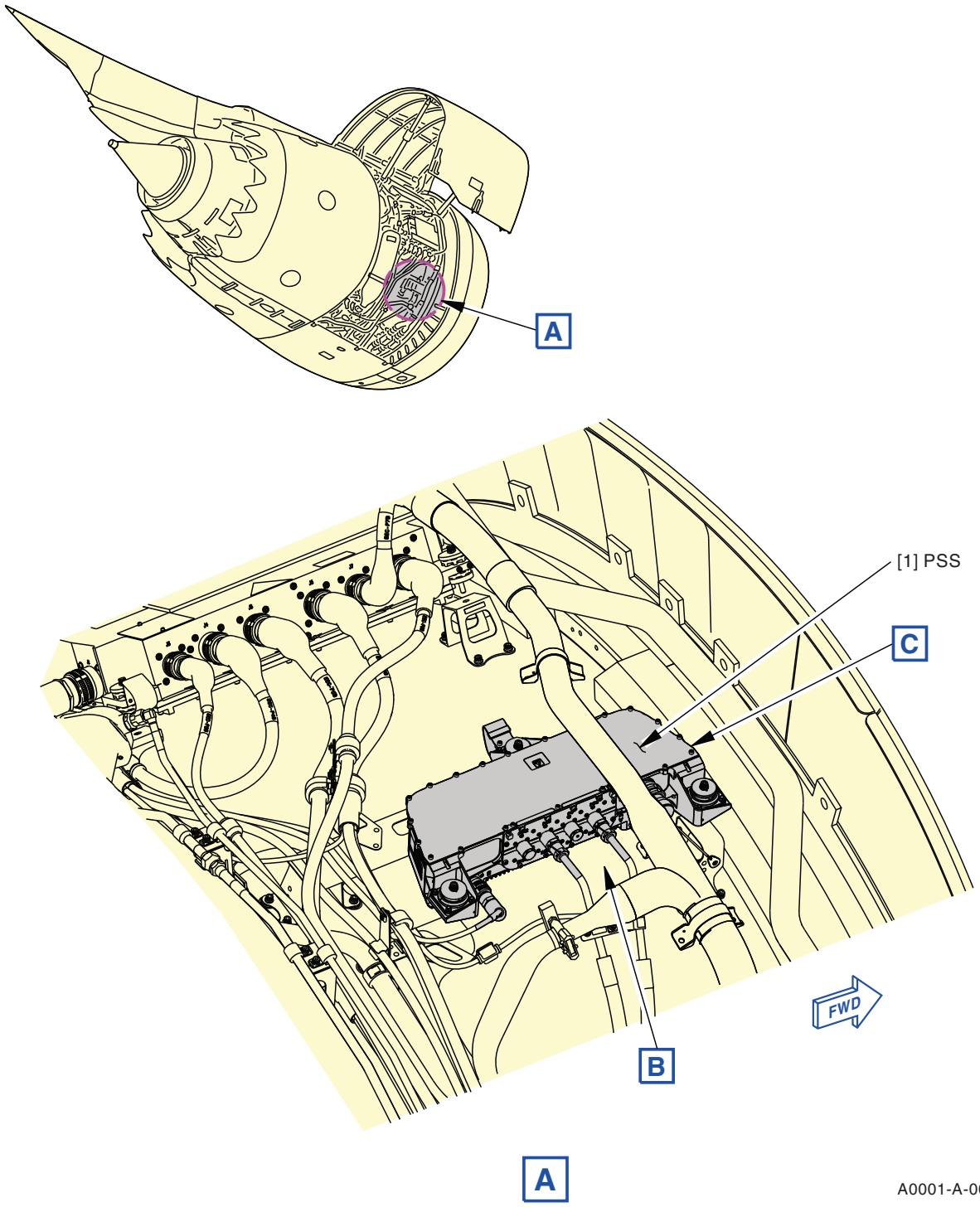
NOTE: This test makes it necessary to check the fault history.

(b) Do this task: Test No. 32 - Engine Idle System Test, TASK 71-00-00-700-808-G00.

NOTE: This test will show as failed if you find PS3 or P3B faults during the test.

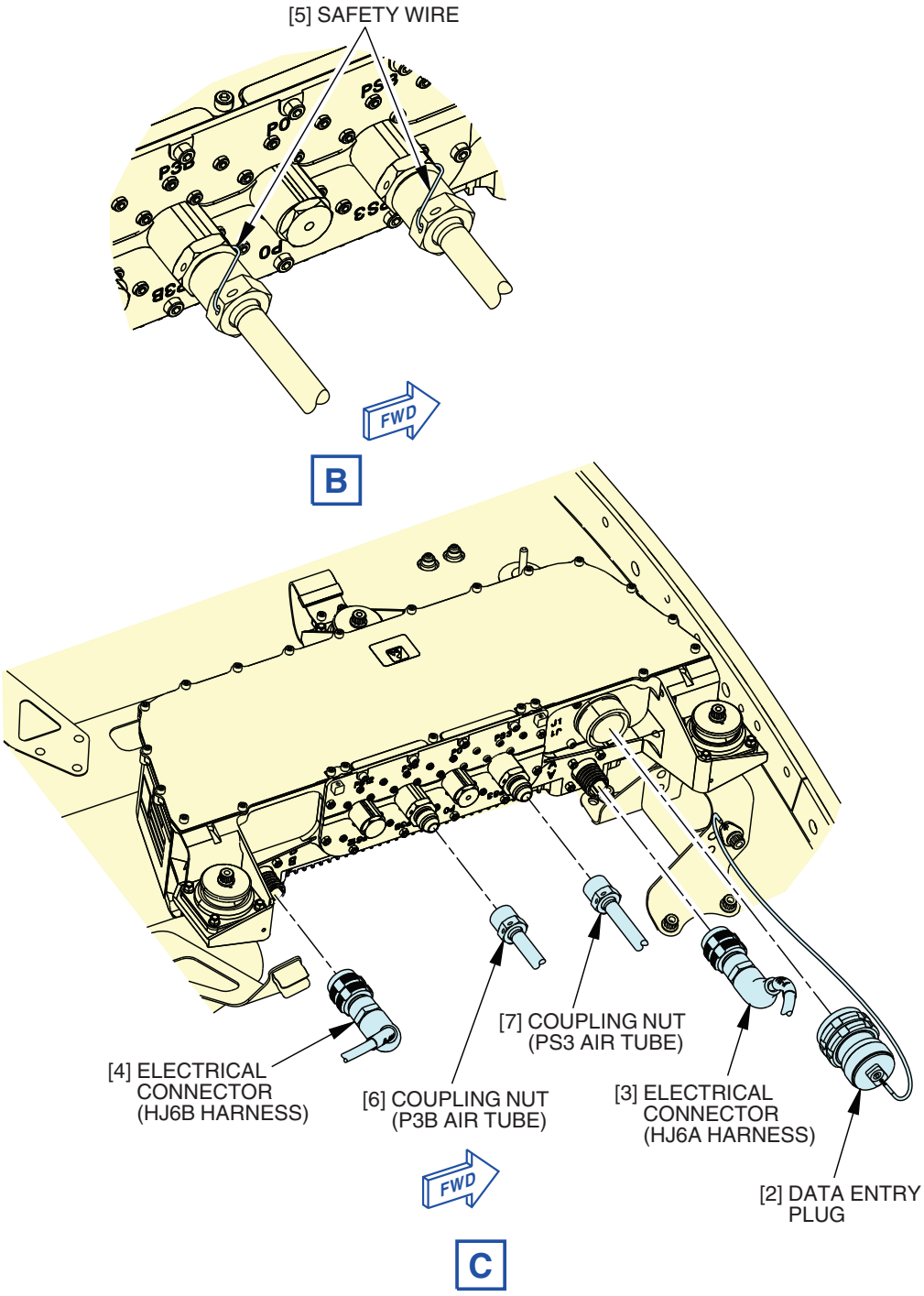
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pressure Sub-System Installation
Figure 201/73-21-11-990-803-G00 (Sheet 1 of 2)

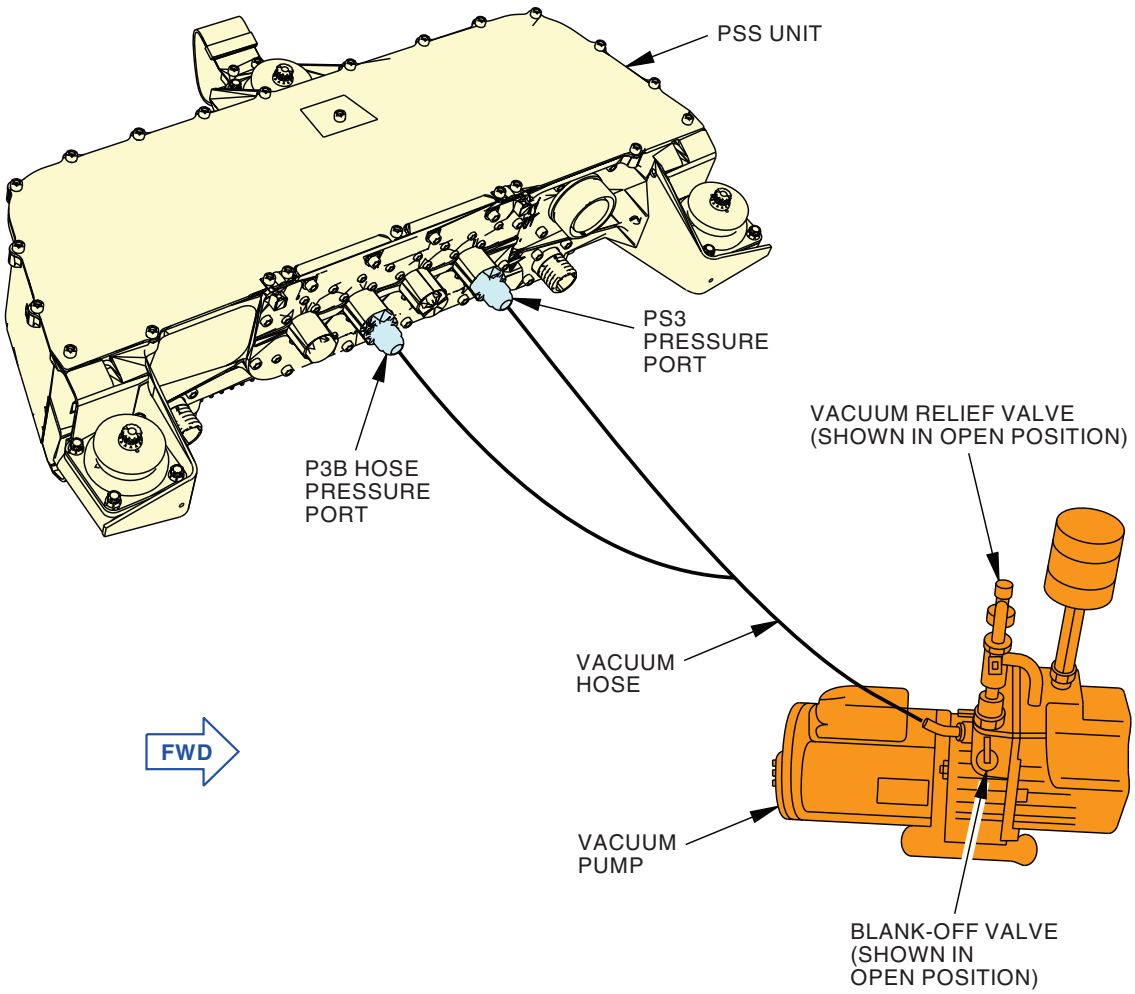
ECCN 9E991 BOEING PROPRIETARY - See title page for details



3060801 S0000824146_V2

Pressure Sub-System Installation
Figure 201/73-21-11-990-803-G00 (Sheet 2 of 2)

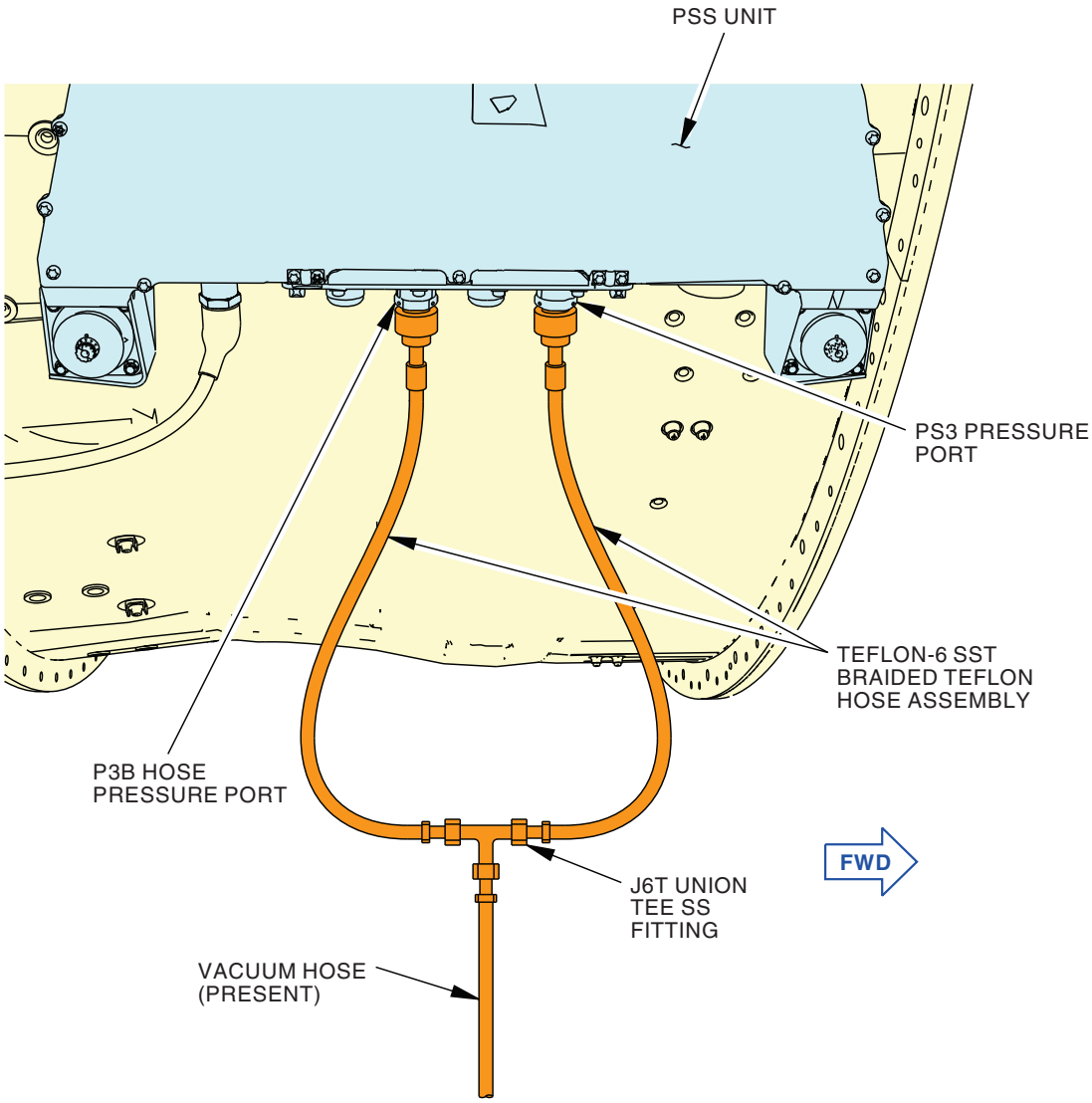
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5054808-00
2830778 S0000656733_V2

Vacuum Pump Tool Connection
Figure 202/73-21-11-990-804-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



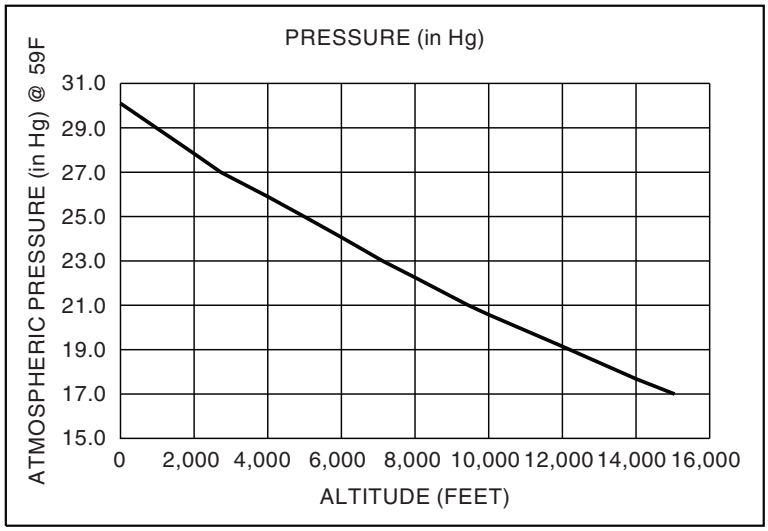
5054809-00

2830782 S0000656734_V1

Vacuum Pump Tool Connection
Figure 202/73-21-11-990-804-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ALTITUDE RANGE (FEET)	PRESSURE RANGE (in Hg)	PRESSURE RANGE (PSIA)
BELOW 5000	25 TO 30	12.3 TO 14.7
5001 TO 10000	20 TO 25	9.9 TO 12.2
10001 TO 15000	15 TO 20	7.4 TO 9.8



NOTE:

ATMOSPHERIC PRESSURE DROPS WITH INCREASE IN ALTITUDE. ATMOSPHERIC PRESSURE (in-Hg) CHANGE FOR ALTITUDE (FEET) IS SHOWN IN THE GRAPH.

FOR THIS PROCEDURE, THE ACCEPTABLE VACUUM INDICATOR GAUGE READING RANGE WITH RESPECT TO ALTITUDE RANGE IS PROVIDED IN THE TABLE.

5054810-00

2830776 S0000656735_V2

Airport Elevation Vacuum Gauge Corrections
Figure 203/73-21-11-990-805-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRESSURE SUB-SYSTEM - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Pressure Sub-System Removal

(2) Pressure Sub-System Installation.

TASK 73-21-11-000-801-G00

2. Pressure Sub-System Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the Pressure Sub-System (PSS).

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-21-05-970-803-G00	Electronic Engine Control (EEC) Data Download (On-Wing)(Selection) (P/B 201)
73-21-09-000-801-G00	Data Entry Plug Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-21-11-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-11-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-11-010-001-G00

(3) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Pressure Sub-System Removal

SUBTASK 73-21-11-010-002-G00

(1) Remove the PSS [1] as follows:

NOTE: Do an EEC NVM and CEOD download if the removed PSS will be sent to the vendor for investigation. Refer to this task: Electronic Engine Control (EEC) Data Download (On-Wing)(Selection), TASK 73-21-05-970-803-G00.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(a) Disconnect the data entry plug [2] from the PSS [1] (TASK 73-21-09-000-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(b) Use the teflon-jawed pliers, STD-664, to disconnect the electrical connector (HJ6A harness) [3] and electrical connector (HJ6B harness) [4] from the PSS [1] as follows (TASK 70-00-01-910-803-G00):

1) Disconnect the electrical connector (HJ6A harness) [3] and electrical connector (HJ6B harness) [4] from the receptacles of the PSS [1].

2) Install the protective covers on the electrical connector (HJ6A harness) [3], electrical connector (HJ6B harness) [4], and receptacles of the PSS [1].

(c) Disconnect the air tubes from the PSS [1] as follows:

1) Cut and remove the safety wire [5] from the coupling nut (P3B) [6] and coupling nut (PS3) [7].

CAUTION USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

2) Disconnect the coupling nut (P3B) [6] and coupling nut (PS3) [7] from the PSS [1].

3) Install the protective caps on the air tube openings.

(d) Remove the PSS [1] as follows:

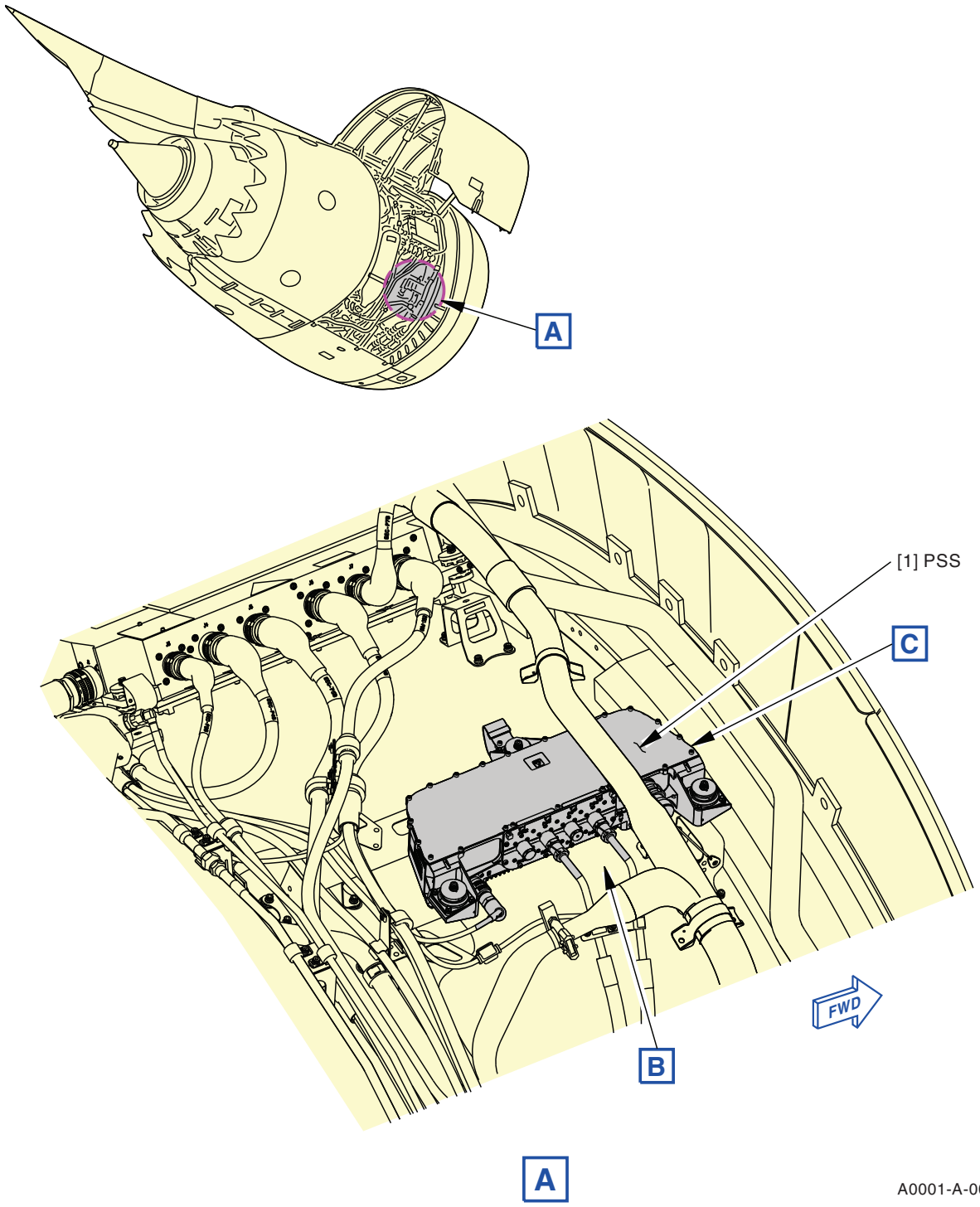
1) Remove the three bolts [8] from the mounting bracket [9].

2) Remove the PSS [1] from the mounting bracket [9].

3) Install the protective covers on the openings.

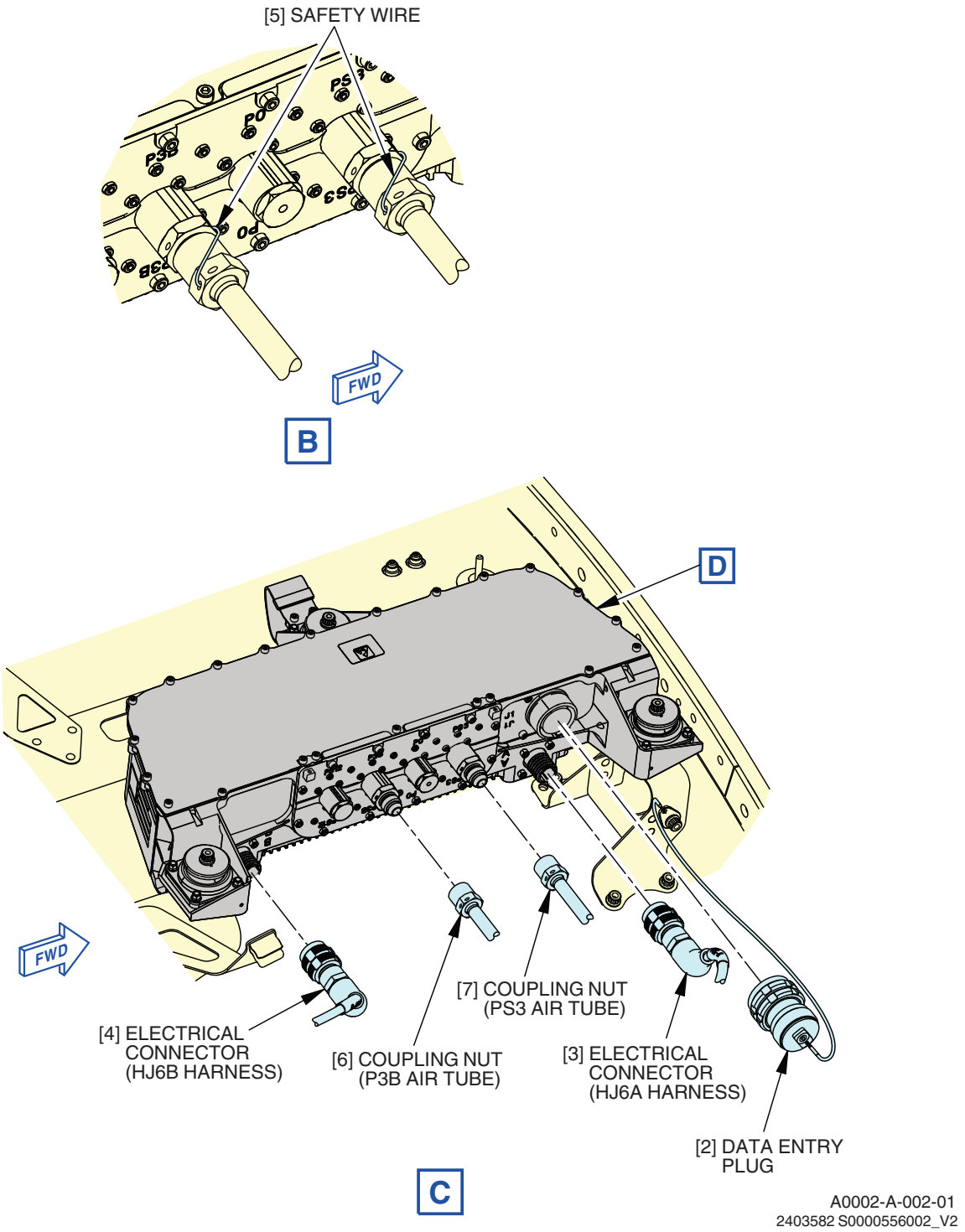
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



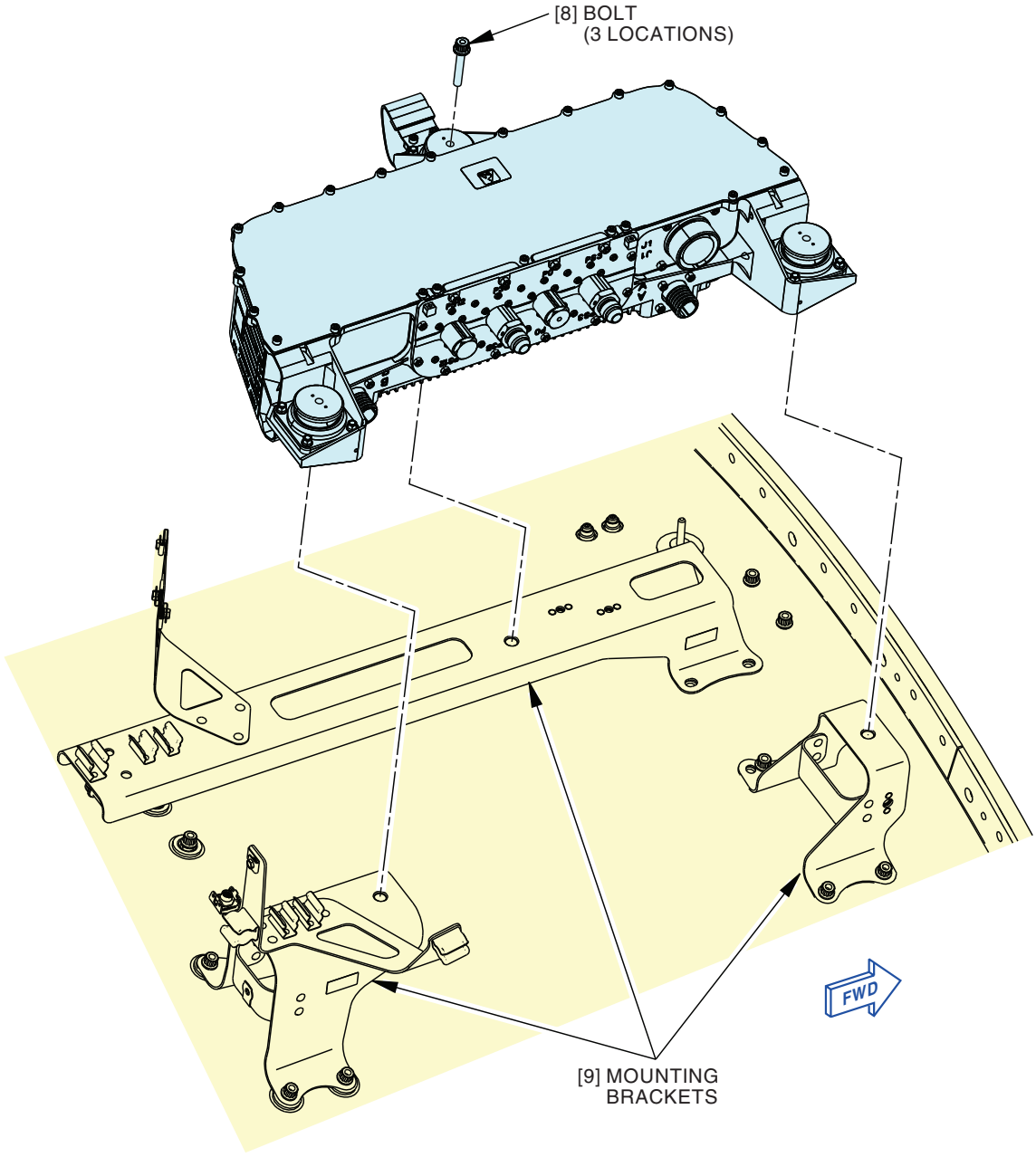
Pressure Sub-System Installation
Figure 401/73-21-11-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pressure Sub-System Installation
Figure 401/73-21-11-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pressure Sub-System Installation
Figure 401/73-21-11-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-11-400-801-G00

3. Pressure Sub-System Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the pressure sub-system.

(2) The pressure sub-system is referred to as the PSS.

(3) You must do the tests that are listed in the power plant test reference table after you install the PSS.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-09-400-801-G00	Data Entry Plug Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE
STD-3940	Air Source - Regulated, Dry Filtered, 0-150 psig

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Pressure Sub-System Installation

SUBTASK 73-21-11-160-001-G00

(1) Clean the coupling nut (PS3) [7] line and coupling nut (P3B) [6] line as follows:

(a) Remove the protective caps from the air tubes (PS3 and P3B).

(b) Use approximately 100 psig (689 kPa) (not to exceed 200 psig (1379 kPa)) to dry the air tubes (PS3 and P3B).

1) Use the regulated air source, STD-3940, to blow air into the air tubes for a minimum of 3 minutes.

2) Continue until you see no moisture exiting the sense line weep hole located at the 6:00 o'clock position in the aft location of the fan compartment.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Do a visual inspection of the weep holes on the air tubes to make sure that they are clear of all blockage.

SUBTASK 73-21-11-420-001-G00

(2) Install the PSS [1] as follows:

(a) Remove the protective covers from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

NOTE: You can lubricate the coupling before or after the installation of the pressure sub-system on the mounting bracket.

(c) Put the PSS [1] on the mounting bracket [9].

(d) Lubricate the threads of the three bolts [8] with graphite mineral vaseline, D00601 [CP2101].

(e) Install the three bolts [8] that attach the PSS [1] to the mounting bracket [9].

(f) Tighten the bolts [8] to 109.25 in-lb (12.34 N·m) - 120.75 in-lb (13.64 N·m).

SUBTASK 73-21-11-420-002-G00

(3) Connect the air tubes to the PSS [1] as follows:

(a) Make sure that the mating faces are clean and clear of unwanted materials.

(b) Lubricate the threads of the coupling nut (P3B) [6] and coupling nut (PS3) [7] with the oil, D00599 [CP2442].

(c) Connect the coupling nut (P3B) [6] and coupling nut (PS3) [7] to the PSS [1].

(d) Tighten the coupling nut (P3B) [6] and coupling nut (PS3) [7] as follows:

CAUTION

USE TWO WRENCHES TO LOOSEN OR TIGHTEN THE COUPLING NUTS. USE ONE TO HOLD THE FITTINGS, AND THE SECOND WRENCH TO LOOSEN OR TIGHTEN THE COUPLING NUT. IF YOU DO NOT USE TWO WRENCHES CORRECTLY, DAMAGE TO THE EQUIPMENT CAN OCCUR.

1) Tighten the two coupling nuts to 270 in-lb (31 N·m) - 300 in-lb (34 N·m).

2) Make sure that the two coupling nuts are fully seated.

3) If the coupling nut(s) are not fully seated, do the steps above again to tighten the coupling nut(s).

(e) Install the safety wire [5] (safety wire, G02345 [CP8001]) between PS3 sense line coupling nut (PS3) [7] and the PS3 pressure port nipple of the PSS [1].

(f) Install the safety wire [5] (safety wire, G02345 [CP8001]) between P3B sense line coupling nut (P3B) [6] and the PS3 pressure port nipple of the PSS [1].

SUBTASK 73-21-11-420-003-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(4) Use the teflon-jawed pliers, STD-664 to connect the electrical connector (HJ6A harness) [3] and electrical connector (HJ6B harness) [4] to the PSS [1] as follows (TASK 70-00-01-910-803-G00):

(a) Remove the protective covers from the electrical connector (HJ6A harness) [3], electrical connector (HJ6B harness) [4], and receptacles of the PSS [1].

(b) Connect the electrical connector (HJ6A harness) [3] and electrical connector (HJ6B harness) [4] to the PSS [1].

(c) Tighten the electrical connector (HJ6A harness) [3] and electrical connector (HJ6B harness) [4].

SUBTASK 73-21-11-420-004-G00

(5) Connect the data entry plug [2] to the PSS [1] (TASK 73-21-09-400-801-G00).

SUBTASK 73-21-11-160-002-G00

(6) Clean the work area and clear the tools and other items.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-11-860-003-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-11-860-004-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-11-410-002-G00

(3) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

I. Pressure Sub-System Installation Test

SUBTASK 73-21-11-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PRESSURE SUB-SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Pressure Sub-System Inspection

TASK 73-21-11-210-801-G00

2. Pressure Sub-System Inspection

(Figure 601)

A. General

(1) This task provides the instructions to inspect the Pressure Sub-System (PSS).

(2) The PSS is located on the right side of the fan case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-11-000-801-G00	Pressure Sub-System Removal (P/B 401)
73-21-11-400-801-G00	Pressure Sub-System Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

F. Prepare for the Inspection

SUBTASK 73-21-11-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-11-860-006-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-11-010-003-G00

(3) On the applicable engine, open the right fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

G. Pressure Sub-System Inspection

SUBTASK 73-21-11-210-001-G00

(1) Inspect the Pressure Sub-System (PSS):

(a) Make sure that the PSS is not loose.

1) If the PSS is loose, tighten the bolts that attach the PSS to the mount brackets (Pressure Sub-System Installation, TASK 73-21-11-400-801-G00).

(b) Make sure that the pressure sense hose fittings are not loose.

1) If a fitting is loose, tighten the fitting (Pressure Sub-System Installation, TASK 73-21-11-400-801-G00).

2) Inspect the P3B and PS3 tube-hose manifolds for general condition and security.

(c) Look for cracks in the PSS housing.

1) If you see cracks, replace the PSS (Pressure Sub-System Removal, TASK 73-21-11-000-801-G00 and Pressure Sub-System Installation, TASK 73-21-11-400-801-G00).

(d) Make sure that the electrical connectors are not loose.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-11-410-003-G00

(1) On the applicable engine, close the right fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
414	Right Fan Cowl, Engine 1
424	Right Fan Cowl, Engine 2

SUBTASK 73-21-11-860-007-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-11-860-008-G00

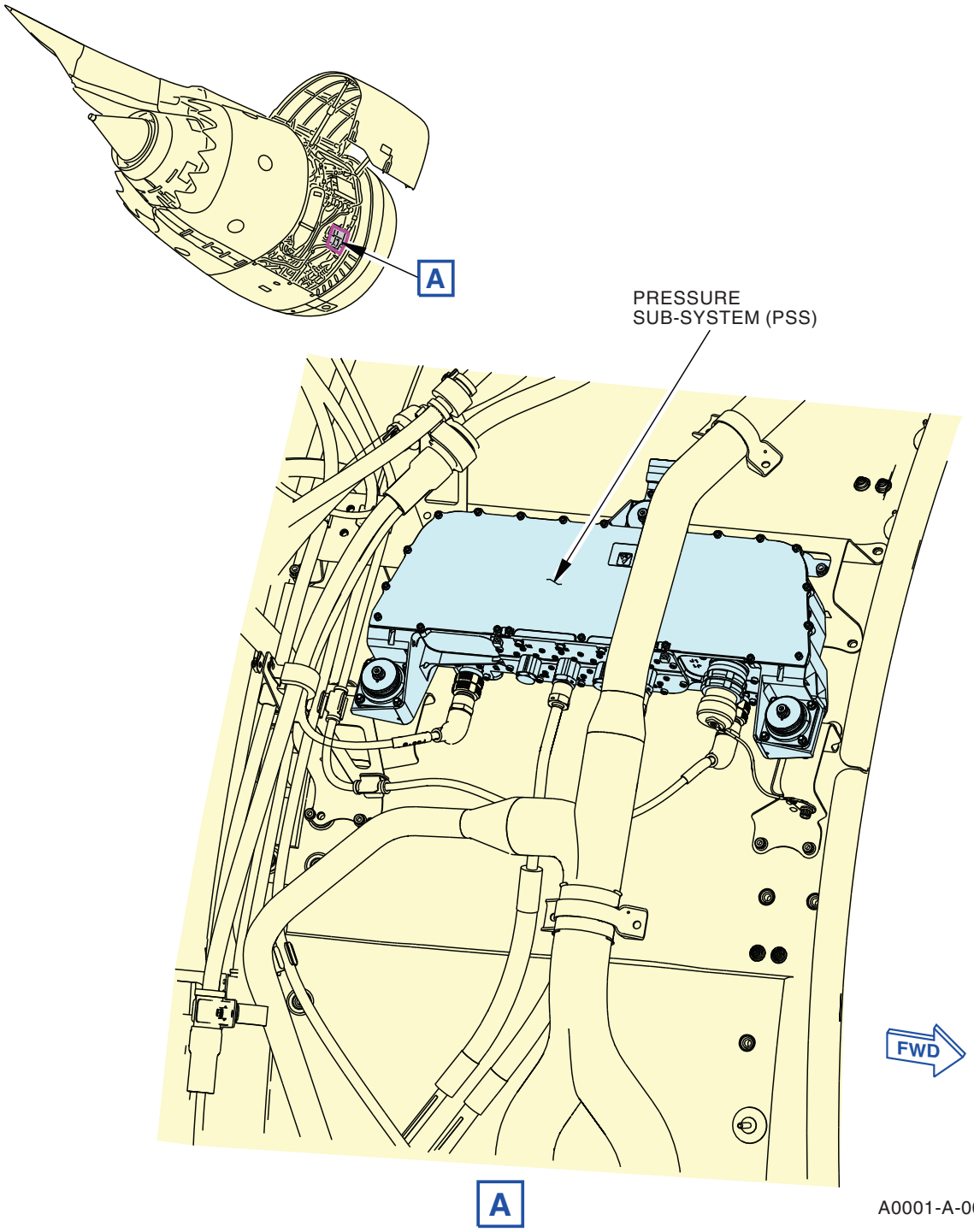
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Pressure Sub-System Inspection
Figure 601/73-21-11-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ENGINE CONTROL MODULE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Engine Control Module Removal

(2) Engine Control Module Installation.

TASK 73-21-12-000-801-G00

2. Engine Control Module Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the engine control module (P5-68).

(2) The engine control module (P5-68) is on the P5 Aft Overhead Panel in the flight compartment.

B. References

Reference	Title
20-40-12-000-802	ESDS Handling for Metal Encased Unit Removal (P/B 201)

C. Location Zones

Zone	Area
211	Flight Compartment - Left

D. Engine Control Module Removal

SUBTASK 73-21-12-860-001-G00

(1) Do these steps:

(a) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	13	C01277	MASTER CAUTION ANNUNCIATOR CONT 3
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

(b) Open these circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

(c) Open these circuit breakers and install safety tags:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-12-910-001-G00

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(2) Do this task: ESDS Handling for Metal Encased Unit Removal, TASK 20-40-12-000-802.

SUBTASK 73-21-12-020-001-G00

(3) Do these steps to remove the engine control module [1]:

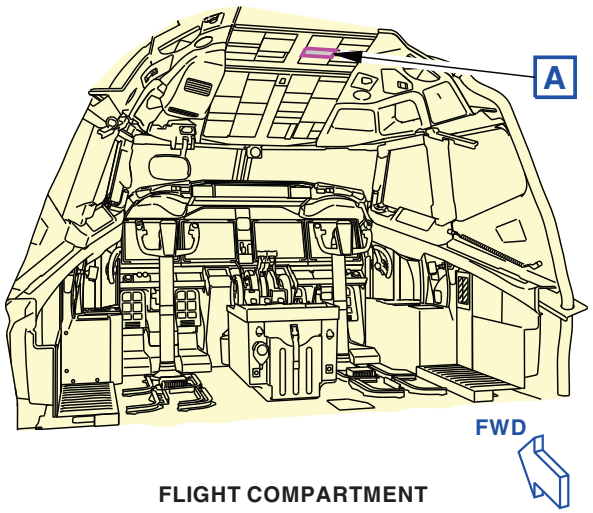
(a) Loosen the four quarter-turn fasteners on the front of the module.

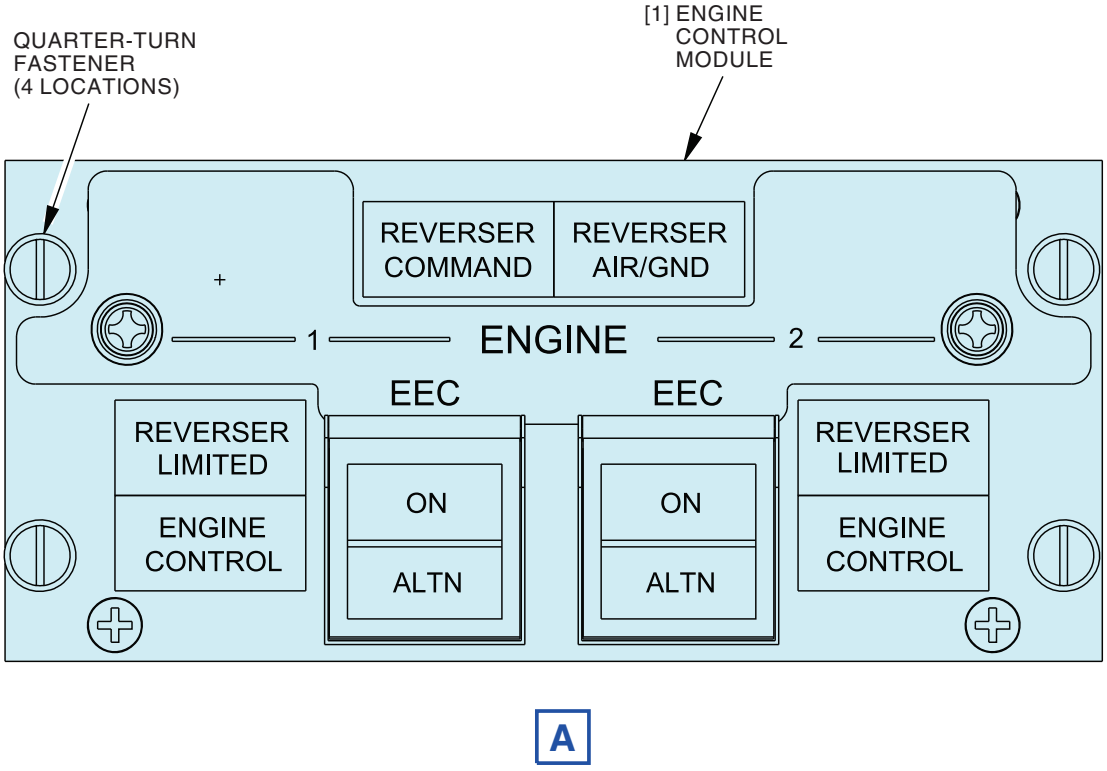
(b) Remove the module from the P5 Aft Overhead Panel.

(c) Remove the two electrical connectors from the rear of the module.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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Engine Control Module Installation
Figure 401/73-21-12-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-12-400-801-G00

3. Engine Control Module Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the engine control module (P5-68).

(2) The engine control module (P5-68) is on the P5 Aft Overhead Panel in the flight compartment.

B. References

Reference	Title
20-40-12-400-802	ESDS Handling for Metal Encased Unit Installation (P/B 201)
71-00-00-700-809-G00	Test No. 33 - Lamp Check Out Special Functions Test (P/B 501)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Engine control module	73-21-12-01-050	PTF ALL

D. Location Zones

Zone	Area
211	Flight Compartment - Left

E. Engine Control Module Installation

SUBTASK 73-21-12-910-002-G00

CAUTION

DO NOT TOUCH THE UNIT BEFORE YOU DO THE PROCEDURE FOR DEVICES THAT ARE SENSITIVE TO ELECTROSTATIC DISCHARGE. ELECTROSTATIC DISCHARGE CAN CAUSE DAMAGE TO THE UNIT.

(1) Do this task: ESDS Handling for Metal Encased Unit Installation, TASK 20-40-12-400-802.

SUBTASK 73-21-12-420-001-G00

(2) Do these steps to install the engine control module [1]:

(a) Connect the two electrical connectors at the rear of the module.

(b) Install the module into the P5 Aft Overhead Panel.

(c) Tighten the four quarter-turn fasteners on the front of the module.

SUBTASK 73-21-12-860-002-G00

(3) Do these steps:

(a) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
C	13	C01277	MASTER CAUTION ANNUNCIATOR CONT 3
F	13	C01179	INDICATOR MASTER DIM SECT 7
F	14	C01180	INDICATOR MASTER DIM SECT 8

(b) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	5	C04007	ENGINE 1 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

(c) Remove the safety tags and close these circuit breakers:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F. Engine Control Module Installation Test

SUBTASK 73-21-12-710-001-G00

(1) For each EEC, do this task: Test No. 33 - Lamp Check Out Special Functions Test, TASK 71-00-00-700-809-G00.

SUBTASK 73-21-12-710-002-G00

(2) Do these steps to examine the EEC ON/ALTN switch:

(a) Do this task to supply power to both ENG 1 and ENG 2 EECs: EEC Maintenance Power Selection, TASK 73-21-00-800-801-G00.

(b) On the engine control module, put both Engine 1 and Engine 2 EEC ON switches to the ON position.

(c) Make sure that both EEC ON lights (white) come on.

(d) Push the Engine 1 EEC ON switch so that ON goes off.

(e) Make sure that the Engine 1 amber ALTN light comes on.

(f) Push the Engine 1 EEC ON switch so that the ON shows.

(g) Make sure that the Engine 1 amber ALTN light goes off.

(h) Push the Engine 2 EEC ON switch so that ON goes off.

(i) Make sure that the Engine 2 amber ALTN light comes on.

(j) Push the Engine 2 EEC ON switch so that the ON shows.

(k) Make sure that the Engine 2 amber ALTN light goes off.

(l) Remove power to both ENG 1 and ENG 2 EECs to put the airplane back to the usual condition (TASK 73-21-00-800-801-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

T25 SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) T25 sensor removal

(2) T25 sensor installation.

TASK 73-21-63-000-801-G00

2. T25 Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the T25 Sensor.

(2) The T25 sensor is installed at the 3:00 o'clock position on the fan hub frame.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-21-63-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-63-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-63-010-001-G00

(3) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

SUBTASK 73-21-63-010-002-G00

(4) Remove the segment shroud 2 (TASK 72-23-04-000-801-G00).

G. T25 Sensor Removal

SUBTASK 73-21-63-030-001-G00

(1) Remove the T25 sensor [3]:

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(a) Use the teflon-jawed pliers, STD-664 to disconnect the channel B electrical connector [1] and the channel A electrical connector [5] from the T25 sensor [3] (TASK 70-00-01-910-803-G00).

1) Disconnect the channel B electrical connector [1] and the channel A electrical connector [5] from the T25 sensor [3].

2) Install the protective covers on the T25 sensor [3] and its receptacles and on the channel B electrical connector [1] and the channel A electrical connector [5].

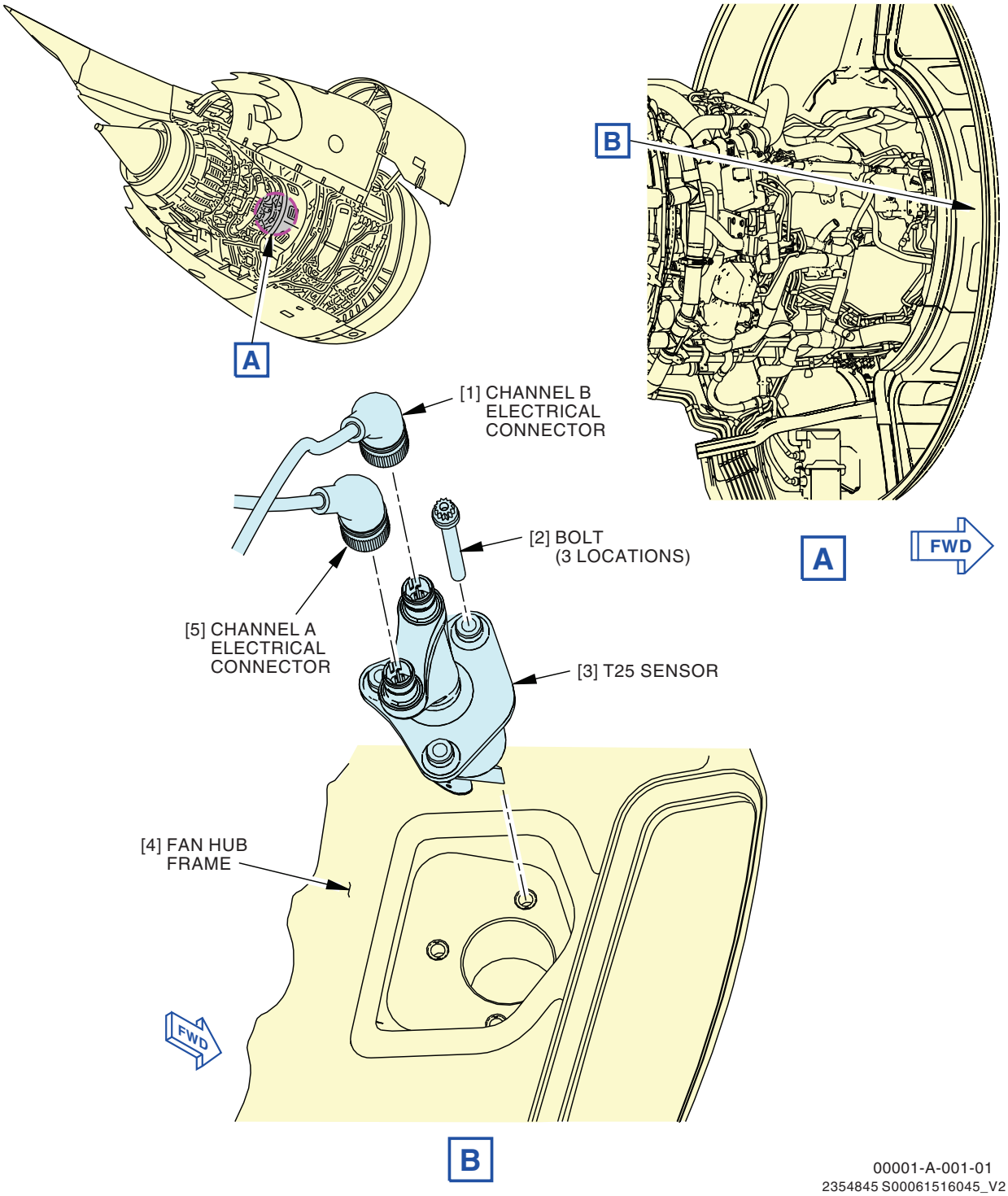
(b) Remove the three bolts [2] that attach the T25 sensor [3] to the fan hub frame [4].

1) Remove the T25 sensor [3].

(c) Install the protective covers on the T25 sensor [3] and the open port on the fan hub frame [4].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



T25 Sensor Installation
Figure 401/73-21-63-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-63-400-801-G00

3. T25 Sensor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the T25 Sensor.

(2) You must do the tests that are listed in the power plant test reference table after you install the T25 sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	T25 sensor	73-21-63-01-010	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

H. T25 Sensor Installation

SUBTASK 73-21-63-090-001-G00

(1) Install the T25 sensor [3]:

(a) Remove the protective covers from the T25 sensor [3] and the open port on the fan hub frame [4].

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(b) Lubricate the threads and friction surfaces of the three bolts [2] with engine oil, D00599 [CP2442].

(c) Put the T25 sensor [3] in its position on the fan hub frame [4].

(d) Install the three bolts [2] that attach the T25 sensor [3] to the fan hub frame [4].

1) Tighten the three bolts [2] to 109 in-lb (12.3 N·m) - 121 in-lb (13.7 N·m).

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(e) Use the teflon-jawed pliers, STD-664 to connect the channel B electrical connector [1] and the channel A electrical connector [5] to the T25 sensor [3] as follows (TASK 70-00-01-910-803-G00):

1) Remove the protective covers from the channel B electrical connector [1], the channel A electrical connector [5] and the T25 sensor [3] receptacles.

2) Connect the channel B electrical connector [1] and the channel A electrical connector [5] to the T25 sensor [3].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-63-410-002-G00

(1) Install the segment shroud 2 (TASK 72-23-04-400-801-G00).

SUBTASK 73-21-63-860-003-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-63-860-004-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

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(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-63-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(4) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

J. T25 Sensor Installation Test

SUBTASK 73-21-63-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

T25 SENSOR - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) T25 Sensor inspection.

TASK 73-21-63-210-801-G00

2. T25 Sensor Inspection

(Figure 601)

A. General

(1) This task gives the instructions to inspect the T25 Sensor.

(2) The T25 sensor is installed at the 3:00 o'clock position on the fan hub frame.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-000-801-G00	Fan Frame Module Segment Shrouds Removal (P/B 401)
72-23-04-400-801-G00	Fan Frame Module Segment Shrouds Installation (P/B 401)
73-21-63-400-801-G00	T25 Sensor Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 73-21-63-860-005-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-63-860-006-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-63-010-003-G00

(3) Do these tasks in sequence to safely open the right thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

CAUTION IF FAN COWLS ARE INSTALLED, MAKE SURE THAT LEFT AND RIGHT FAN COWLS ARE IN THE FULL OPEN POSITION. MAKE SURE THAT THE SPRING DOOR OPENING-SYSTEM (SDOS) AND HOLD OPEN RODS (HOR) ARE LOCKED IN THEIR POSITION. IF YOU DO NOT, STRUCTURAL DAMAGE TO THE FAN COWL AND THRUST REVERSER CAN OCCUR.

(b) Open both fan cowl panels (TASK 71-11-04-010-801-G00).

NOTE: Because the center line of the thrust reversers is off 6:00 o'clock position, both fan cowl panels must be opened to prevent damaging the fan cowl panel if either thrust reverser needs to be opened.

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1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable right thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

SUBTASK 73-21-63-010-004-G00

(4) Remove the segment shroud 2 (Fan Frame Module Segment Shrouds Removal, TASK 72-23-04-000-801-G00).

G. T25 Sensor Inspection

SUBTASK 73-21-63-210-001-G00

(1) Inspect the T25 sensor:

(a) Make sure that the T25 sensor is not loose.

1) If the sensor is loose, tighten the bolts that attach the sensor to the fan hub frame (T25 Sensor Installation, TASK 73-21-63-400-801-G00).

(b) Make sure that the electrical connector is not loose.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-63-410-003-G00

(1) Install the segment shroud 2 (Fan Frame Module Segment Shrouds Installation, TASK 72-23-04-400-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-21-63-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(2) Do these tasks in sequence to safely close the right thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
416	Right Thrust Reverser, Engine 1
426	Right Thrust Reverser, Engine 2

(b) Close the fan cowl panels (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-21-63-860-007-G00

(3) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-63-860-008-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

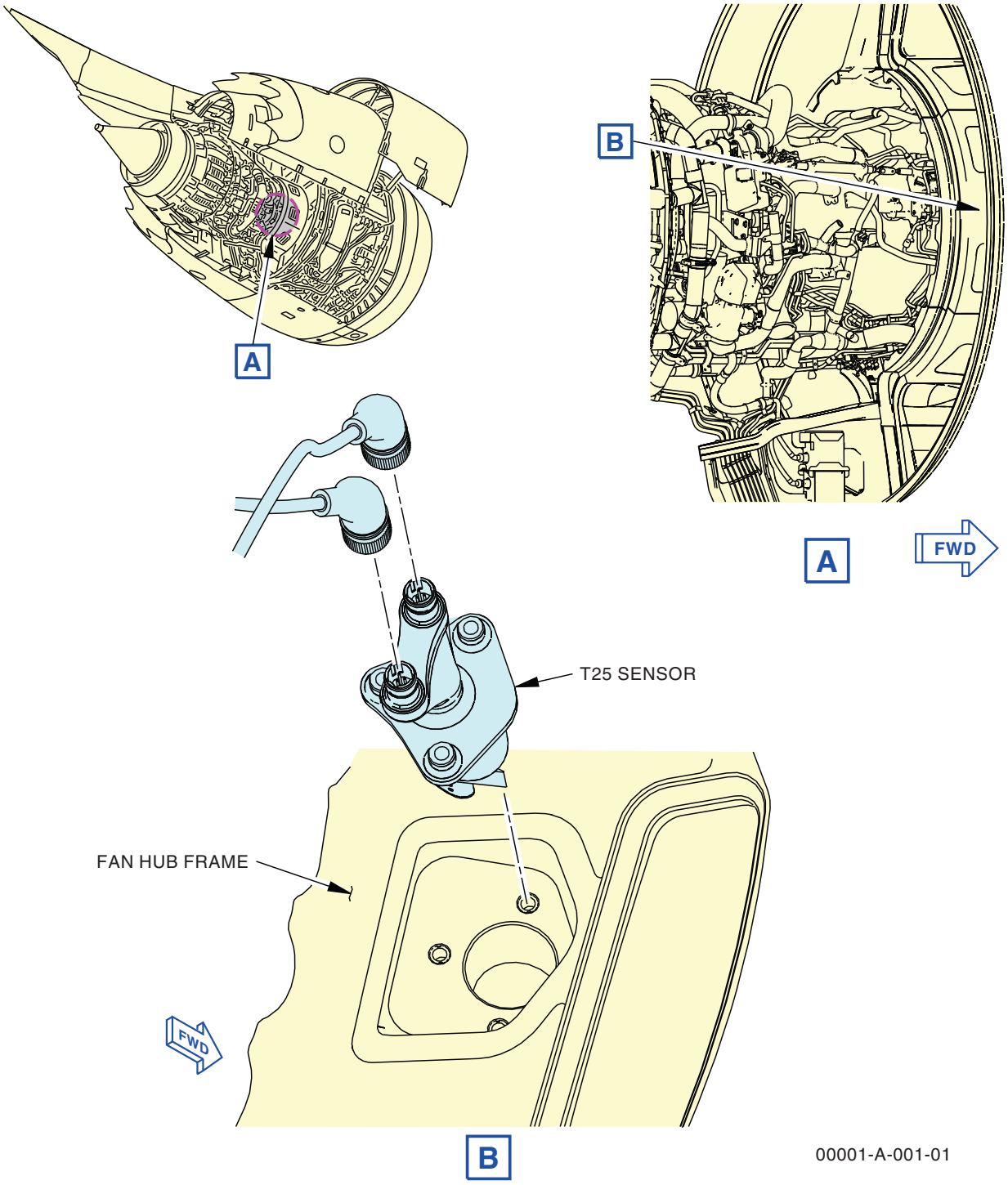
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



T25 Sensor Inspection
Figure 601/73-21-63-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

T3 SENSOR - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) T3 Sensor Removal

(2) T3 Sensor Installation.

TASK 73-21-64-000-801-G00

2. T3 Sensor Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the T3 sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-10718	Fiberscope (Meets General Electric or CFM Standard Practices Manual Specification)

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-21-64-010-004-G00

(1) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. T3 Sensor Removal

SUBTASK 73-21-64-020-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use a teflon-jawed pliers, STD-664, to disconnect the electrical connectors as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the HJ8A electrical connector CH A [21] from the T3 sensor [7] CH A.

(b) Disconnect the HJ8B electrical connector CH B [22] from the T3 sensor [7] CH B.

(c) Install protective covers on the HJ8A electrical connector CH A [21] and HJ8B electrical connector CH B [22].

(d) Install a protective cover on the T3 sensor [7] and its receptacles.

SUBTASK 73-21-64-030-001-G00

(2) Remove the T3 sensor [7] as follows:

(a) Remove the bolt [8] that attaches the loop clamp [9] to the support bracket [20].

(b) Remove the bolt [18] that attaches the loop clamp [19] to the support bracket [10].

(c) Remove the bolt [11] that attaches the loop clamp [12] to the support bracket [17].

(d) Remove the safety cable or lockwire that secures the T3 sensor [7] to the combustor case threaded insert adapter [16].

(e) Disconnect the T3 sensor [7] nut from the combustor case threaded insert adapter [16].

(f) Remove the bolt [13] that attaches the loop clamp [14] to the support bracket [15].

(g) Remove the HJ8A electrical harness [26] from the clip on the support bracket [20].

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(h) Remove the bolt [25] that attaches the loop clamp [24] and HJ8A electrical harness [26] to the support bracket [20].

(i) Remove the two bolts [27] that attach the T3 sensor [7] to the support bracket [28].

(j) Remove the T3 sensor [7] with the attached support bracket [20] from the engine.

NOTE: Do not bend the sensor cable when you remove T3 sensor [7].

(k) Remove the two bolts [23] that attach the T3 sensor [7].

(l) Examine the T3 sensor [7] inlet for loose metal particles.

1) If you see metal particles on the inlet, remove the particles for more analysis.

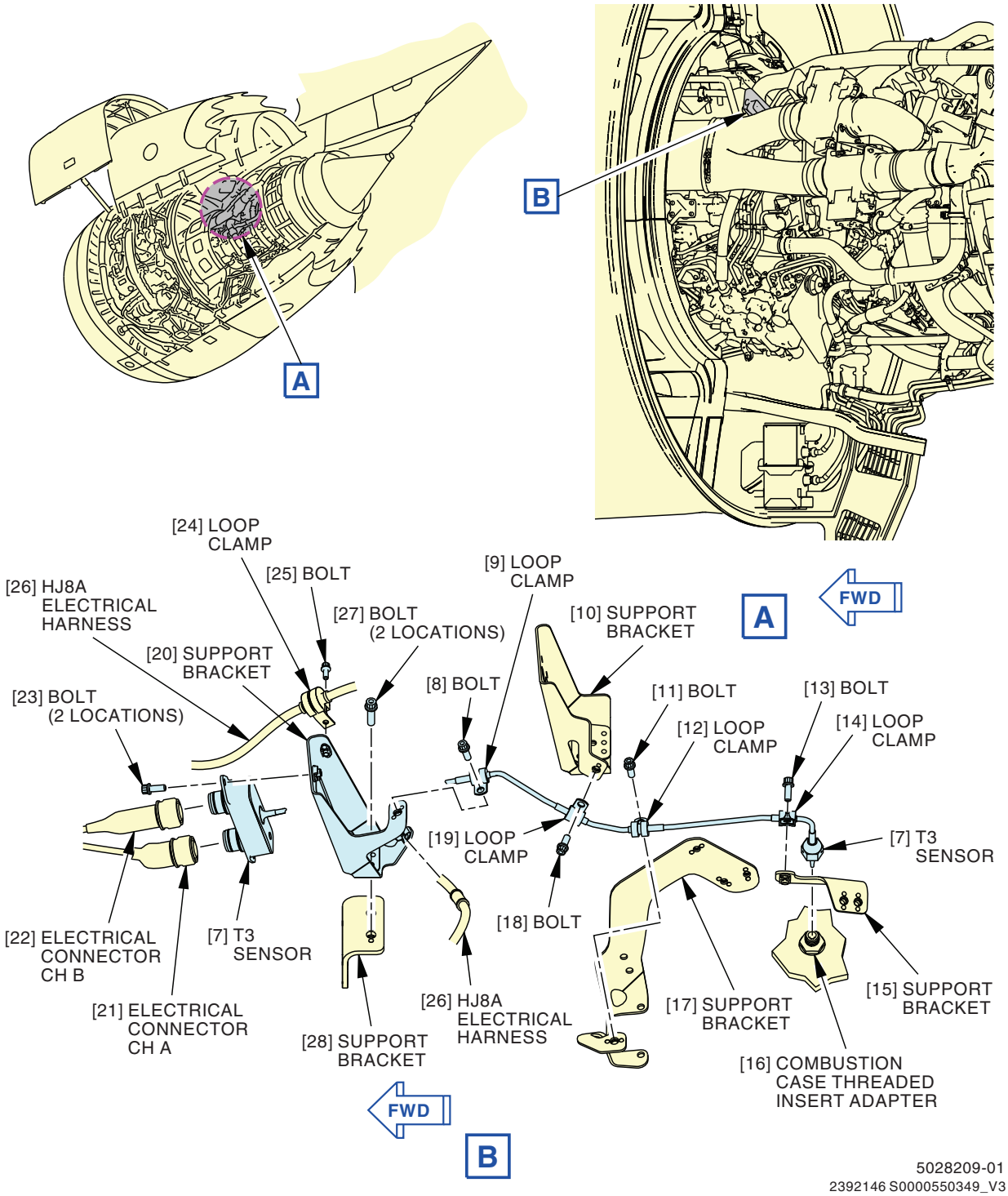
2) Examine the inner section of the T3 sensor [7] with a fiberscope, STD-10718, after you remove the particles.

3) If you see metal particles in the sensor, remove the particles for more analysis.

(m) Install protective covers on the T3 sensor [7] and combustor case threaded insert adapter [16].

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



T3 Sensor Installation
Figure 401/73-21-64-990-801-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-64-400-801-G00

3. T3 Sensor Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the T3 sensor.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-41-00-400-801-G00	Lockwire Installation (P/B 201)
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D50034 [CP2619]	Compound - Antiseize - Pure Nickel Special, Never - Seez	
D50355 [CP2961]	Lubricant - Anti-Seize Compound	D6Y31C2
G02495 [CP8002]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS5689
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

G. T3 Sensor Installation

SUBTASK 73-21-64-420-001-G00

(1) Install the T3 sensor [7] as follows:

(a) Remove the protective cover from the T3 sensor [7] and combustor case threaded insert adapter [16].

(b) Put a light layer of oil, D00599 [CP2442], on the threads of the two bolts [23].

(c) Install the two bolts [23] that attach the T3 sensor [7] to the support bracket [20].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that you do not bend the sensor cable when you attach the T3 sensor [7] to the bracket.

(d) Put the T3 sensor [7] with the support bracket [20] into its position on the support bracket [28].

(e) Put a light layer of engine oil, D00599 [CP2442], on the threads of the two bolts [27].

(f) Install the two bolts [27] that attach the T3 sensor [7] to the support bracket [28].

1) Do not tighten at this time.

(g) Put a light layer of Tiodize T8E-H grease, D50355 [CP2961], or pure nickel special compound, D50034 [CP2619] (alternate), on the threads of the combustor case threaded insert adapter [16].

(h) Connect the T3 sensor [7] nut on the combustor case threaded insert adapter [16].

(i) Put a light layer of engine oil, D00599 [CP2442], on the threads of the bolt [8], bolt [11], and bolt [13].

(j) Put a light layer of engine oil, D00599 [CP2442], on the threads of the bolt [18] and bolt [25].

(k) Install the bolt [25] that attaches the loop clamp [24] and HJ8A electrical harness [26] to the support bracket [20].

1) Do not tighten at this time.

(l) Install the HJ8A electrical harness [26] in the clip on the support bracket [20].

(m) Install the bolt [13] that attaches the loop clamp [14] to the support bracket [15].

(n) Install the bolt [11] that attaches the loop clamp [12] to the support bracket [17].

(o) Install the bolt [18] that attaches the loop clamp [19] to the support bracket [10].

(p) Install the bolt [8] that attaches the loop clamp [9] to the support bracket [20].

(q) Tighten the T3 sensor [7] nut to 262.0 in-lb (29.60 N·m) - 308.0 in-lb (34.80 N·m).

(r) Tighten the two bolts [23] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(s) Tighten the two bolts [27] to 106.0 in-lb (11.98 N·m) - 124.0 in-lb (14.01 N·m).

(t) Tighten the bolt [25] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

(u) Tighten the bolt [8] and bolt [11] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

(v) Tighten the bolt [13] and bolt [18] to 60.0 in-lb (6.8 N·m) - 70.0 in-lb (7.9 N·m).

(w) Install a safety wire, G02495 [CP8002], or safety cable, G50065 [CP8006], on the T3 sensor [7] and combustor case threaded insert adapter [16] (TASK 70-41-00-400-801-G00 or TASK 70-41-00-400-802-G00).

SUBTASK 73-21-64-420-002-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTOR AND RECEPTACLE ARE CLEAN WHEN YOU CONNECT THEM. DIRTY CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT USE PLIERS THAT HAVE METAL JAWS TO TIGHTEN THE ELECTRICAL CONNECTOR. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Use a teflon-jawed pliers, STD-664, to connect these electrical connectors (TASK 70-00-01-910-803-G00):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Remove the protective covers from the HJ8A electrical connector CH A [21], HJ8B electrical connector CH B [22], and T3 sensor [7] receptacles.

(b) Connect and tighten the HJ8A electrical connector CH A [21] to the T3 sensor [7] CH A.

(c) Connect and tighten the HJ8B electrical connector CH B [22] to the T3 sensor [7] CH B.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-64-410-004-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

I. T3 Sensor Installation Test

SUBTASK 73-21-64-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

T3 SENSOR - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) T3 Sensor Inspection.

TASK 73-21-64-210-801-G00

2. T3 Sensor Inspection

(Figure 601)

A. General

(1) This task gives the instructions to inspect the T3 Sensor.

(a) The T3 sensor is located on the left side of the HPC case.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-64-000-801-G00	T3 Sensor Removal (P/B 401)
73-21-64-400-801-G00	T3 Sensor Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 73-21-64-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-64-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-64-010-005-G00

(3) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. T3 Sensor Inspection

SUBTASK 73-21-64-210-001-G00

(1) Inspect the T3 sensor:

(a) Look for signs of air leaks.

1) If you see signs of air leaks, tighten the sensor nut and if it is necessary, the combustor case adapter (T3 Sensor Installation, TASK 73-21-64-400-801-G00).

(b) Make sure that the T3 sensor is not loose.

1) If the T3 sensor is loose, tighten the sensor nut and if it is necessary, the combustor case adapter (T3 Sensor Installation, TASK 73-21-64-400-801-G00).

(c) Look for cracks.

1) If you see cracks, replace the T3 sensor (T3 Sensor Removal, TASK 73-21-64-000-801-G00 and T3 Sensor Installation, TASK 73-21-64-400-801-G00).

(d) Look for signs of chaffing or wear on the electrical lead.

1) If you see chaffing or wear, do the steps in these tasks to replace the electrical lead for the T3 sensor (T3 Sensor Removal, TASK 73-21-64-000-801-G00 and T3 Sensor Installation, TASK 73-21-64-400-801-G00).

(e) Make sure that the electrical connector is not loose.

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-64-410-005-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-21-64-860-003-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-21-64-860-004-G00

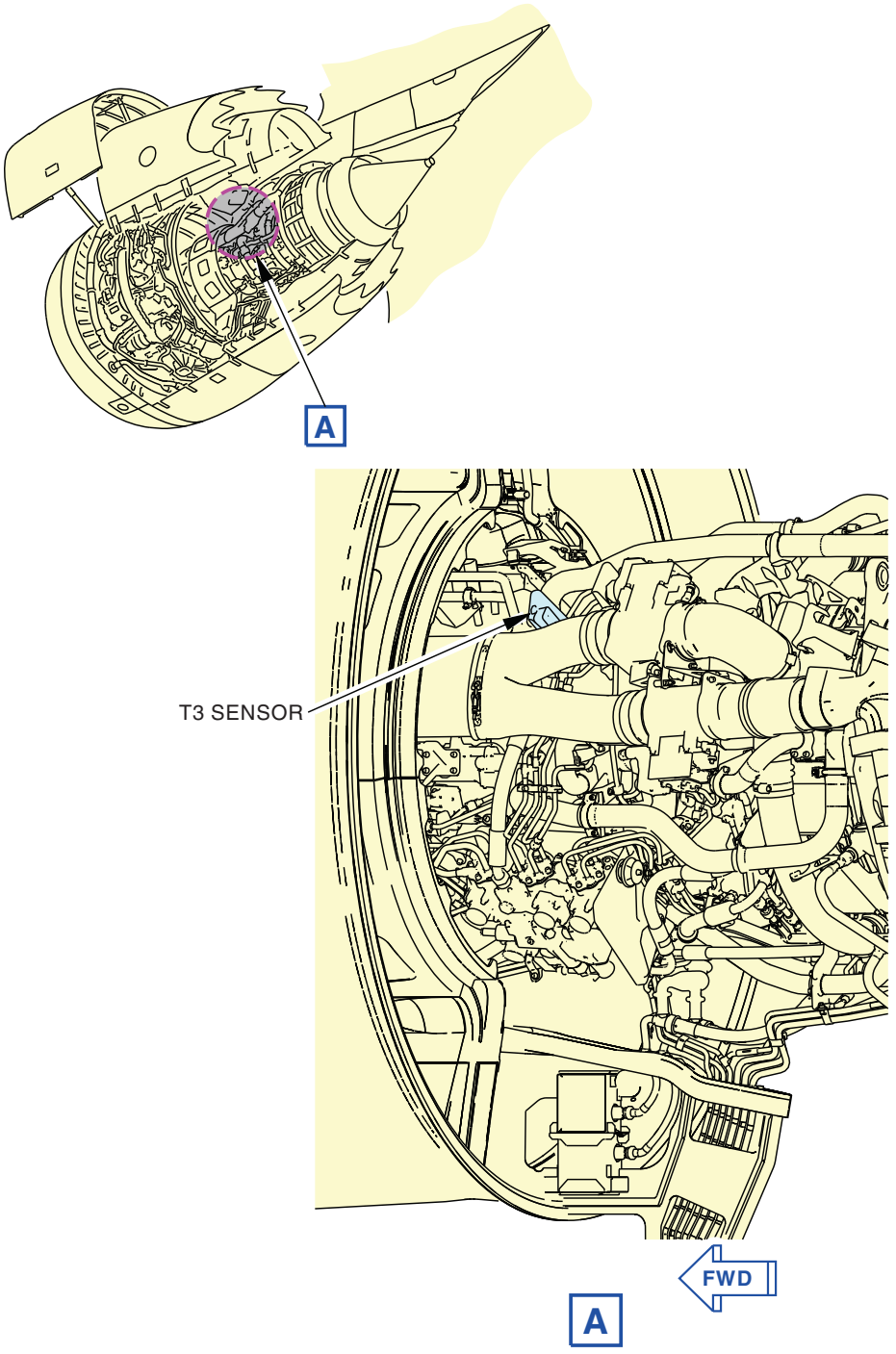
(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



T3 Sensor Inspection
Figure 601/73-21-64-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL MANIFOLD PRESSURE TRANSDUCER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fuel manifold pressure transducer removal

(2) Fuel manifold pressure transducer Installation.

TASK 73-21-67-000-801-G00

2. Fuel Manifold Pressure Transducer Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fuel manifold pressure transducer.

(2) The fuel manifold pressure transducer is referred to as the FMP transducer.

(3) The FMP transducer is installed on the split control unit at the 7:00 o'clock position.

(4) You must open the left thrust reverser to get access to the FMP transducer.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-21-67-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-67-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-21-67-010-001-G00

(3) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. Fuel Manifold Pressure Transducer Removal

SUBTASK 73-21-67-020-001-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the electrical connector [4] from the FMP transducer [6] (TASK 70-00-01-910-803-G00).

(a) Disconnect the electrical connector [4] from the FMP transducer [6].

(b) Install the protective covers on the electrical connector [4] and the electrical receptacle on the FMP transducer [6].

SUBTASK 73-21-67-020-002-G00

(2) Remove the FMP transducer [6] as follows:

(a) Remove the two bolts [5] that attach FMP transducer [6] to the split control unit [1].

(b) Remove the FMP transducer [6].

(c) Remove the packing retainers [3] and the preformed packing [2] from the FMP transducer [6].

1) Discard the preformed packing [2].

2) Examine the two packing retainers [3] for damage.

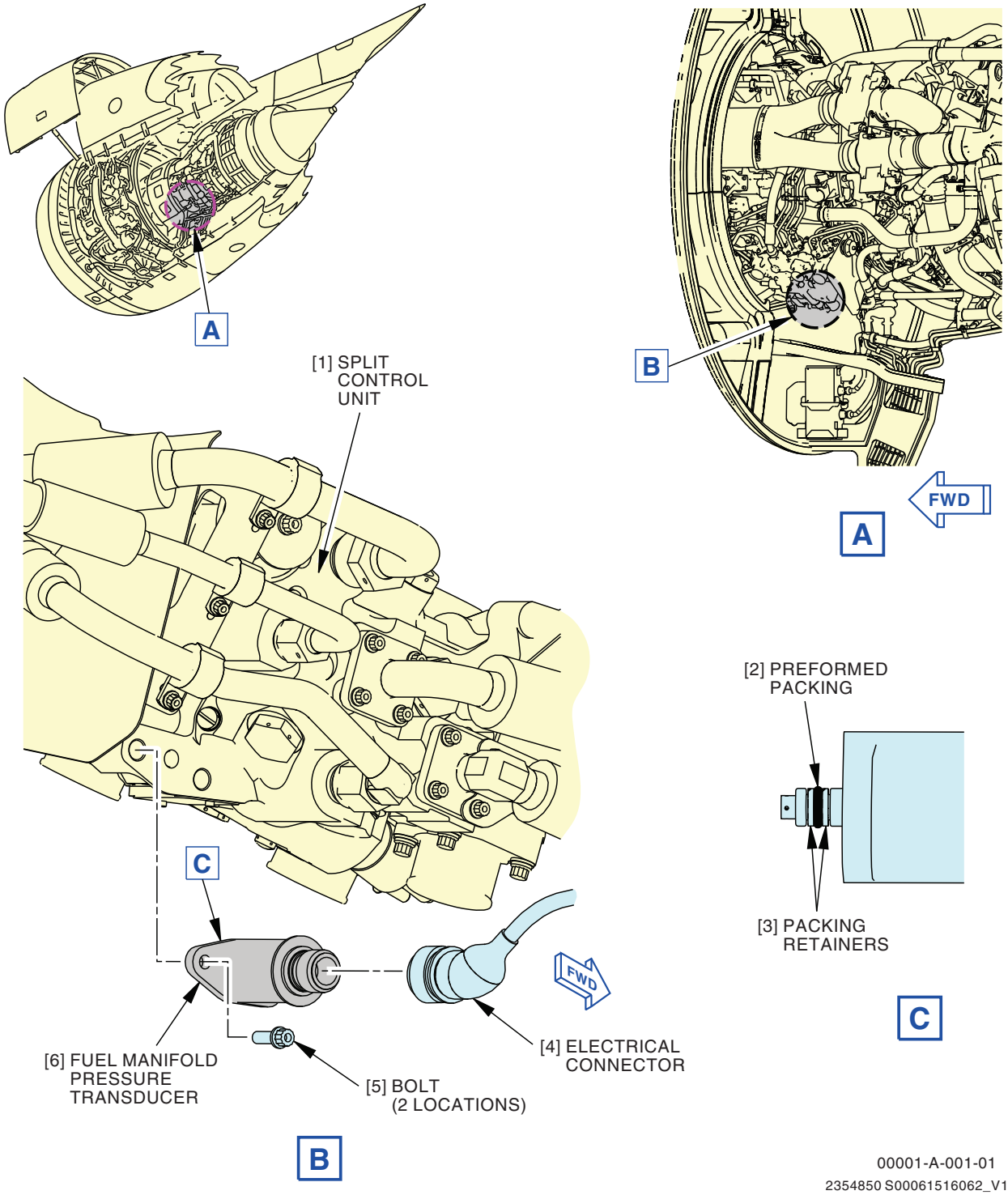
a) If the two packing retainers [3] are not damaged, keep them for the subsequent installation.

b) If the two packing retainers [3] are damaged, replace the packing retainers [3].

(d) Install the protective covers on the split control unit [1] and the FMP transducer [6].

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Manifold Pressure Transducer Installation
Figure 401/73-21-67-990-802-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-21-67-400-801-G00

3. Fuel Manifold Pressure Transducer Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fuel manifold pressure transducer.

(2) The fuel manifold pressure transducer is referred to as the FMP transducer.

(3) You must do the tests that are listed in the power plant test reference table after you install the FMP transducer.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
2	Preformed packing	73-21-67-01-040	PTF ALL
3	Packing retainer	73-21-67-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

H. Fuel Manifold Pressure Transducer Installation

SUBTASK 73-21-67-420-001-G00

(1) Install the FMP transducer [6] as follows:

(a) Remove the protective covers from the FMP transducer [6] and the split control unit [1].

(b) Lubricate the new preformed packing [2] with clean engine oil, D00599 [CP2442].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION INSTALL THE PACKING RETAINERS AND THE PREFORMED PACKING IN THE CORRECT SEQUENCE. IF YOU DO NOT INSTALL THE RETAINERS AND PACKING CORRECTLY, FUEL LEAKS CAN OCCUR. FUEL CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION DO NOT CAUSE DAMAGE TO THE PREFORMED PACKINGS AND PACKING RETAINERS WHEN YOU INSTALL THE FUEL SYSTEM COMPONENTS. A DAMAGED PACKING CAN CAUSE ENGINE FUEL LEAKS.

(c) Install the two packing retainers [3] and the new preformed packing [2] in this sequence.

NOTE: If you open the packing retainer more than is necessary, you can permanently change the shape or break the packing retainer.

1) Install the first packing retainer [3] on the FMP transducer [6].

a) Twist the first packing retainer [3] open and carefully slide it down the fitting into the groove.

2) Install the new preformed packing [2] on the FMP transducer [6].

3) Install the second packing retainer [3] on the FMP transducer [6].

a) Twist the second packing retainer [3] open and carefully slide it down the fitting into the groove.

(d) Lubricate the threads of the two bolts [5] with engine oil, D00599 [CP2442].

(e) Put the FMP transducer [6] in the correct position on the split control unit [1].

(f) Install the FMP transducer [6] on the split control unit [1] with the two bolts [5].

1) Tighten the two bolts [5] to 106.0 in-lb (12.0 N·m)-124.0 in-lb (14.0 N·m).

SUBTASK 73-21-67-420-002-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(2) Use the teflon-jawed pliers, STD-664 to connect the electrical connector [4] to the FMP transducer [6] (TASK 70-00-01-910-803-G00).

(a) Remove the protective covers from the electrical connector [4] and the electrical receptacle on the FMP transducer [6].

(b) Connect the electrical connector [4] to the FMP transducer [6].

1) Tighten the electrical connector [4].

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-67-860-004-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-21-67-860-005-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-21-67-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

J. Fuel Manifold Pressure Transducer Installation Test

SUBTASK 73-21-67-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL MANIFOLD PRESSURE TRANSDUCER - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fuel Manifold Pressure Transducer Inspection

TASK 73-21-67-210-801-G00

2. Fuel Manifold Pressure Transducer Inspection

(Figure 601)

A. General

(1) This task gives the instructions to inspect the fuel manifold pressure transducer.

(2) The fuel manifold pressure transducer is located on the split control unit at the 7:00 o'clock position.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-67-000-801-G00	Fuel Manifold Pressure Transducer Removal (P/B 401)
73-21-67-400-801-G00	Fuel Manifold Pressure Transducer Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 73-21-67-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-21-67-860-008-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-21-67-010-002-G00

(3) Do these tasks in sequence to safely open the left thrust reverser on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left fan cowl panel (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left thrust reverser (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

G. Fuel Manifold Pressure Transducer Inspection

SUBTASK 73-21-67-210-001-G00

(1) Inspect the fuel manifold pressure transducer.

(a) Look for signs of fuel leaks.

1) If you see signs of fuel leaks, do the steps in these tasks to replace the preformed packings (Fuel Manifold Pressure Transducer Removal, TASK 73-21-67-000-801-G00 and Fuel Manifold Pressure Transducer Installation, TASK 73-21-67-400-801-G00).

(b) Make sure that the fuel manifold pressure transducer is not loose.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) If the transducer is loose, tighten the bolts that attach the transducer to the Split Control Unit (SCU) (Fuel Manifold Pressure Transducer Installation, TASK 73-21-67-400-801-G00).

(c) Make sure that the electrical connector is not loose.

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

1) If an electrical connector is loose, use a teflon-jawed pliers, STD-664 to tighten the electrical connector.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-21-67-410-002-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the left thrust reverser:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

(b) Close the applicable left Fan Cowl Panel (TASK 71-11-04-410-801-G00).

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-21-67-860-009-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-21-67-860-010-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

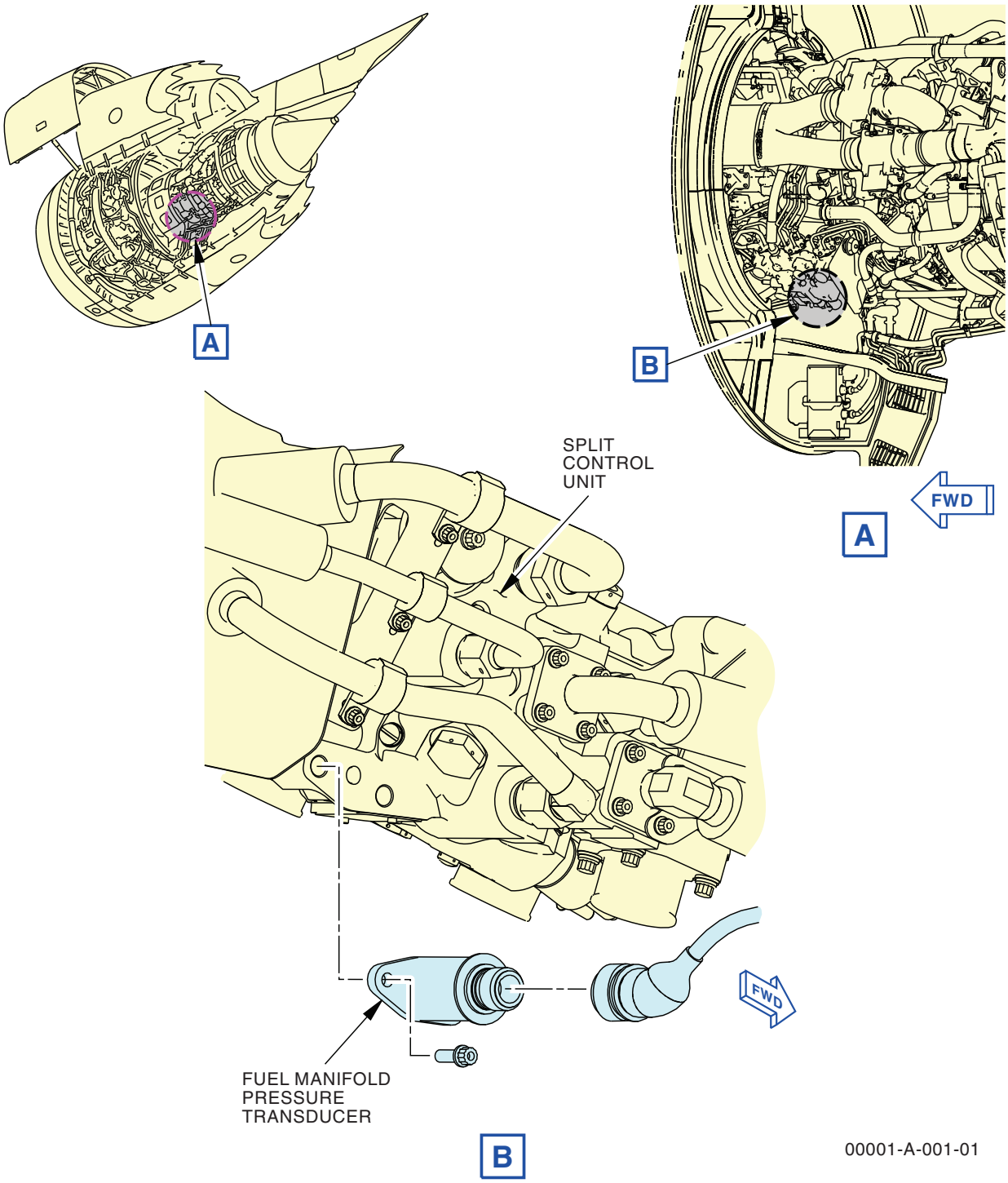
ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Manifold Pressure Transducer Inspection
Figure 601/73-21-67-990-803-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELECTRONIC ENGINE CONTROL (EEC) ELECTRICAL HARNESSES - MAINTENANCE PRACTICES

1. General

A. This procedure has this task:

(1) Electronic Engine Control (EEC) Electrical Harness (HJ8A) Tape Wrap.

TASK 73-22-01-390-802-G00

2. Electronic Engine Control (EEC) Electrical Harness (HJ8A) Tape Wrap

(Figure 201)

A. General

(1) This task is only applicable to the Electronic Engine Control (EEC) electrical harness (HJ8A) P/N 364-083-811-0. Other EEC electrical harness (HJ8A) part numbers do not require the tape wrap.

(2) This task gives the instructions to apply sealant and aluminum tape to the EEC electrical harness (HJ8A). The tape wrap is required to prevent fluid leaks from migrating down the wire harness through the firewall at the 6 o'clock location.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A00335	Adhesive - Silicone Rubber, 2 Part, RTV	BAC5010 Type 68
B00130	Alcohol - Isopropyl	TT-I-735
B50073	Alcohol - Isopropyl	ASTM D 770
C00511	Primer - Adhesion, Momentive SS4155	BAC5010 Type 68 Class P
D00599 [CP2442]	Oil - Engine	
G00034	Cotton Wiper - Process Cleaning Absorbent Wiper (Cheesecloth, Gauze)	AMS3819 Class 1 Grade A or B Form 1 (Supersede BMS15-5 CL A)
G50739	Tape - High Temperature Aluminum Foil/Glass Cloth Tape, (3M 363 Tape)	
G51293	Cotton Wiper - Knit, Process Cleaning Absorbent Wiper	BMS15-5 Class B

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Tape Wrap

SUBTASK 73-22-01-860-108-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-109-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-860-110-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-111-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-036-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Electronic Engine Control (EEC) Electrical Harness (HJ8A) Tape Wrap

SUBTASK 73-22-01-080-001-G00

(1) Carefully remove the TBV and start bleed/booster anti-ice valve ID tags.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-020-029-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(2) Use the teflon-jawed pliers, STD-664 to disconnect the EEC electrical harness (HJ8A) [1] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ8A) [1] connector from the TBV receptacle [2].

(b) Disconnect the EEC electrical harness (HJ8A) [1] connector from the start bleed/booster anti-ice valve receptacle [3].

(c) Install the protective covers on all of the harness connectors and the component receptacles.

SUBTASK 73-22-01-020-030-G00

(3) Remove the bolt [4] that attaches the EEC electrical harness (HJ8A) [1] to the tube bracket.

SUBTASK 73-22-01-020-031-G00

(4) Remove the bolt [5] from the clamp [6] that attaches the EEC electrical harness (HJ8A) [1] to the fan case bracket.

SUBTASK 73-22-01-100-001-G00

(5) Clean the external of the EEC electrical harness (HJ8A) [1] wire bundle metallic overbraid with alcohol, B00130 or alcohol, B50073.

(a) Dry the EEC electrical harness (HJ8A) [1] with a cotton wiper, G00034 or knit cotton wiper, G51293.

SUBTASK 73-22-01-390-004-G00

(6) Apply an adhesion primer, C00511 to the external surface of the EEC electrical harness (HJ8A) [1] wire bundle metallic overbraid, extend 0.25 in. (6.35 mm) past overbraid.

(a) Let the primer, C00511 dry before you continue.

SUBTASK 73-22-01-390-005-G00

(7) Use a brush to apply 0.02 in. (0.51 mm) - 0.03 in. (0.76 mm) thick of adhesive, A00335 to the external surface of the EEC electrical harness (HJ8A) [1] wire bundle metallic overbraid.

SUBTASK 73-22-01-390-006-G00

(8) Do these steps to apply 3M 363 Tape, G50739 onto the EEC electrical harness (HJ8A) [1] wire bundle metallic overbraid:

(a) Start at the electrical plug backshell connectors, spiral wrap over adhesive 2.0 in. (50.8 mm) wide with 3M 363 Tape, G50739.

(b) Start with full wrap perpendicular to the wire harness centerline and over lap a minimum of 0.5 in. (12.7 mm) per wrap.

(c) Make sure that you pull the tape tightly onto the wire bundle metallic overbraid.

(d) At the end of the spiral wrap, use one full wrap perpendicular to the wire harness centerline.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Tape down edges of all tape until adhesive has cured.

(f) Wipe clean any excess adhesive.

(g) As necessary, do the procedure again for the remaining two sections of the EEC electrical harness (HJ8A) [1] wire bundle metallic overbraid.

SUBTASK 73-22-01-420-029-G00

(9) Install the EEC electrical harness (HJ8A) [1] as follows:

(a) Lubricate the threads of the bolt [4] and bolt [5] with oil, D00599 [CP2442].

(b) Install the bolt [4] that attaches the EEC electrical harness (HJ8A) [1] to the tube bracket.

(c) Install the bolt [5] to the clamp [6] that attaches the EEC electrical harness (HJ8A) [1] to the fan case bracket.

(d) Tighten the bolt [4] to 110 in-lb (12 N·m) - 120 in-lb (14 N·m).

(e) Tighten the bolt [5] to 32 in-lb (4 N·m) - 38 in-lb (4 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(f) Use the teflon-jawed pliers, STD-664 to connect the EEC electrical harness (HJ8A) [1] as follows (TASK 70-00-01-910-803-G00):

1) Remove the protective covers from the connectors and receptacles.

2) Connect the EEC electrical harness (HJ8A) [1] connector to the TBV receptacle [2].

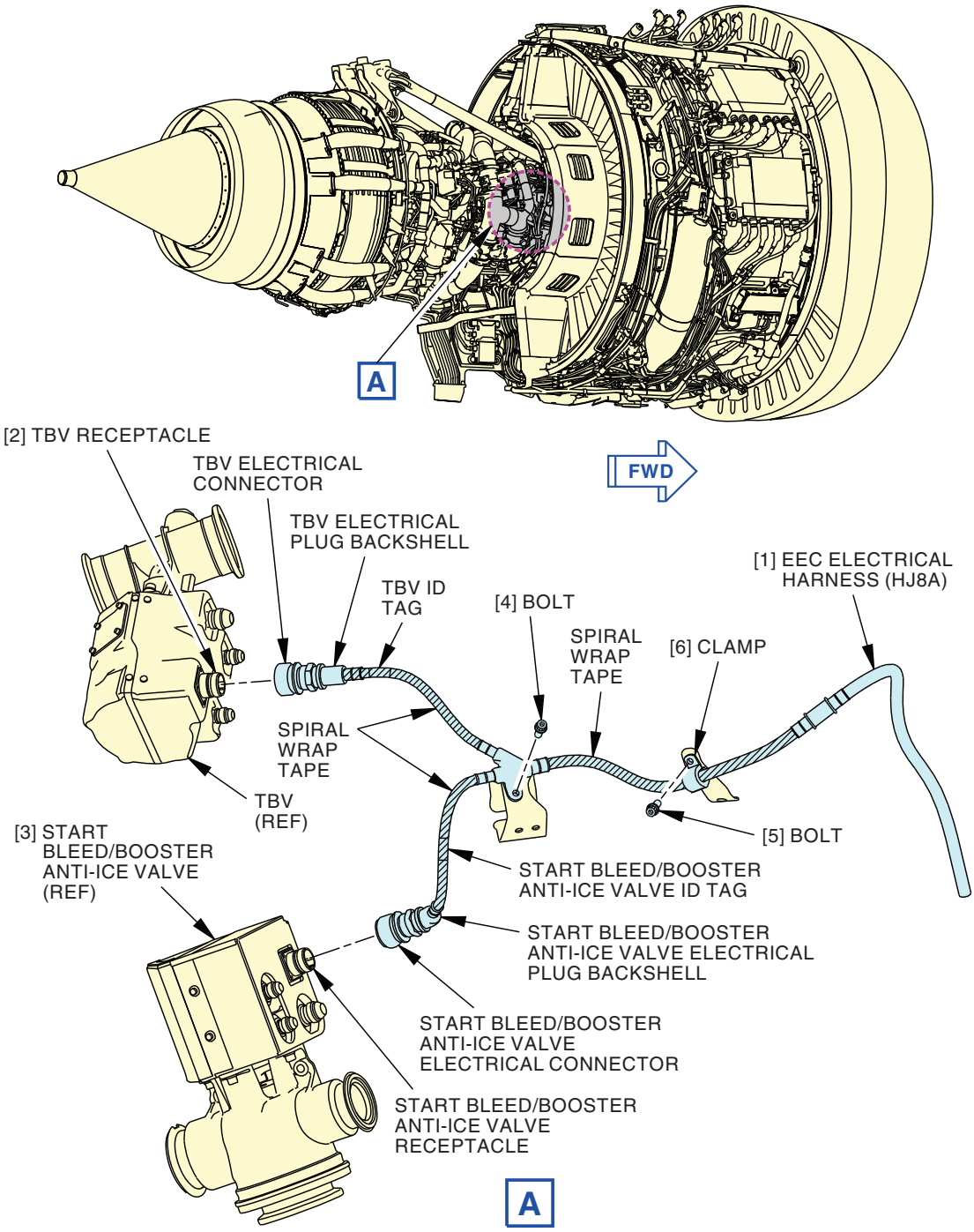
3) Connect the EEC electrical harness (HJ8A) [1] connector to the start bleed/booster anti-ice valve receptacle [3].

SUBTASK 73-22-01-480-001-G00

(10) Install the TBV and start bleed/booster anti-ice valve ID tags.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Electronic Engine Control (EEC) Electrical Harness (HJ8A) Tape Wrap
Figure 201/73-22-01-990-865-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

ELECTRONIC ENGINE CONTROL (EEC) ELECTRICAL HARNESSES - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Electronic Engine Control (EEC) Electrical Harness (HJ3A) Removal
(2) Electronic Engine Control (EEC) Electrical Harness (HJ3A) Installation
(3) Electronic Engine Control (EEC) Electrical Harness (HJ3B) Removal
(4) Electronic Engine Control (EEC) Electrical Harness (HJ3B) Installation
(5) Electronic Engine Control (EEC) Electrical Harness (HJ5A) Removal
(6) Electronic Engine Control (EEC) Electrical Harness (HJ5A) Installation
(7) Electronic Engine Control (EEC) Electrical Harness (HJ5B) Removal
(8) Electronic Engine Control (EEC) Electrical Harness (HJ5B) Installation
(9) Electronic Engine Control (EEC) Electrical Harness (HJ6A) Removal
(10) Electronic Engine Control (EEC) Electrical Harness (HJ6A) Installation
(11) Electronic Engine Control (EEC) Electrical Harness (HJ6B) Removal
(12) Electronic Engine Control (EEC) Electrical Harness (HJ6B) Installation
(13) Electronic Engine Control (EEC) Electrical Harness (HJ7) Removal
(14) Electronic Engine Control (EEC) Electrical Harness (HJ7) Installation
(15) Electronic Engine Control (EEC) Electrical Harness (HJ8A) Removal
(16) Electronic Engine Control (EEC) Electrical Harness (HJ8A) Installation
(17) Electronic Engine Control (EEC) Electrical Harness (HJ8B) Removal
(18) Electronic Engine Control (EEC) Electrical Harness (HJ8B) Installation
(19) Electronic Engine Control (EEC) Electrical Harness (W4A) Removal
(20) Electronic Engine Control (EEC) Electrical Harness (W4A) Installation
(21) Electronic Engine Control (EEC) Electrical Harness (W4B) Removal
(22) Electronic Engine Control (EEC) Electrical Harness (W4B) Installation
(23) Electronic Engine Control (EEC) Electrical Harness (W2) Removal
(24) Electronic Engine Control (EEC) Electrical Harness (W2) Installation.

TASK 73-22-01-000-801-G00

2. Electronic Engine Control (EEC) Electrical Harness (HJ3A) Removal

(Figure 401)

A. General

(1) This task gives the instructions on how to remove the electronic engine control (EEC) electrical harness (HJ3A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Reference	Title
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-22-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-002-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

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F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-003-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-004-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-001-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ3A) Removal

(Figure 401)

SUBTASK 73-22-01-030-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the EEC electrical harness (HJ3A) [1] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ3A) [1] connector from the EEC channel A J3 receptacle [2].

(b) Disconnect the EEC electrical harness (HJ3A) [1] connector from the PMA channel A receptacle [3].

(c) Remove the nut [4] and the bolt [5] from the clamp [6] that attach the EEC electrical harness (HJ3A) [1] to the engine.

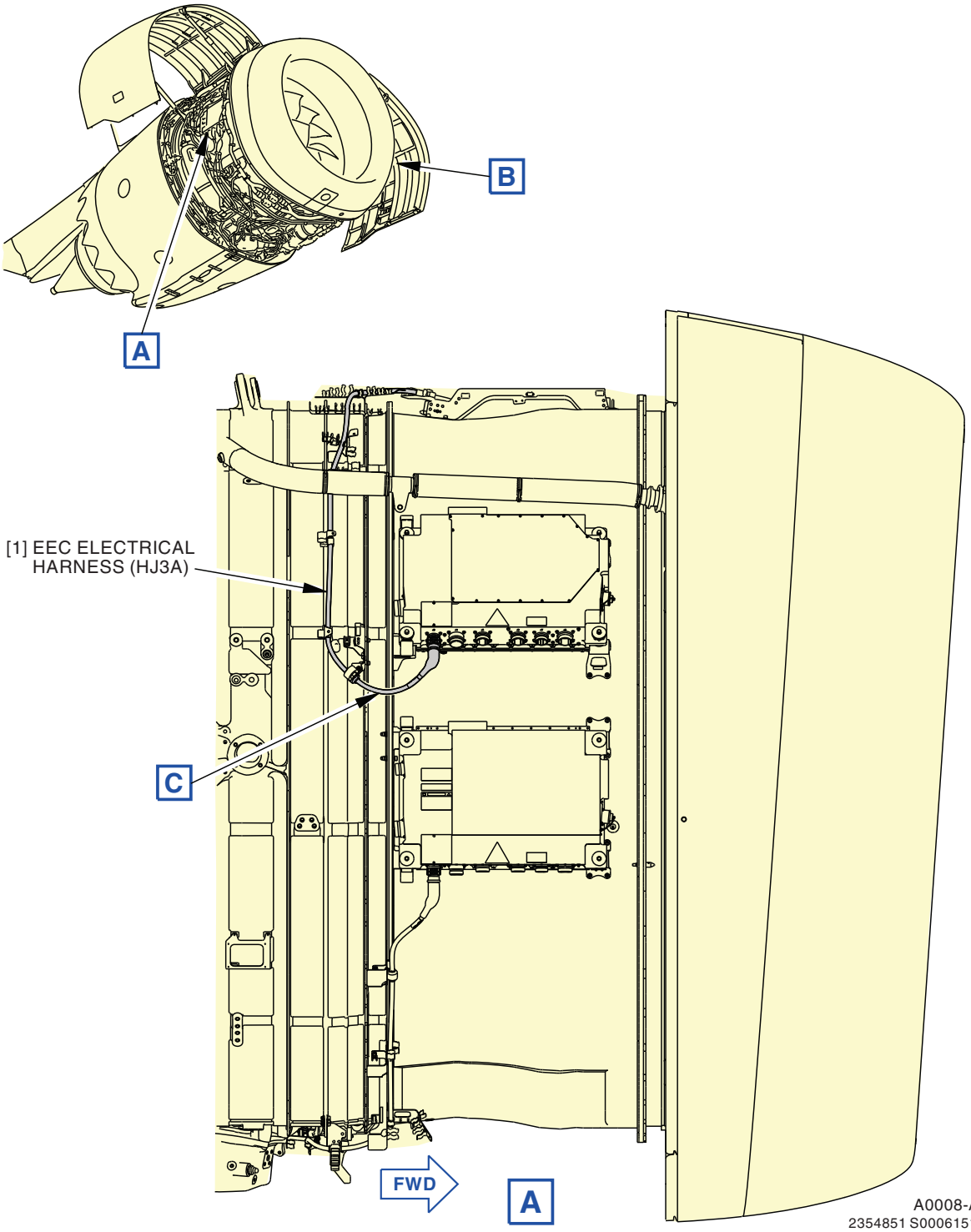
(d) Remove the EEC electrical harness (HJ3A) [1] from the 15 clips [7] along its routing.

(e) Carefully remove the EEC electrical harness (HJ3A) [1] from the engine.

(f) Install the protective covers on all of the harness connectors and the component receptacles.

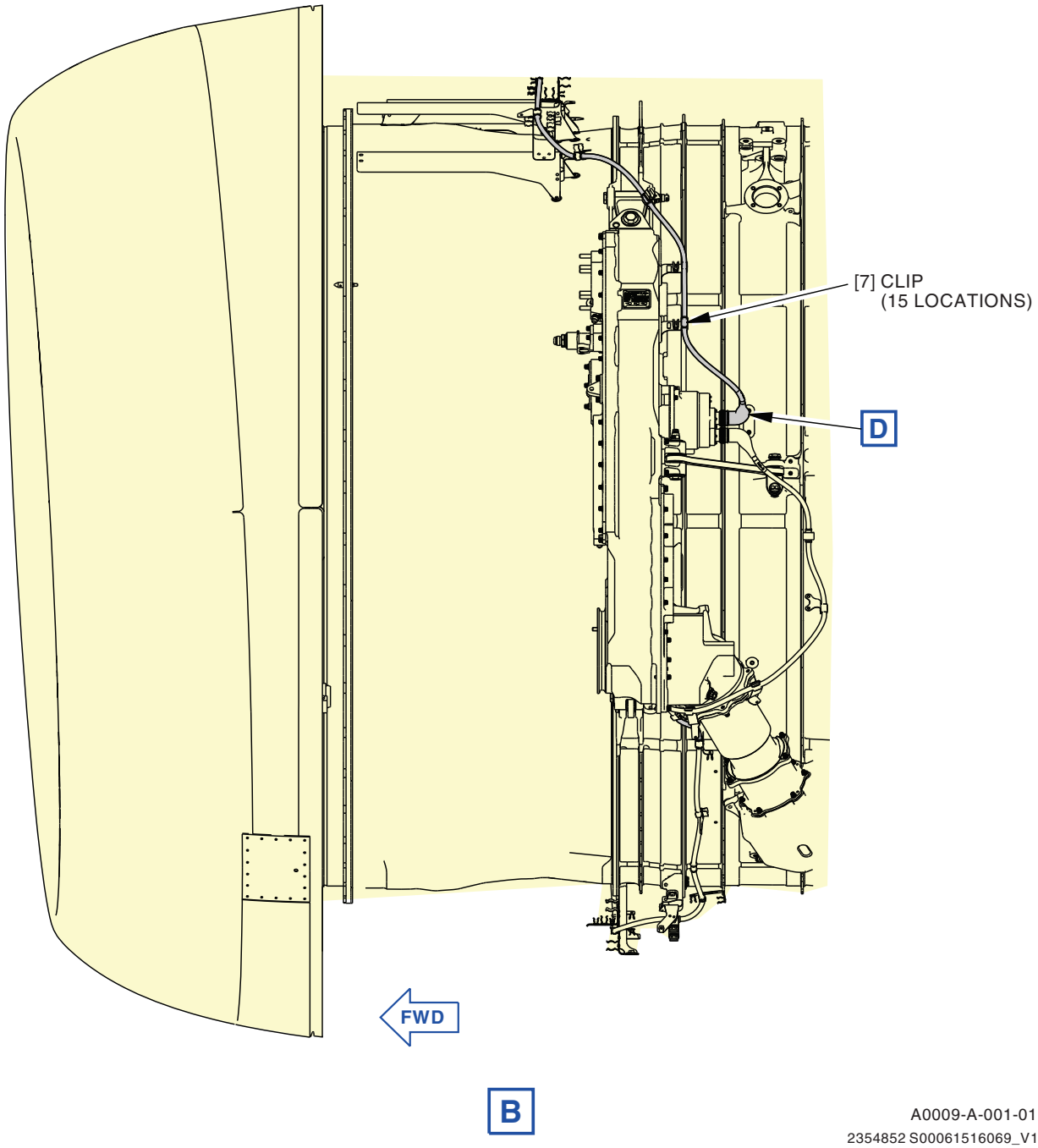
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



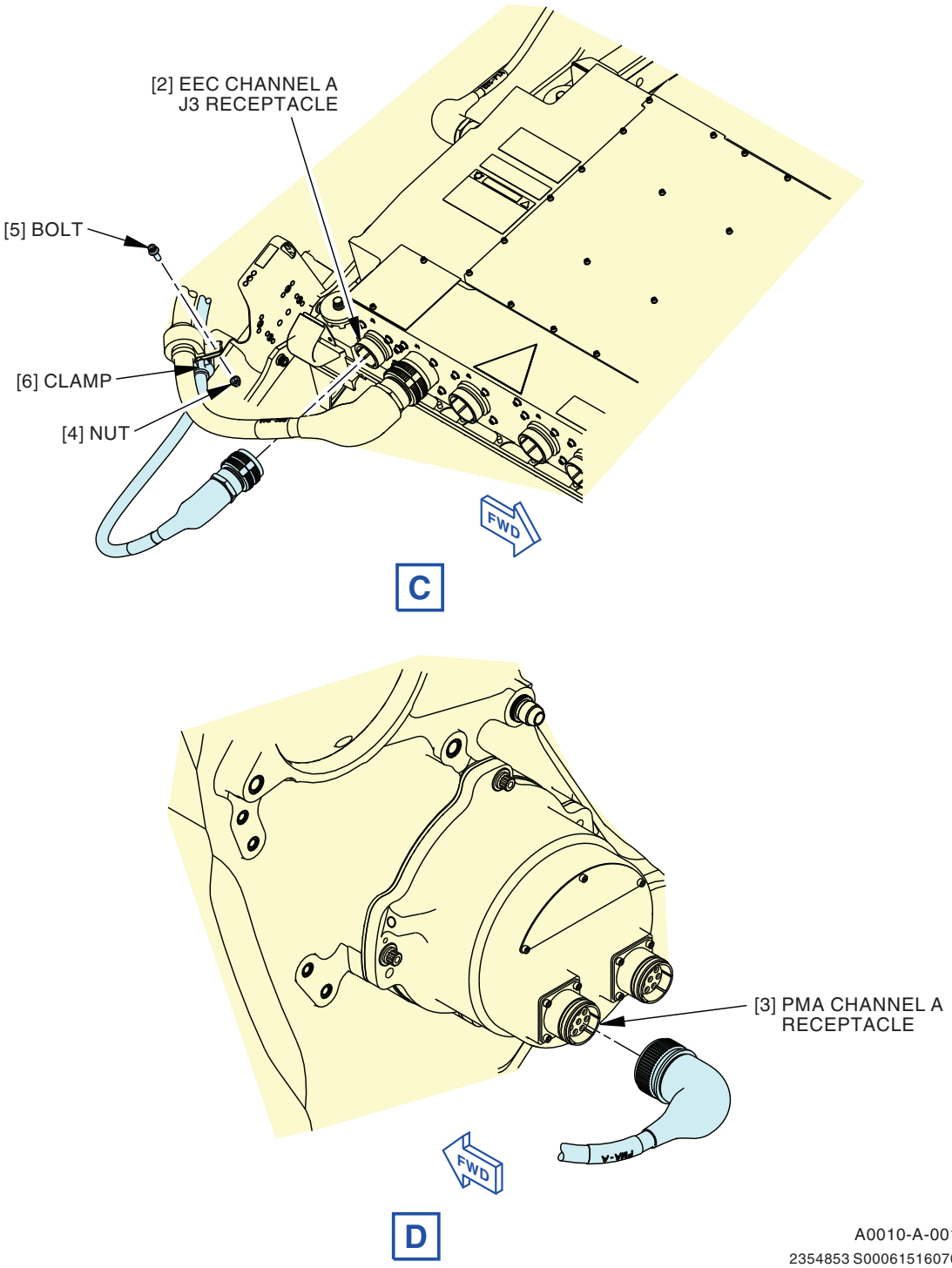
EEC Electrical Harness (HJ3A) Installation
Figure 401/73-22-01-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ3A) Installation
Figure 401/73-22-01-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ3A) Installation
Figure 401/73-22-01-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-22-01-400-801-G00

3. Electronic Engine Control (EEC) Electrical Harness (HJ3A) Installation

(Figure 401)

A. General

(1) This task gives the instructions on how to install the electronic engine control (EEC) electrical harness (HJ3A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Electronic Engine Control (EEC) Electrical Harness (HJ3A) Installation

(Figure 401)

SUBTASK 73-22-01-410-021-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Install the EEC electrical harness (HJ3A) [1] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the EEC electrical harness (HJ3A) [1] in a correct position with the 15 clips [7] along its routing.

(d) Install the bolt [5], the nut [4], and the clamp [6] that attach the EEC electrical harness (HJ3A) [1] to the engine.

(e) Tighten the nut [4] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(f) Use the teflon-jawed pliers, STD-664 to connect the EEC electrical harness (HJ3A) [1] to the EEC CH A as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (HJ3A) [1] connector to the PMA channel A receptacle [3].

2) Connect the EEC electrical harness (HJ3A) [1] connector to the EEC channel A J3 receptacle [2].

(g) Make sure that the work area is clean of all tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-022-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-005-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-006-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-007-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-860-008-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

H. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ3A) Installation Test

SUBTASK 73-22-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-22-01-000-802-G00

4. Electronic Engine Control (EEC) Electrical Harness (HJ3B) Removal

(Figure 402)

A. General

(1) This task gives the instructions on how to remove the electronic engine control (EEC) electrical harness (HJ3B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-22-01-860-009-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-010-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-011-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-012-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-024-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ3B) Removal

(Figure 402)

SUBTASK 73-22-01-020-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the EEC electrical harness (HJ3B) [28] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ3B) [28] connector from the EEC channel B J3 receptacle [30].

(b) Remove the bolt [32] and the clamp [31] that attaches the EEC electrical harness (HJ3B) [28] to the engine.

(c) Disconnect the EEC electrical harness (HJ3B) [28] connector from the PMA channel B receptacle [33].

(d) Remove the bolt [36] and the clamp [35] that attaches the EEC electrical harness (HJ3B) [28] to the oil tube [34].

(e) Remove the EEC electrical harness (HJ3B) [28] from the 10 clips [7] along its routing.

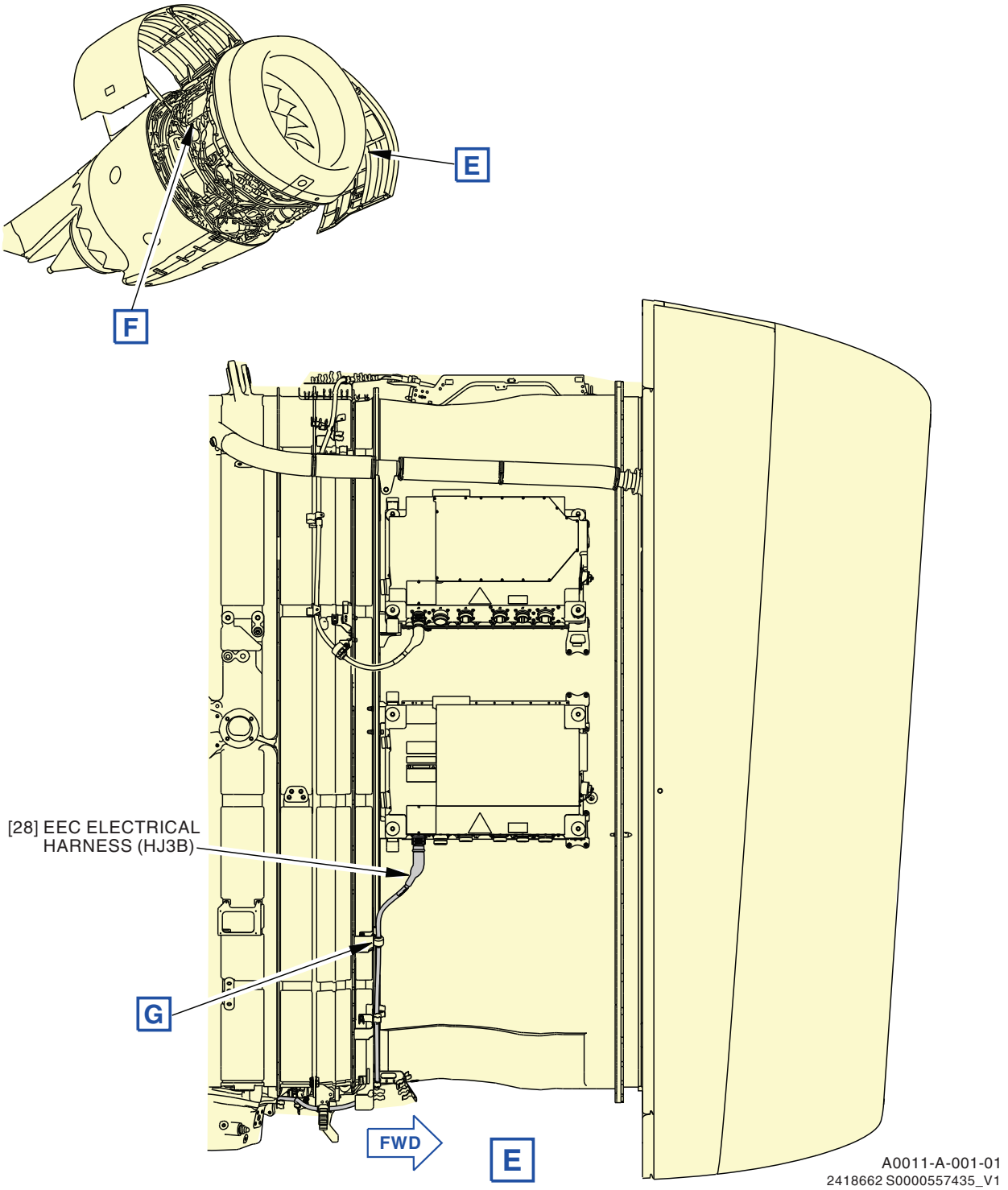
(f) Carefully remove the EEC electrical harness (HJ3B) [28] from the engine.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Install the protective covers on all of the harness connectors and the component receptacles.

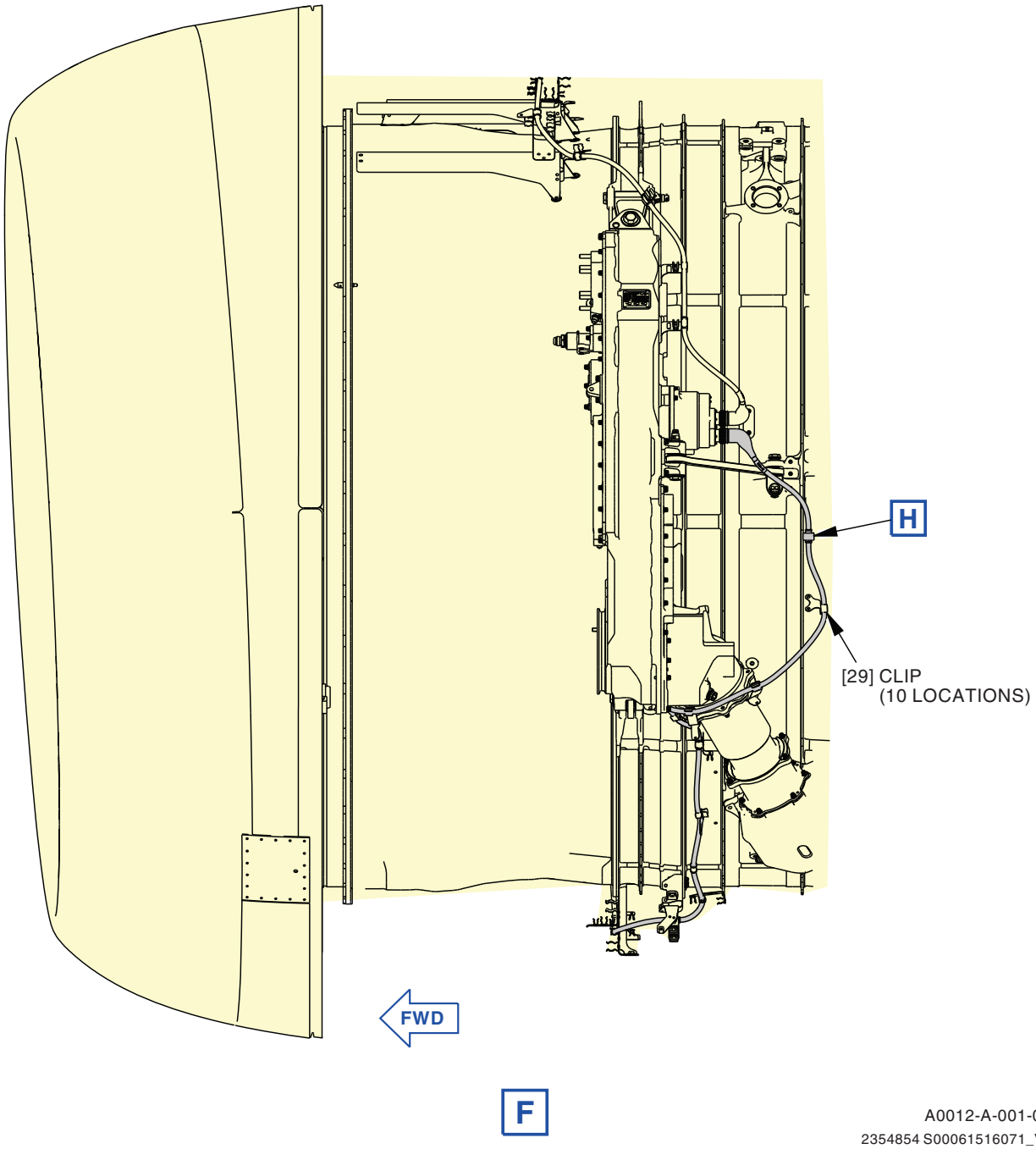
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



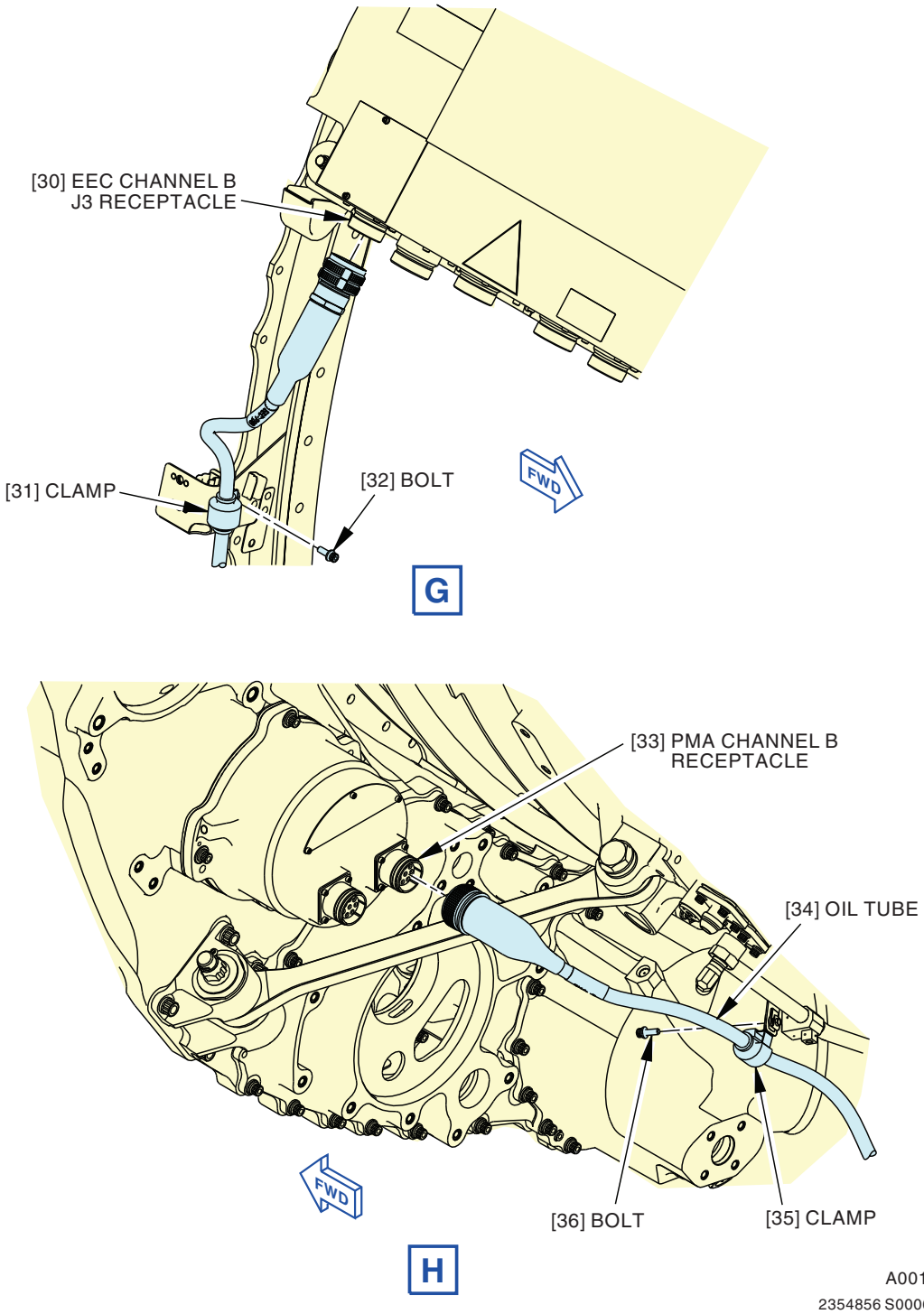
EEC Electrical Harness (HJ3B) Installation
Figure 402/73-22-01-990-860-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ3B) Installation
Figure 402/73-22-01-990-860-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ3B) Installation
Figure 402/73-22-01-990-860-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-22-01-400-802-G00

5. Electronic Engine Control (EEC) Electrical Harness (HJ3B) Installation

(Figure 402)

A. General

(1) This task gives the instructions on how to install the electronic engine control (EEC) electrical harness (HJ3B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Electronic Engine Control (EEC) Electrical Harness (HJ3B) Installation

(Figure 402)

SUBTASK 73-22-01-420-001-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(1) Install the EEC electrical harness (HJ3B) [28] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the EEC electrical harness (HJ3B) [28] in a correct position with the 10 clips [7] along its routing.

(d) Install the bolt [36] and the clamp [35] that attach the EEC electrical harness (HJ3B) [28] to the oil tube [34].

(e) Tighten the bolt [36] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(f) Install the bolt [32] and the clamp [31] that attach the EEC electrical harness (HJ3B) [28] to the engine.

(g) Tighten the bolt [32] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(h) Use the teflon-jawed pliers, STD-664 to connect the EEC electrical harness (HJ3B) [28] to the EEC CH B as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (HJ3B) [28] connector to the PMA channel B receptacle [33].

2) Connect the EEC electrical harness (HJ3B) [28] connector to the EEC channel B J3 receptacle [30].

(i) Make sure that the work area is clean of all tools and other items.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-023-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-013-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-014-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-015-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-860-016-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

H. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ3B) Installation Test

SUBTASK 73-22-01-730-002-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-22-01-000-803-G00

6. Electronic Engine Control (EEC) Electrical Harness (HJ5A) Removal

(Figure 403, Figure 404, and Figure 405)

A. General

(1) This task gives the instructions to remove the electronic engine control (EEC) electrical harness (HJ5A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
72-23-06-000-801-G00	Fan Frame Strut Panels Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-22-01-860-017-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-018-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-019-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-860-020-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-025-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ5A) Removal

SUBTASK 73-22-01-020-002-G00

(1) Remove the segment shroud 2 [68].

SUBTASK 73-22-01-020-003-G00

(2) Remove the segment shroud 3 [69].

SUBTASK 73-22-01-020-004-G00

(3) Remove the segment shroud 4 [71].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-020-005-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(4) Use the teflon-jawed pliers, STD-664, to disconnect the EEC electrical harness (HJ5A) [67] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ5A) [67] connector from the EEC channel A J5 receptacle [72].

(b) Disconnect the EEC electrical harness (HJ5A) [67] connector from the fuel flow transmitter meter receptacle [75].

(c) Disconnect the EEC electrical harness (HJ5A) [67] connector from the FMU channel A receptacle [76].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

(d) Disconnect the EEC electrical harness (HJ5A) [67] from the lower overheat sensor [114].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0178 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

(e) Disconnect the EEC electrical harness (HJ5A) [67] from the bracket assembly [115].

PTF ALL

(f) Remove the EEC electrical harness (HJ5A) [67] as follows:

1) Remove the bolt [74] from the clamp [73] that attaches the EEC electrical harness (HJ5A) [67] to the fan case bracket.

2) Remove the bolt [78] from the clamp [77] that attaches the EEC electrical harness (HJ5A) [67] to the fan case bracket.

(g) Disconnect the connector socket [82] of the EEC electrical harness (HJ5A) [67] as follows:

1) Disconnect the EEC electrical harness (HJ6B) [80] from the EEC HJ6A harness receptacle [79].

2) Disconnect the HJ5B engine wiring harness electrical connector [84] (at 5:00 o'clock) from the connector socket [82].

NOTE: If the HJ5B engine wiring harness has been removed, do not do this step.

3) Remove the four screws [81] that attach the connector socket [82] to the bracket [83].

4) Disconnect the connector socket [82] from the bracket [83].

(h) Disconnect the EEC electrical harness (HJ5A) [67] connector from the T25 sensor channel A receptacle [86].

(i) Disconnect the EEC electrical harness (HJ5A) [67] connector from the N1 speed sensor Channel A receptacle [147A].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(j) Disconnect the EEC electrical harness (HJ5A) [67] connector from the fuel manifold transducer receptacle [93].

(k) Disconnect the EEC electrical harness (HJ5A) [67] connector from the SCU/SVA channel A receptacle [96].

(l) Disconnect the EEC electrical harness (HJ5A) [67] connector from the left VBV actuator receptacle [97].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

(m) Disconnect the EEC electrical harness (HJ5A) [67] connector from the left MTC actuator receptacle [98].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

(n) Disconnect the EEC electrical harness (HJ5A) [67] connector from the electrical stow bracket [117].

PTF ALL

(o) Disconnect the EEC electrical harness (HJ5A) [67] connector from the left VSV actuator receptacle [99].

(p) Remove the bolt [88] from the clamp [87] that attaches the EEC electrical harness (HJ5A) [67] to the fan hub bracket.

(q) Remove the bolt [90] from the clamp [89] that attaches the EEC electrical harness (HJ5A) [67] to the fan hub bracket.

(r) Remove the bolt [91] from the clamp [92] that attaches the EEC electrical harness (HJ5A) [67] to the fan case bracket.

(s) Remove the bolt [94] from the clamp [95] that attaches the EEC electrical harness (HJ5A) [67] to the fan frame module bracket.

(t) Remove the bolt [100] from the clamp [101] that attaches the EEC electrical harness (HJ5A) [67] to the fan frame module bracket.

(u) Remove the EEC electrical harness (HJ5A) [67] from the 24 clips [70] along its routing.

(v) Remove blanket [190] from the strut 5 outer fitting [109] (View B, Figure 404).

(w) Remove the EEC electrical harness (HJ5A) [67] from the harness strut (6:00 o'clock) [104] as follows:

1) Remove the Fan Frame Strut 5 Panel (left) (TASK 72-23-06-000-801-G00).

PTF ALL; AIRPLANES WITH PRE LEAP-1B SERVICE BULLETIN 72-0462

2) Remove the bolt [102] from the clamp [103] that attaches the EEC electrical harness (HJ5A) [67] to the harness strut (6:00 o'clock) [104].

PTF ALL; AIRPLANES WITH POST LEAP-1B SERVICE BULLETIN 72-0462

3) Remove the bolt [102] and nut [105] from the clamp [103] that attaches the EEC electrical harness (HJ5A) [67] to the harness strut (6:00 o'clock) [104].

PTF ALL

4) Remove the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] from the strut 5 outer fitting [109] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Remove the five bolts [106] that attach the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] to the strut 5 outer fitting [109].

b) Cut and remove the sealant between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] to the strut 5 outer fitting [109].

c) Move the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] rearward from the electrical harnesses.

d) Remove the two nuts [107] and two bolts [108] to disassemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

e) Cut and remove the sealant on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and around the harnesses.

f) Remove the harnesses from the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

g) Remove any remaining sealant residue.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

5) Remove the harness support block [504], harness support block [505], harness support block [506] from the harness strut (6:00 o'clock) [104] as follows:

a) Remove the four bolts [111] and four nuts [110] that attach the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

b) Cut and remove the sealant between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

c) Move the assembly of the harness support block [504], harness support block [505], and harness support block [506] upward and away from the harness strut (6:00 o'clock) [104].

d) Remove the two nuts [112] and two bolts [113] to disassemble the harness support block [504], harness support block [505], and harness support block [506].

e) Cut and remove the sealant on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses.

f) Remove the harnesses from the harness support block [504], harness support block [505], and harness support block [506].

g) Remove any remaining sealant residue.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

6) Remove the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] from the harness strut (6:00 o'clock) [104] as follows:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

a) Remove the four bolts [124] that attach the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

b) Cut and remove the sealant between the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

c) Move the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] upward and away from the harness strut (6:00 o'clock) [104].

d) Remove the four nuts [126] and four bolts [125] to disassemble the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

e) Cut and remove the sealant on the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and around the harnesses.

f) Remove the harnesses from the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

g) Remove any remaining sealant residue.

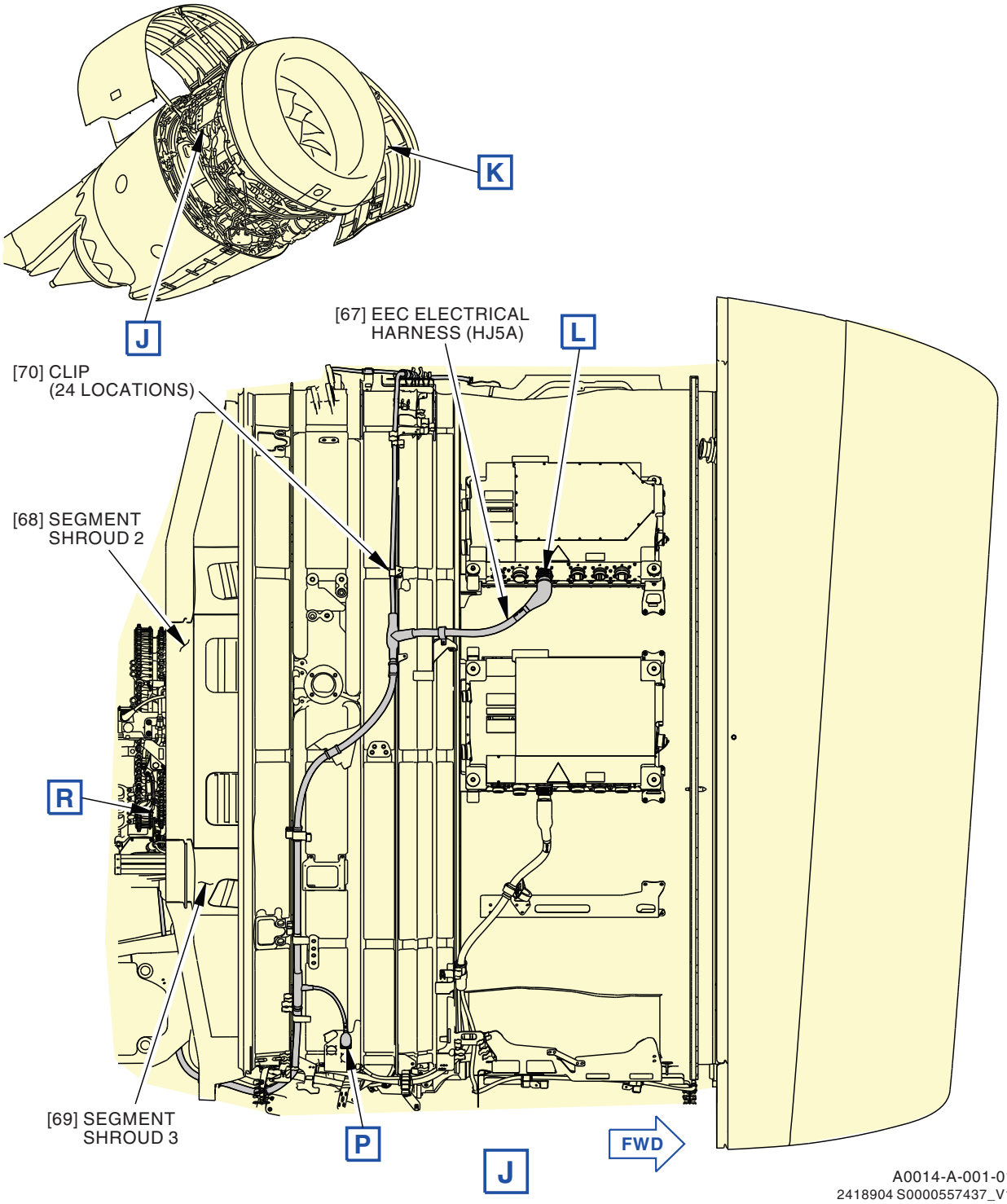
PTF ALL

(x) Carefully remove the EEC electrical harness (HJ5A) [67] from the engine.

(y) Install the protective covers on all of the harness connectors and the component receptacles.

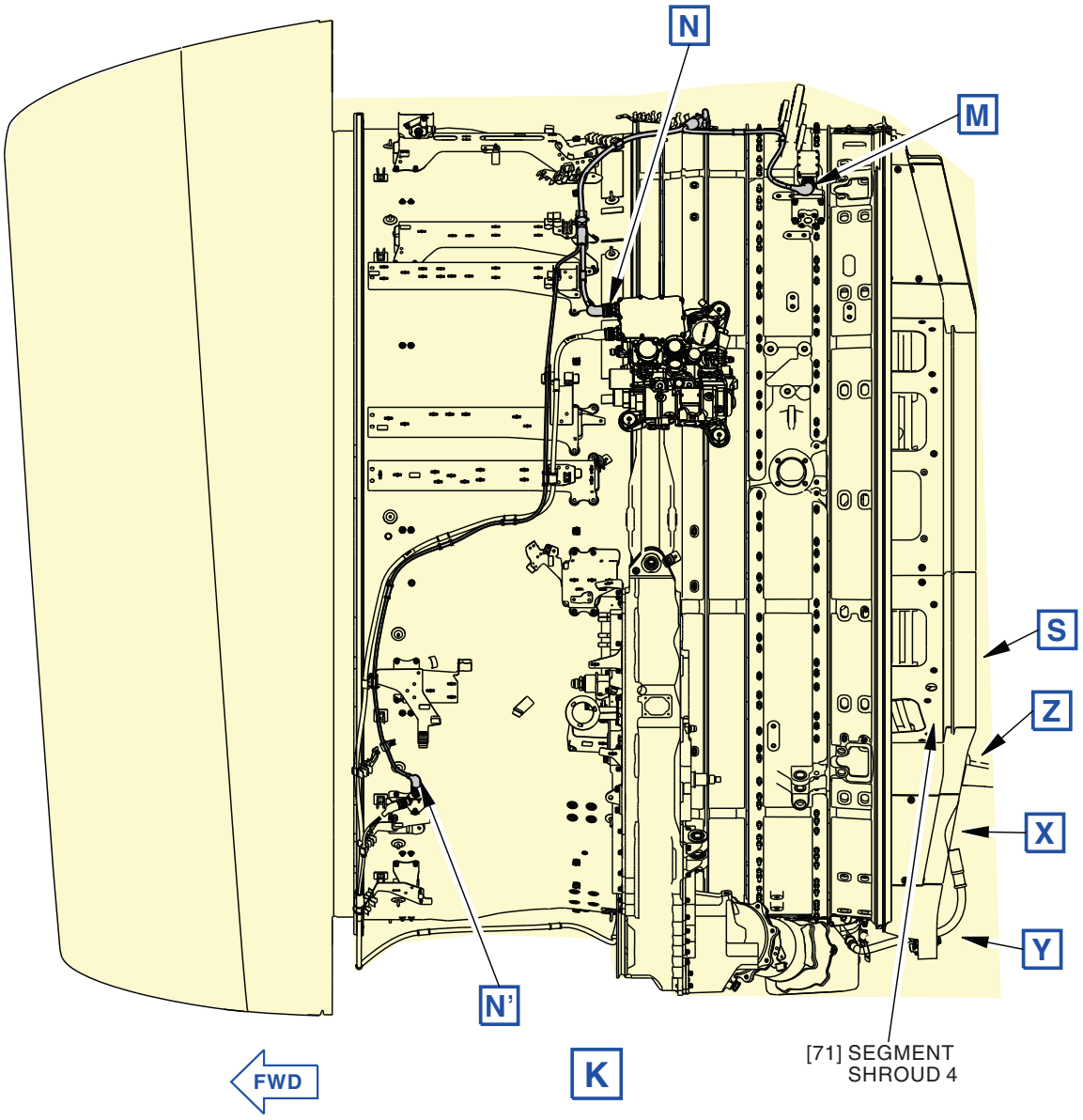
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 1 of 12)

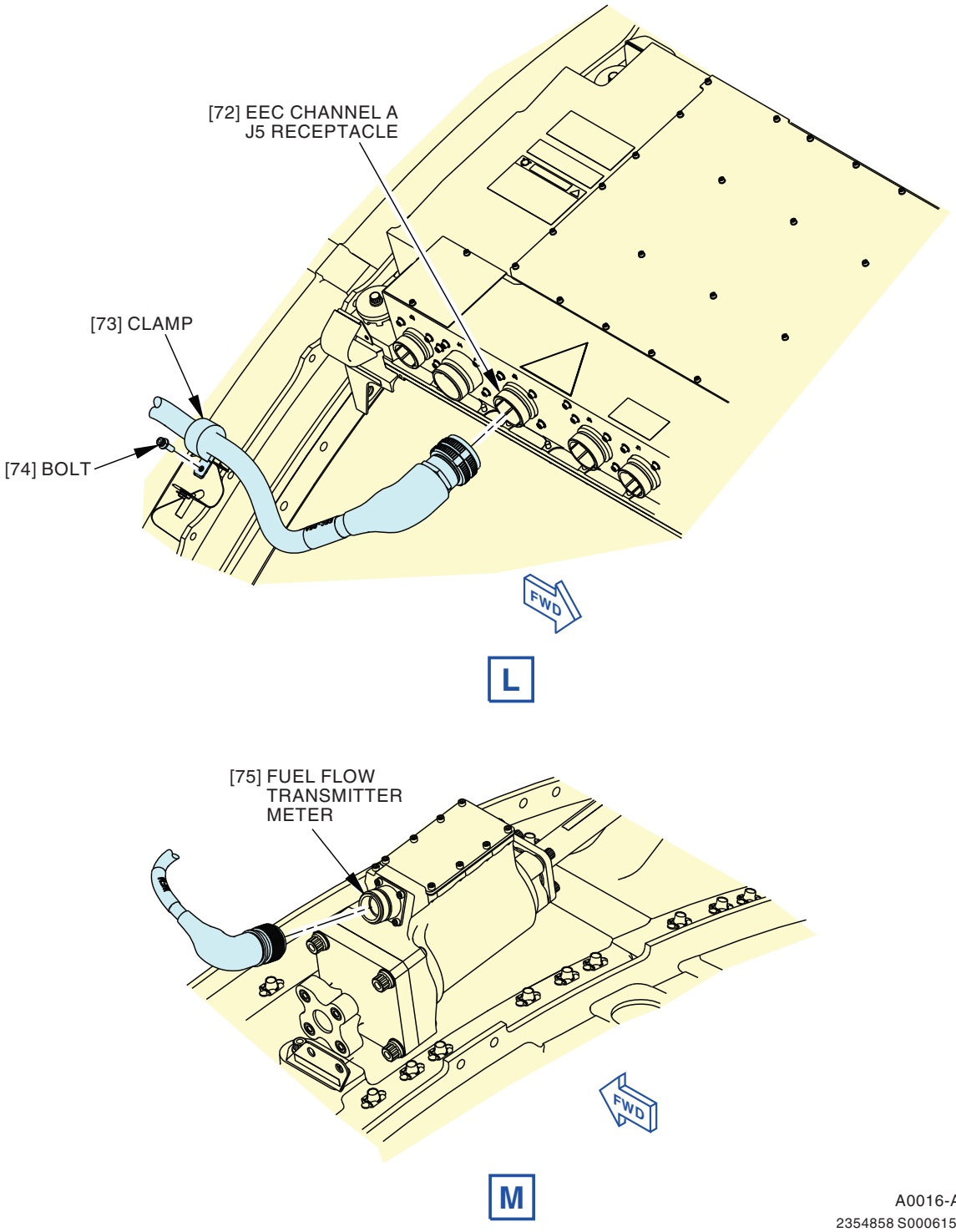
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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2354857 S00061516073_V2

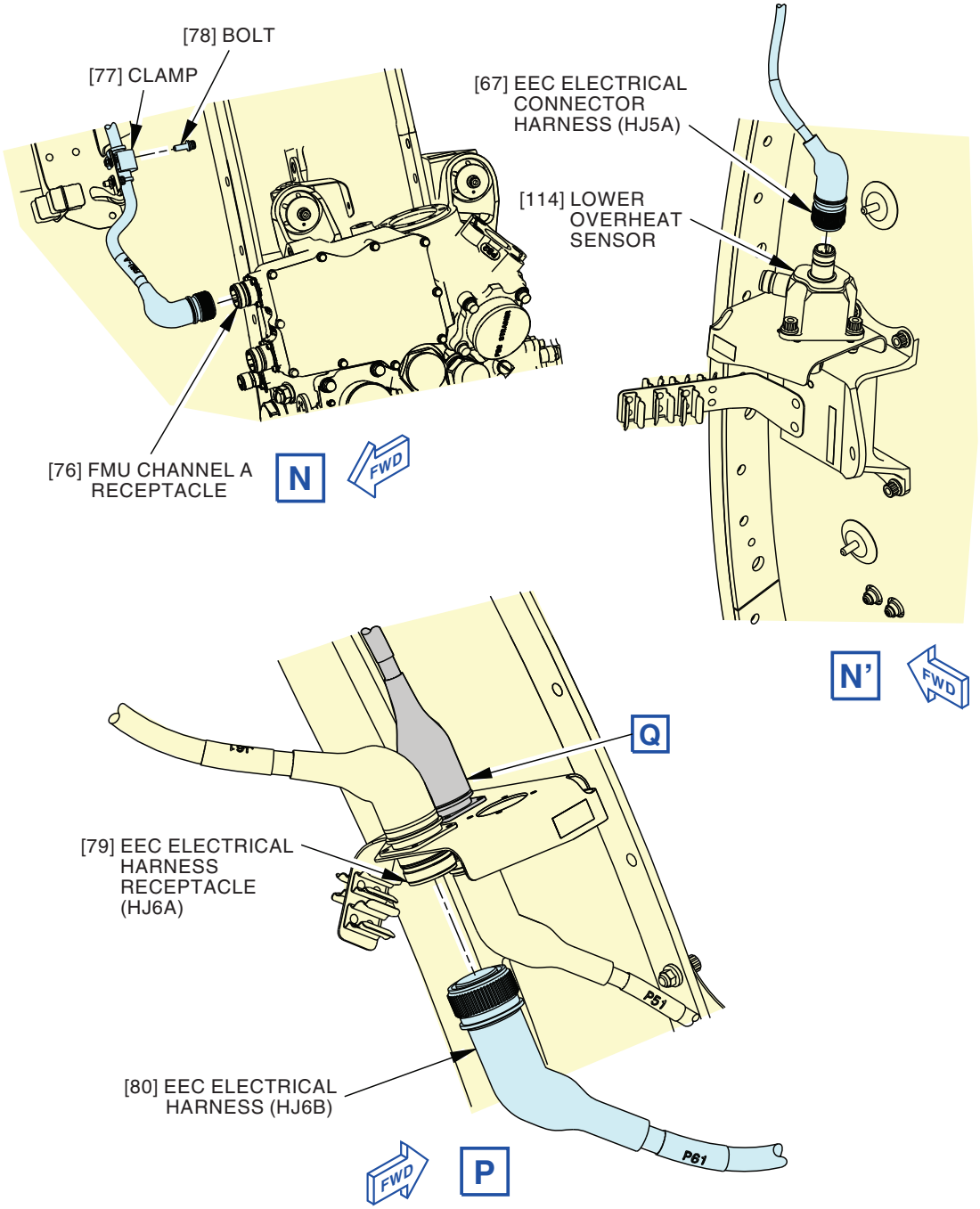
EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 2 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 3 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

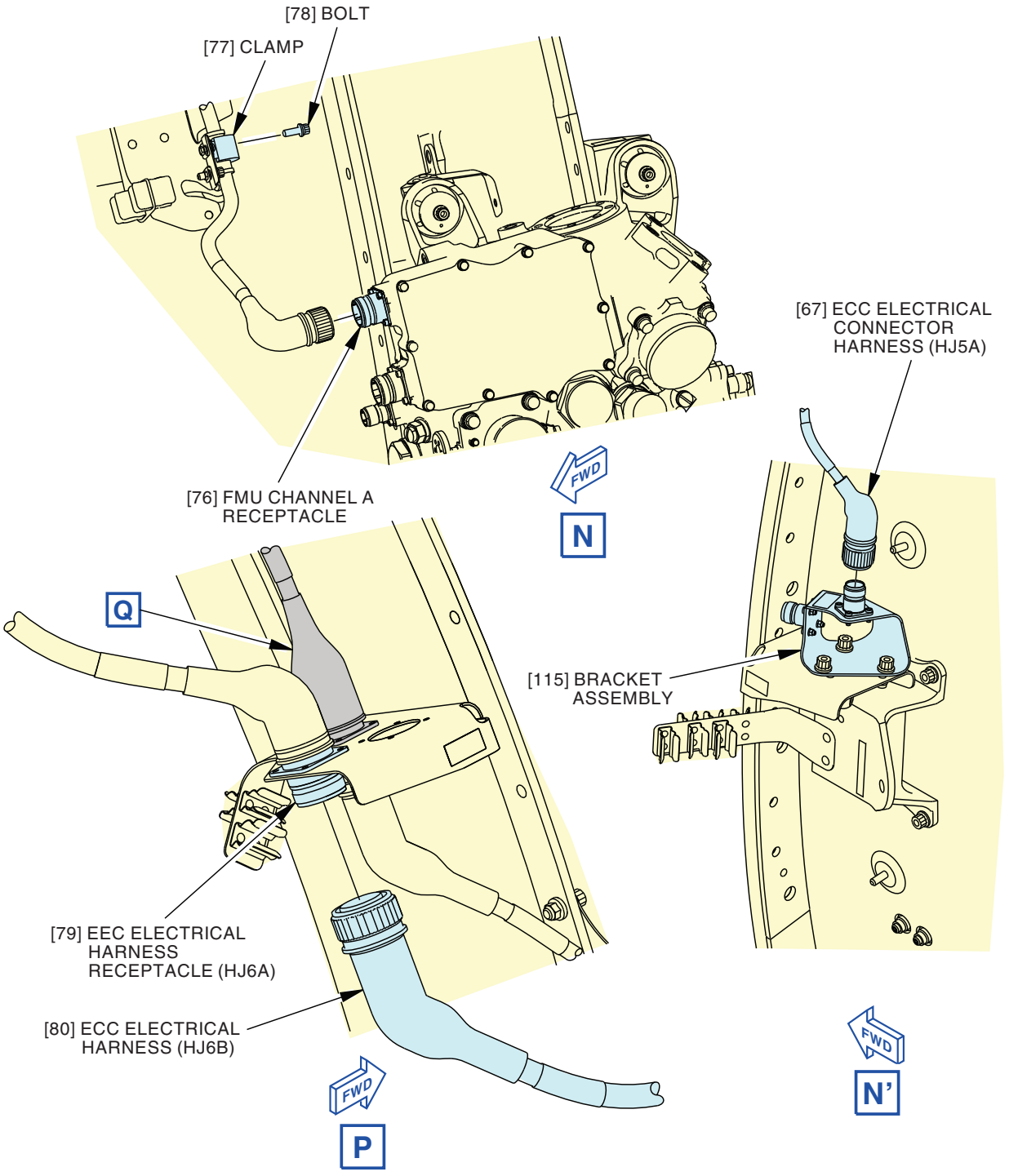


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2354859 S00061516075_V2

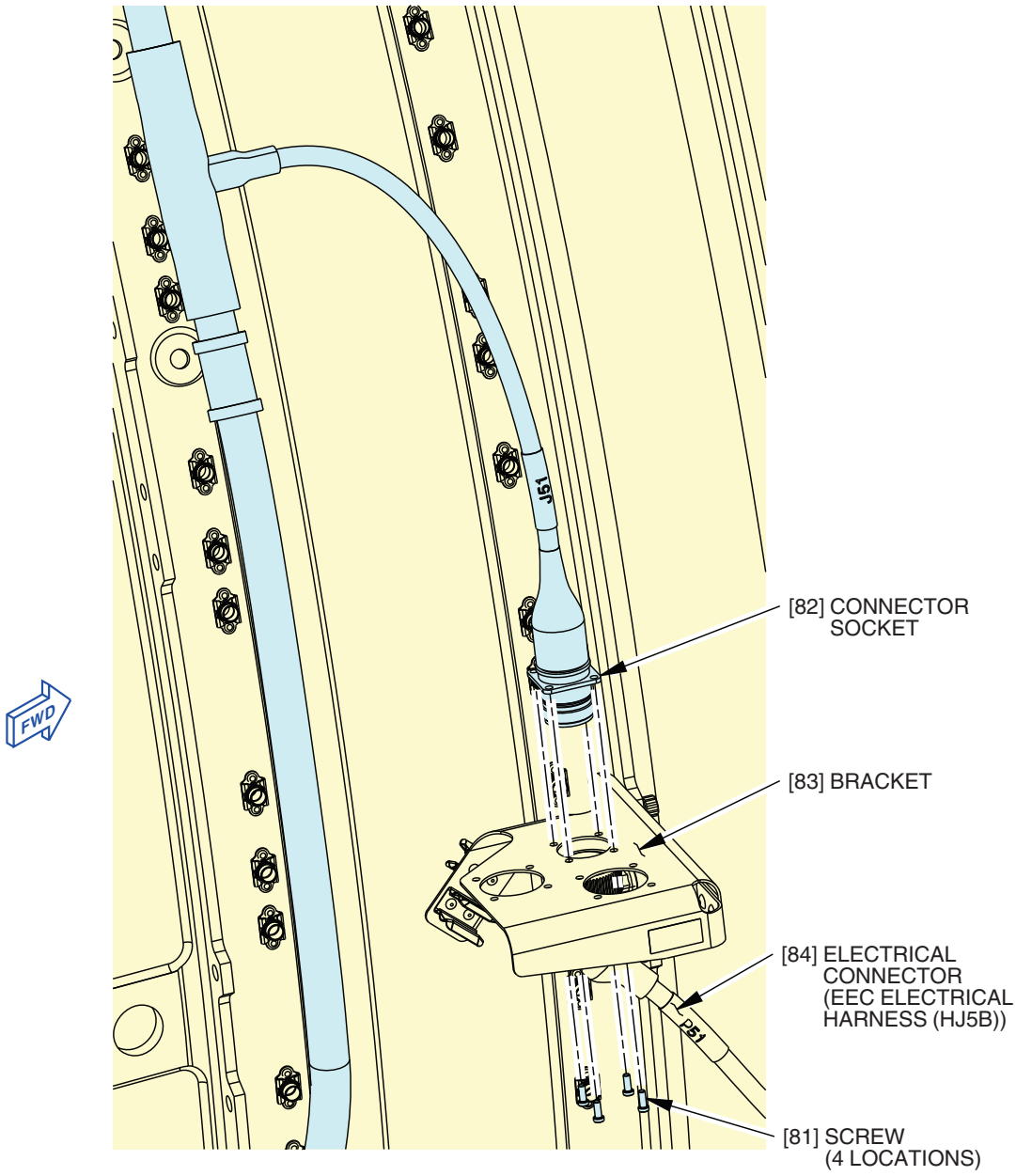
EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 4 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



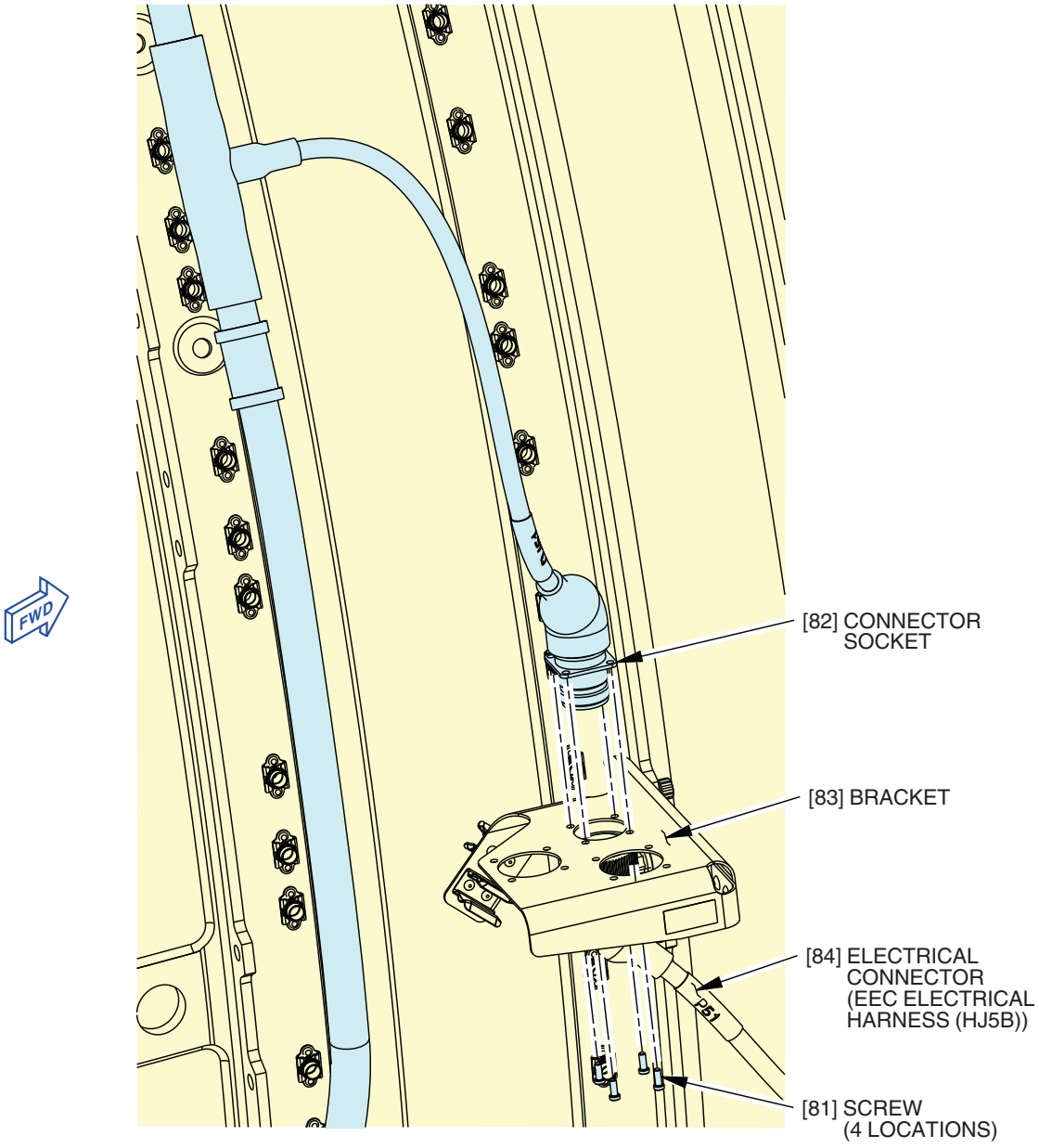
EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 5 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 6 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



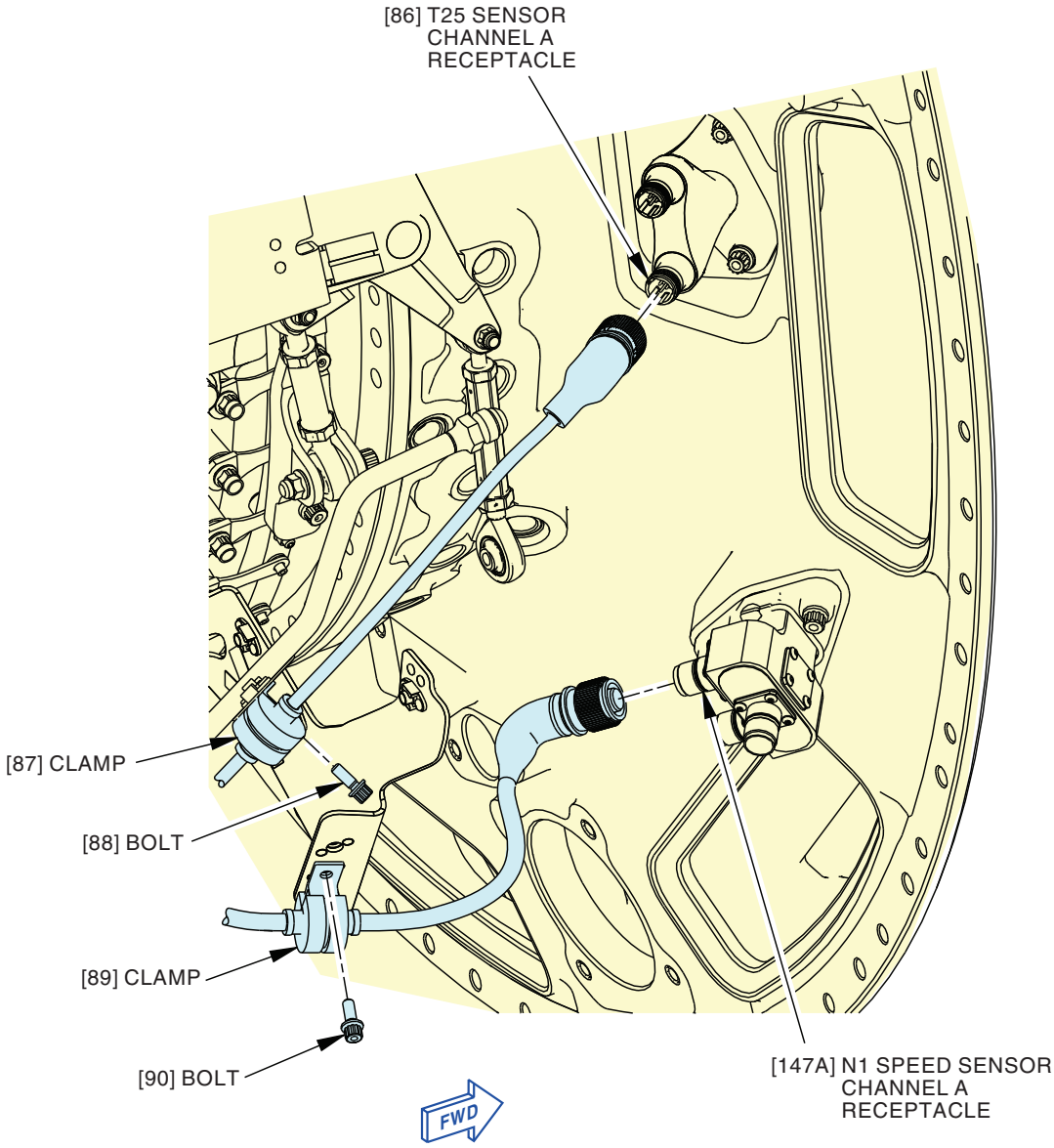
Q

ICNLEAP1B732201D5882800132A00101

3120939 S0000855764_V1

EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 7 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

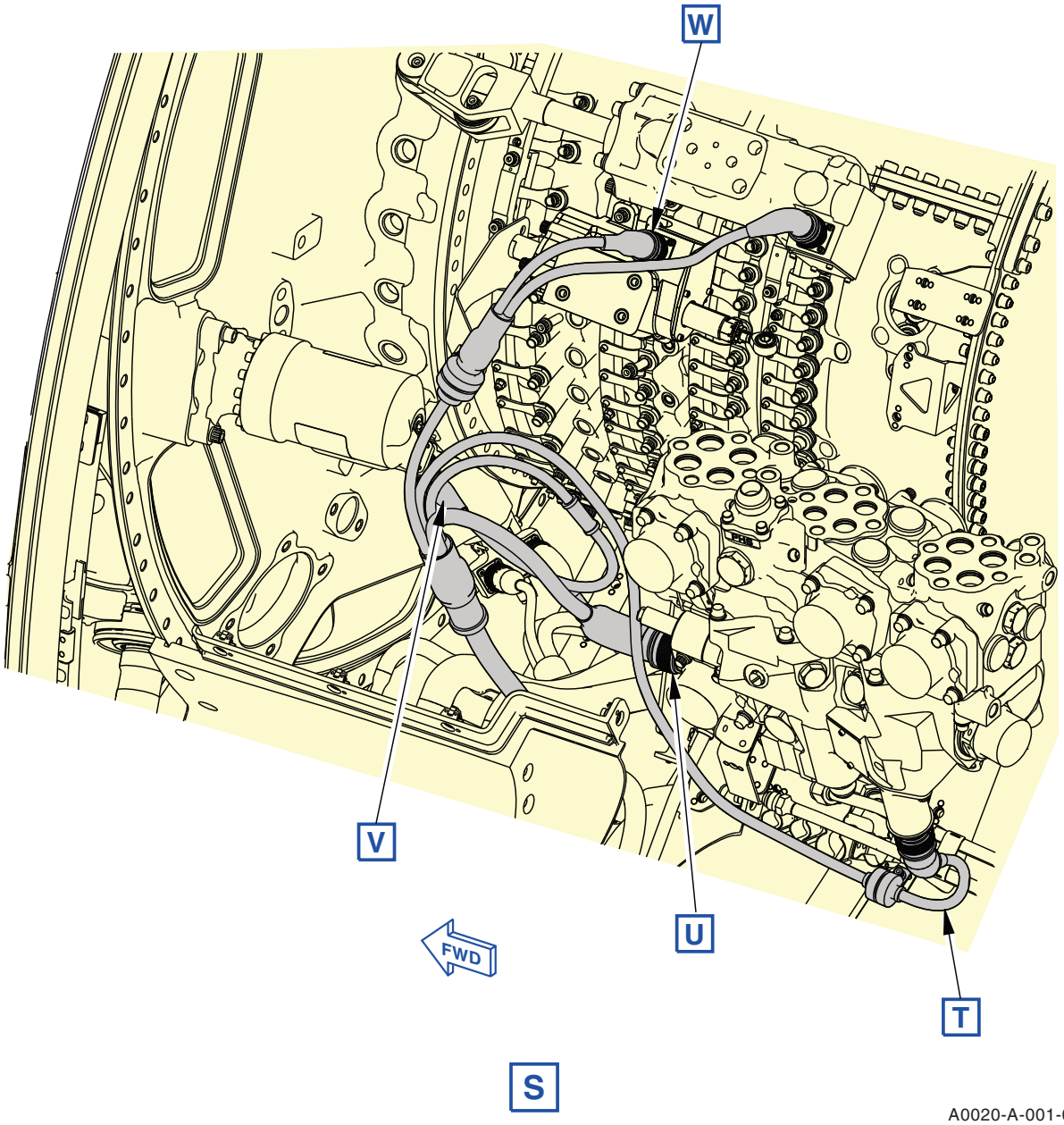


R

A0019-A-001-01
2354861 S00061516077_V1

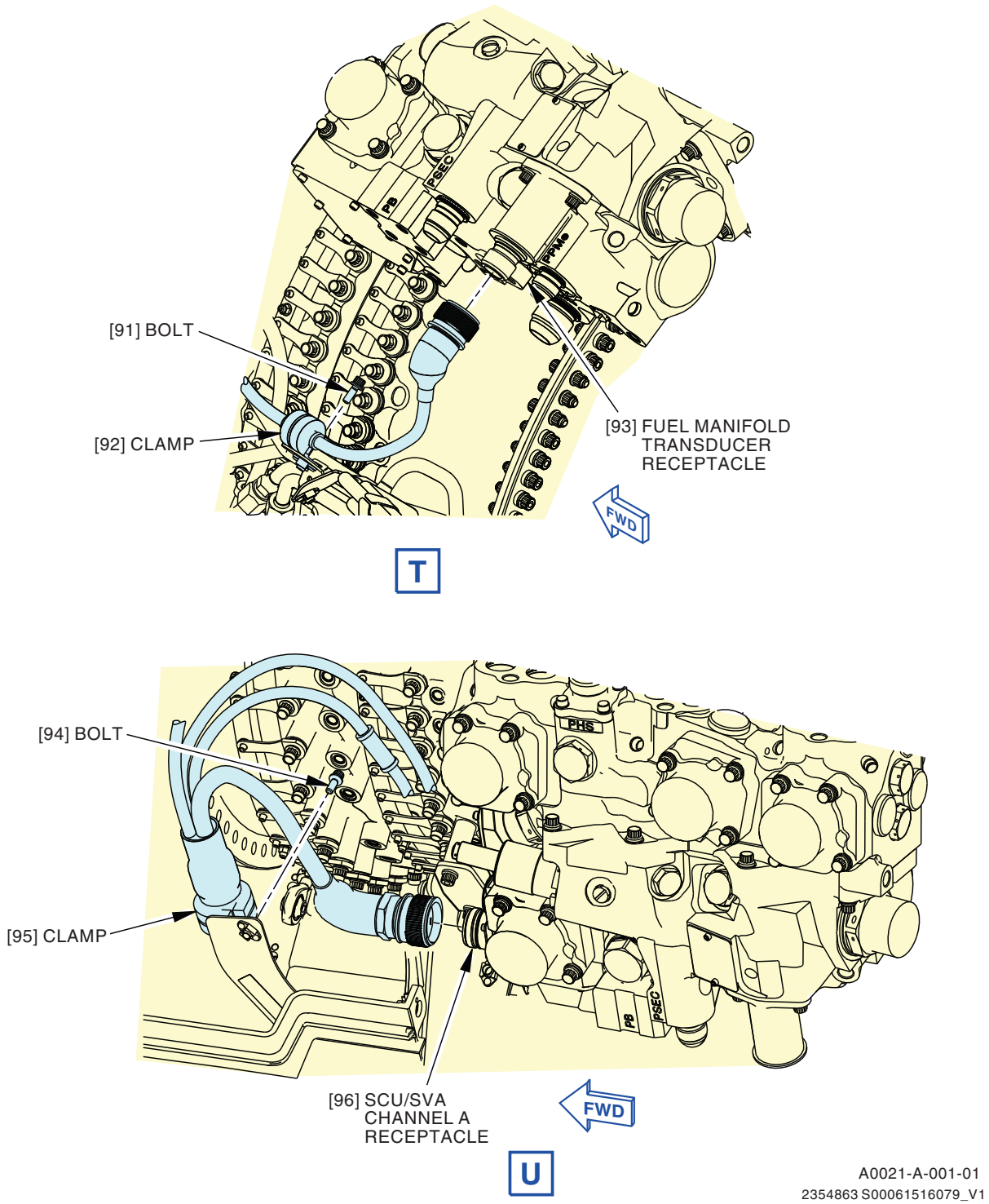
EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 8 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



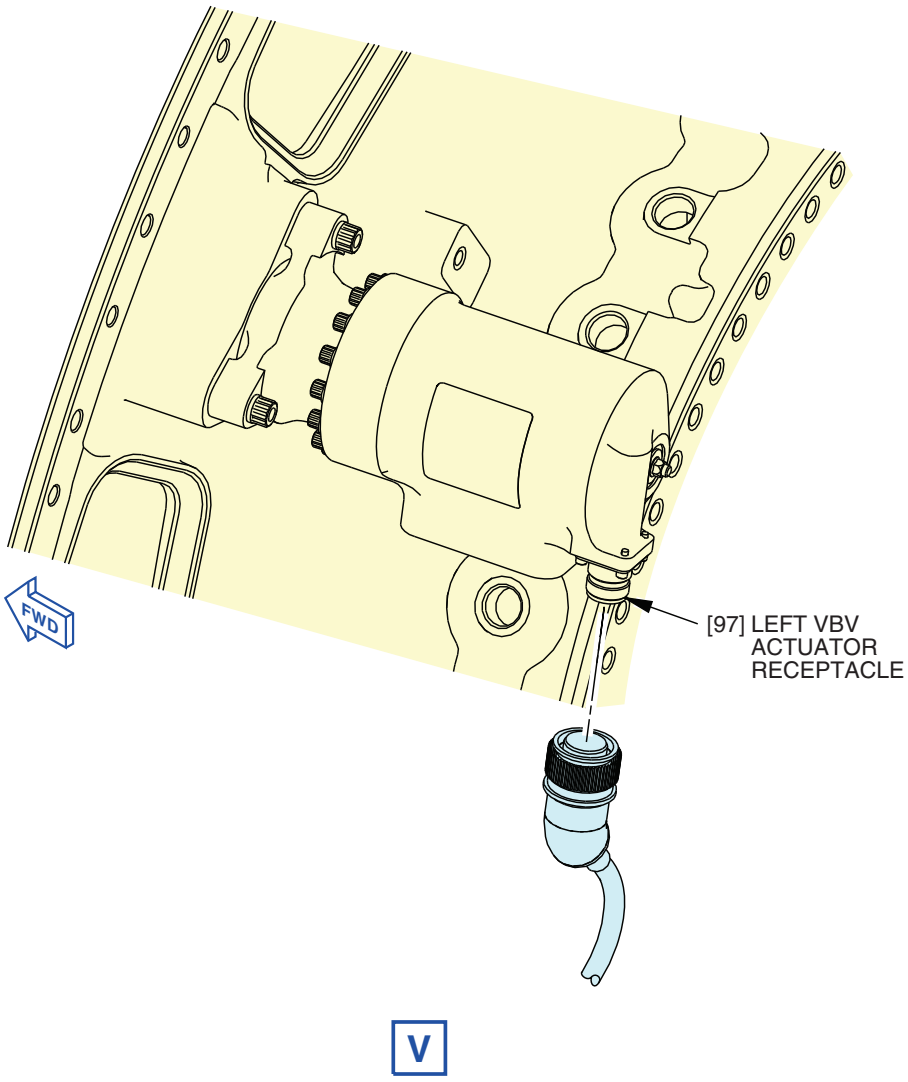
EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 9 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 10 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

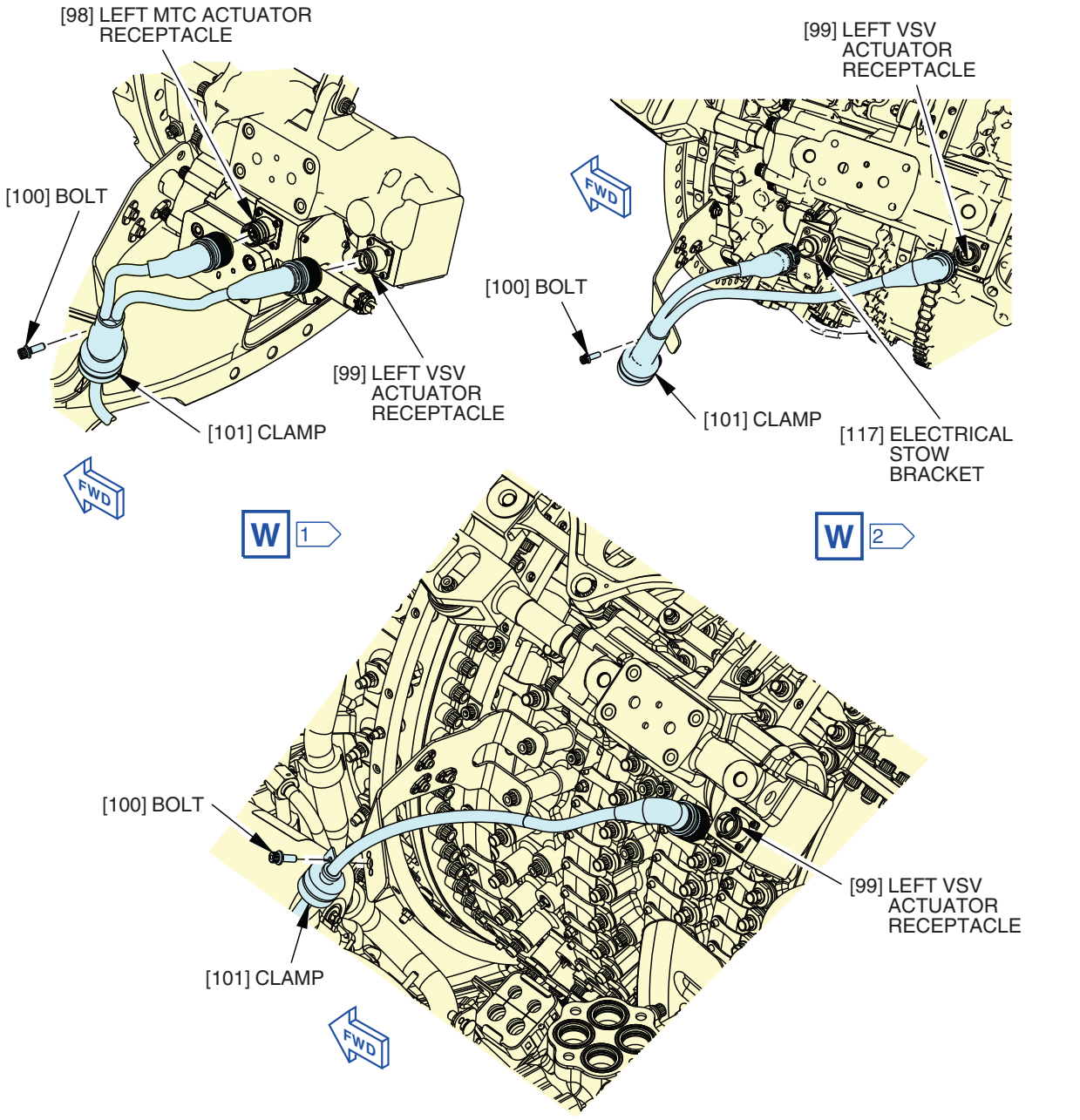


5057792-00

2354864 S00061516080_V4

EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 11 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



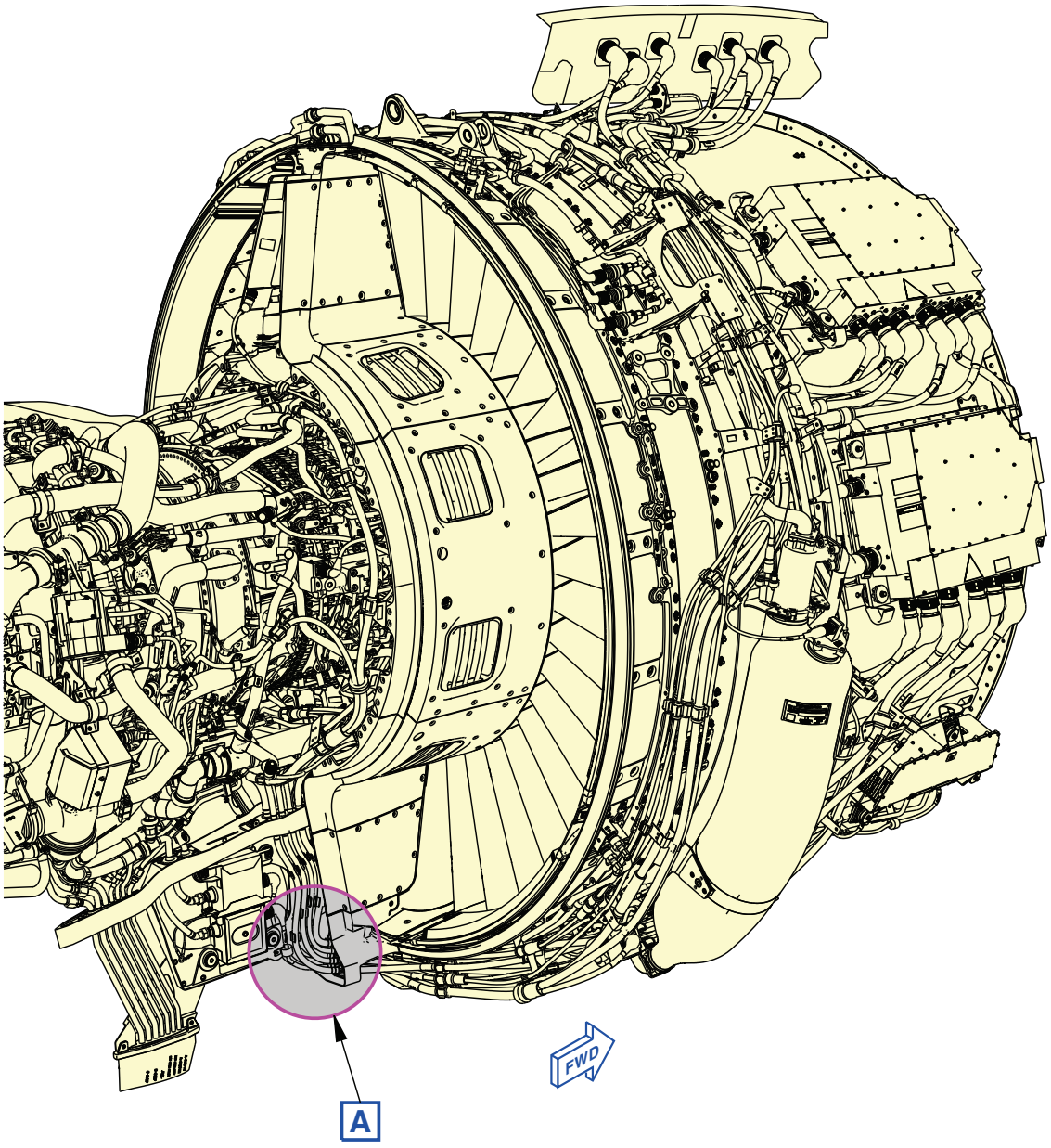
1 PRE LEAP-1B SERVICE BULLETIN 72-0124
2 POST LEAP-1B SERVICE BULLETIN 72-0124 AND PRE SB 73-0019
3 POST LEAP-1B SERVICE BULLETIN 73-0019

ICNLEAP1B732201D5882800166A00101

3120942 S0000855766_V1

EEC Electrical Harness (HJ5A) Installation
Figure 403/73-22-01-990-861-G00 (Sheet 12 of 12)

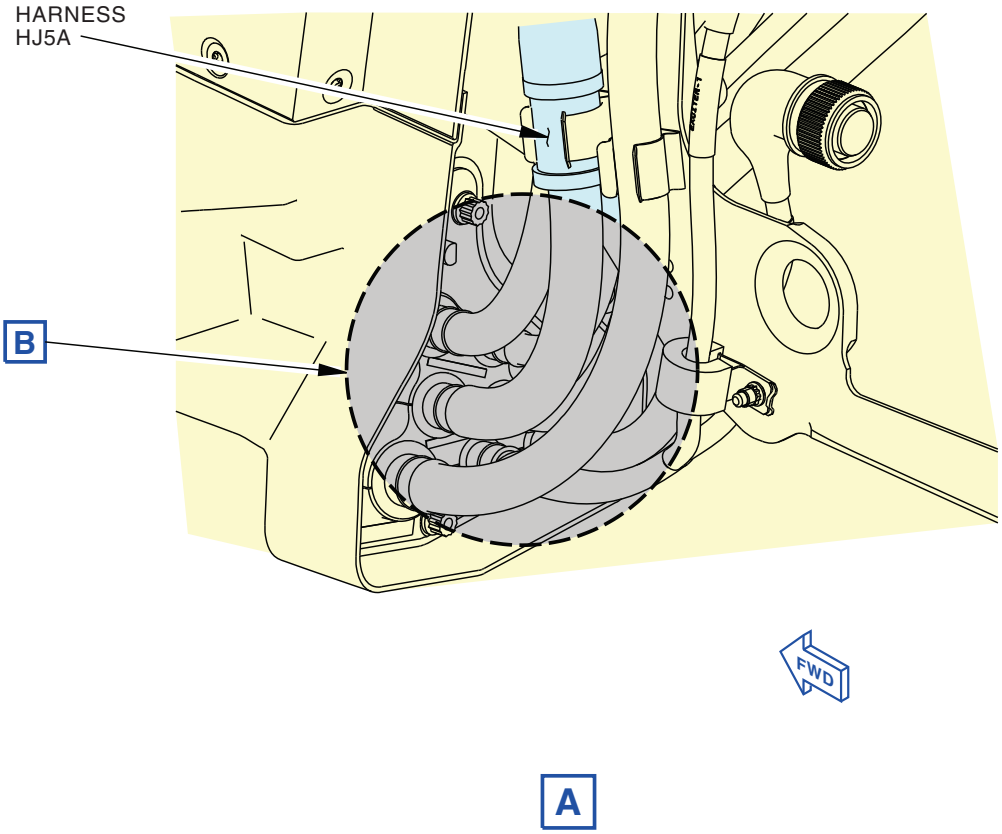
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937794 S0000713383_V3

Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 1 of 9)

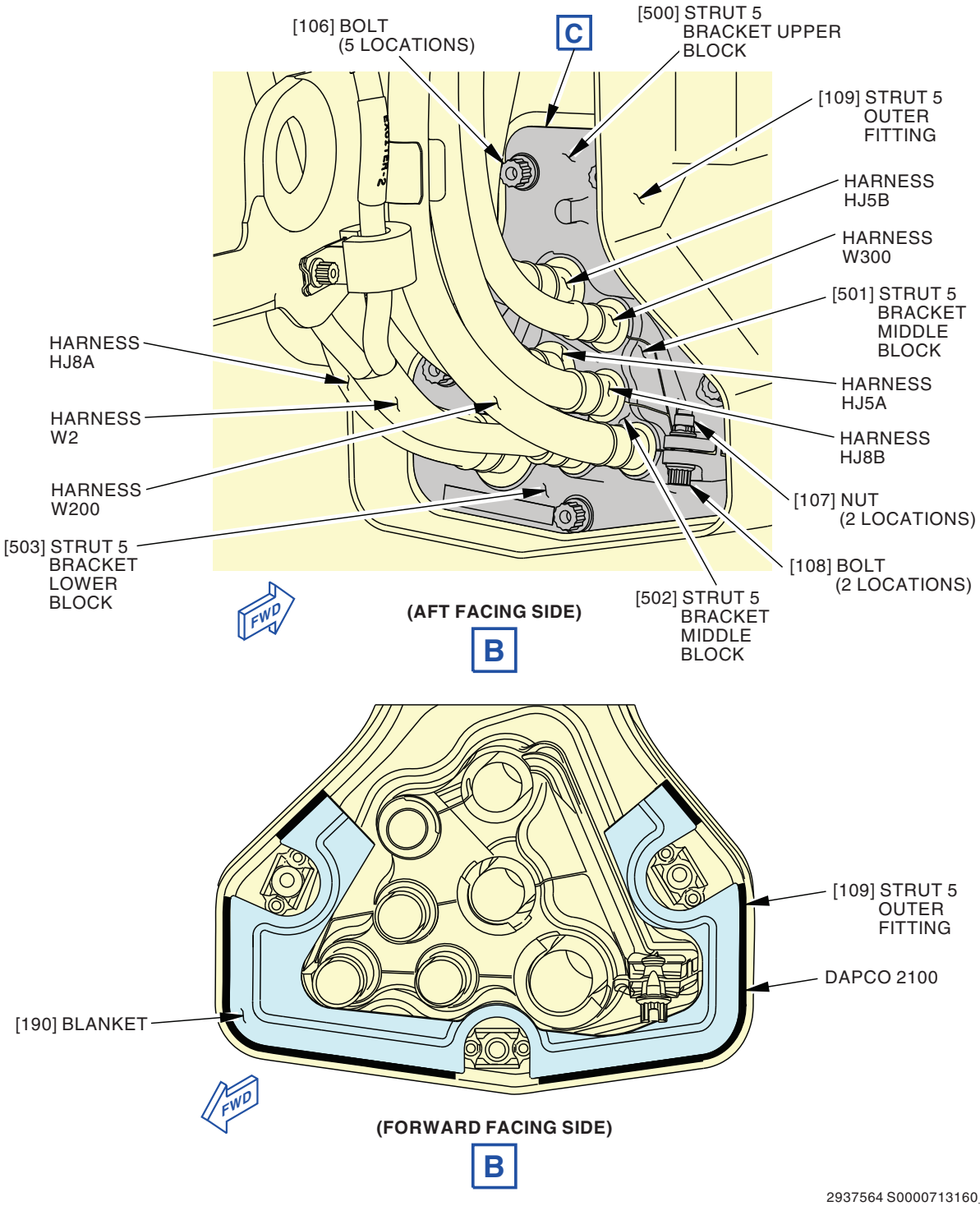
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937795 S0000713384_V2

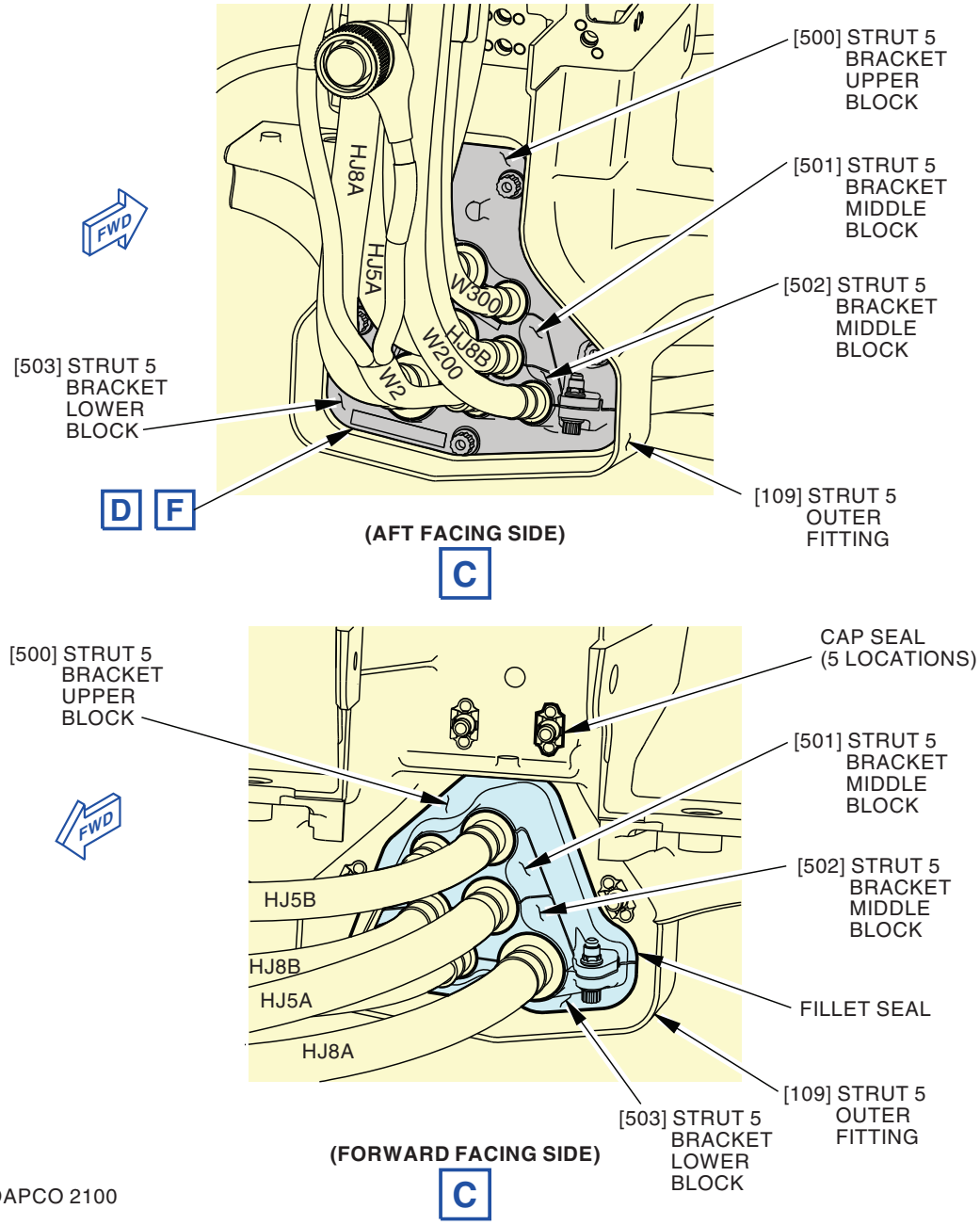
Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

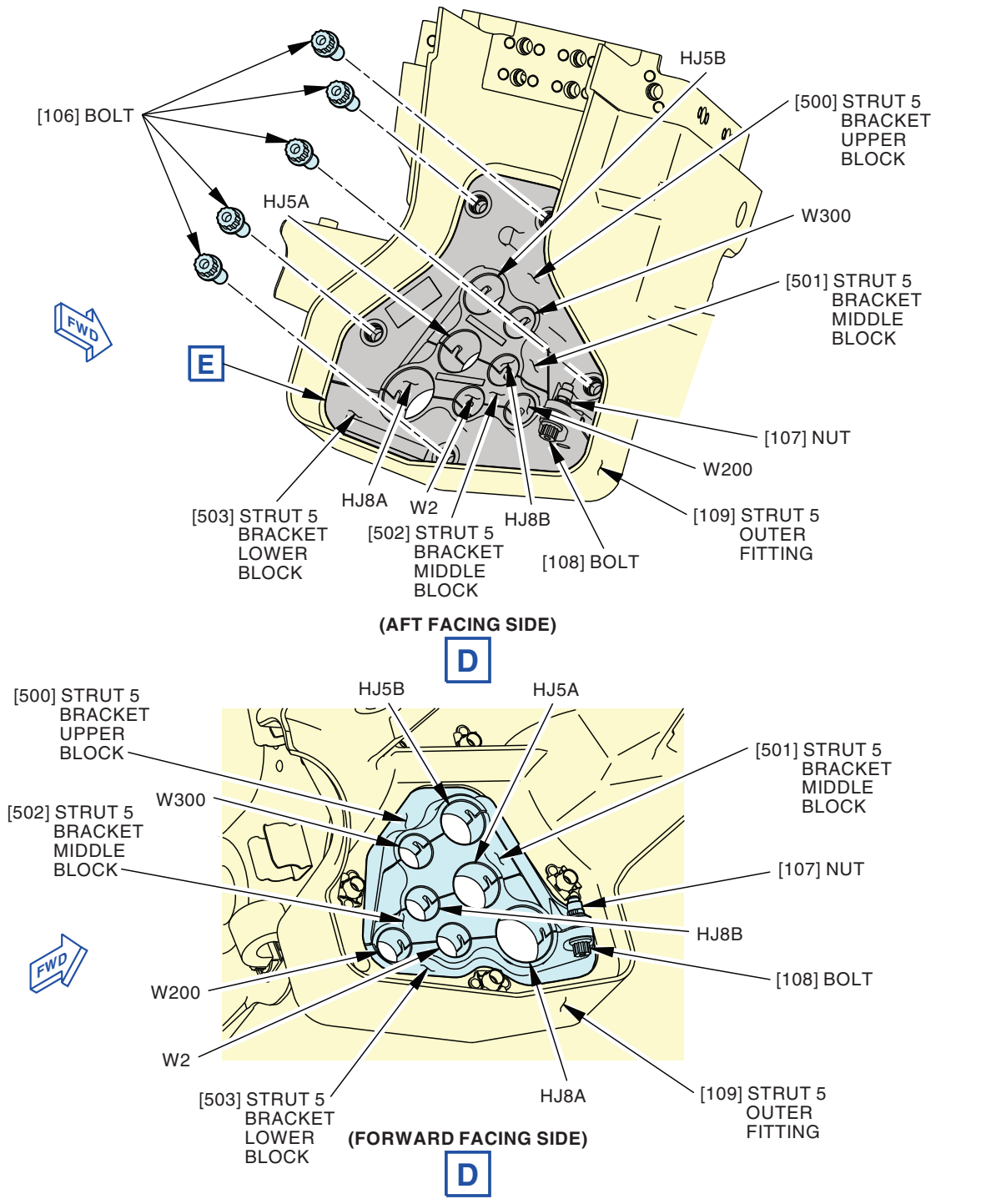
FAY SEAL PER AMM 51-22-00 ON INNER CONTOUR OF THE FITTING (AFT SIDE). SEALANT SQUEEZE-OUT SHOULD BE CONTINUOUS AND EVEN.

IT IS ACCEPTABLE TO APPLY THE FILLET SEAL ALONG INNER CONTOUR OF THE FITTING ON THE FORWARD SIDE.

2937801 S0000713390_V3

Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



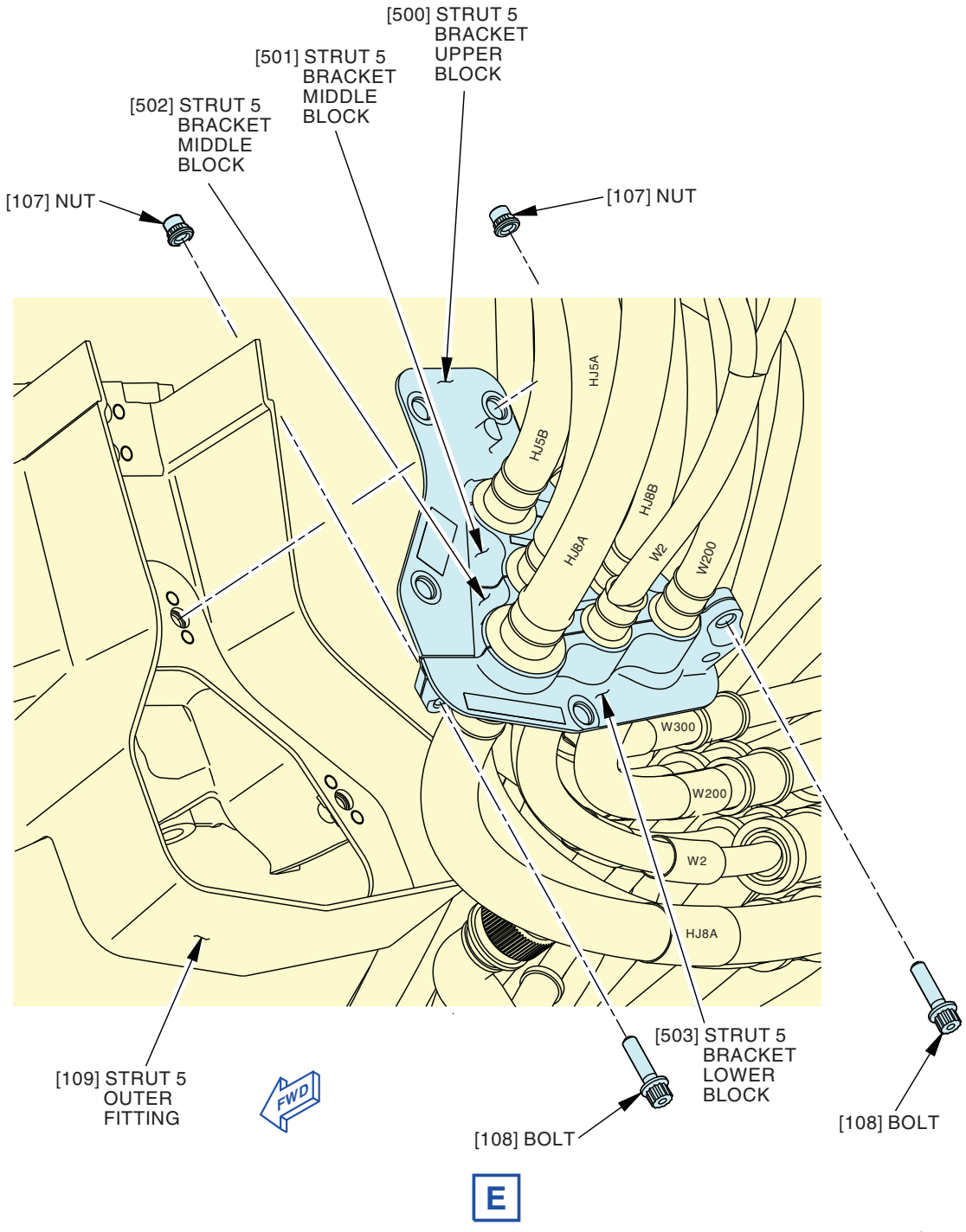
NOTE:

HARNESSES NOT SHOWN

2943980 S0000720369_V2

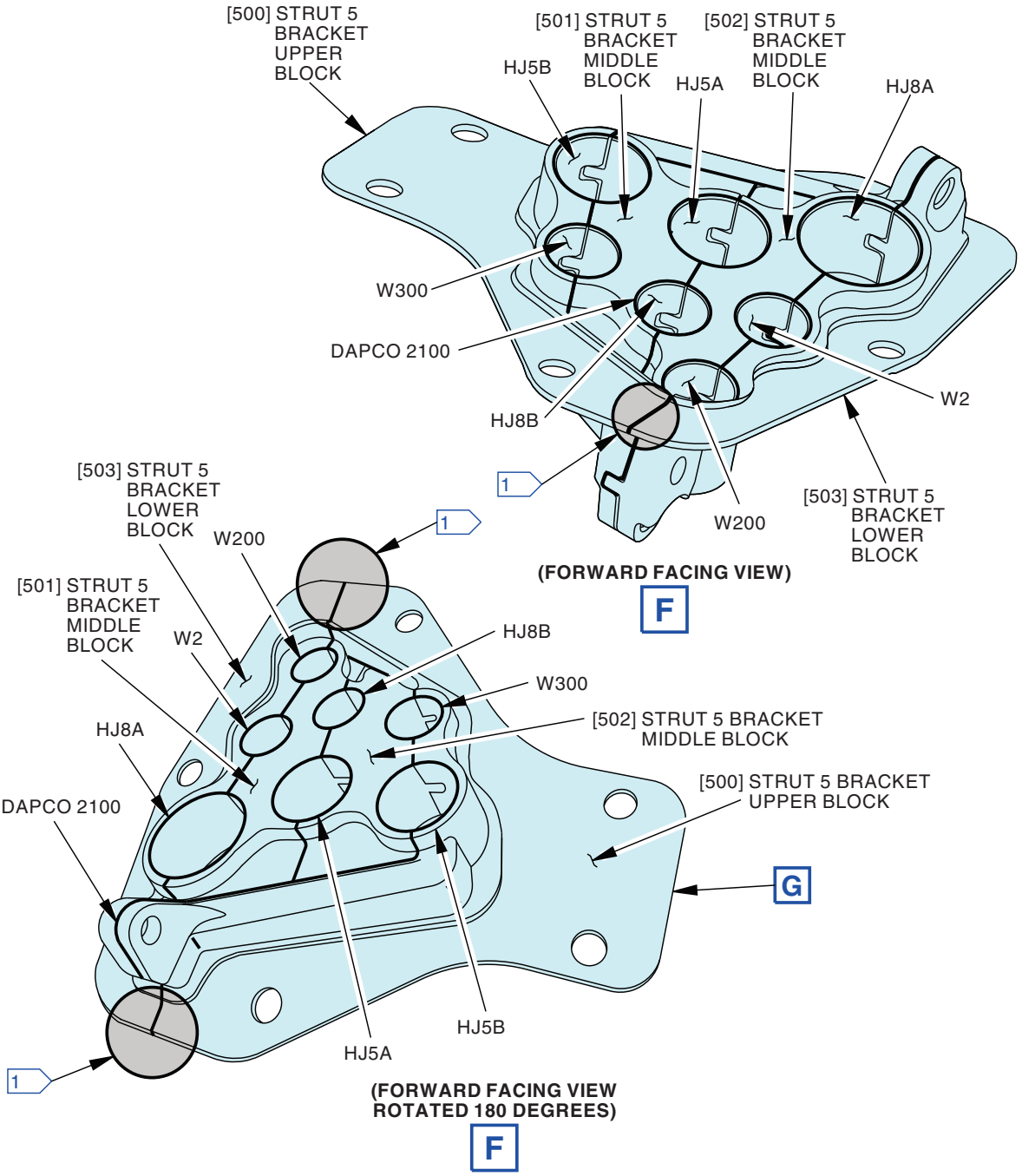
Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

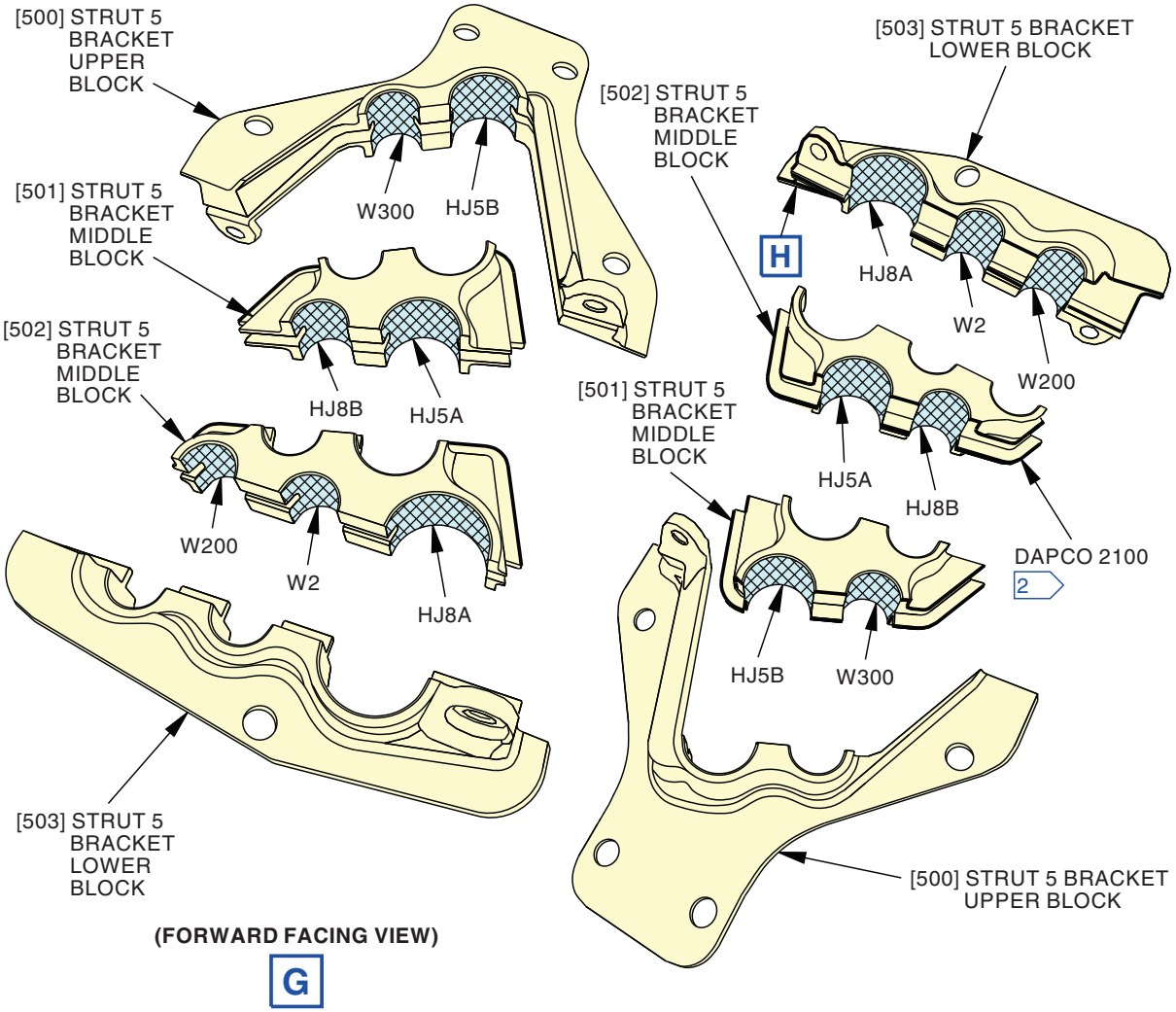
HARNESSES NOT SHOWN.

1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

2937800 S0000713389_V3

Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

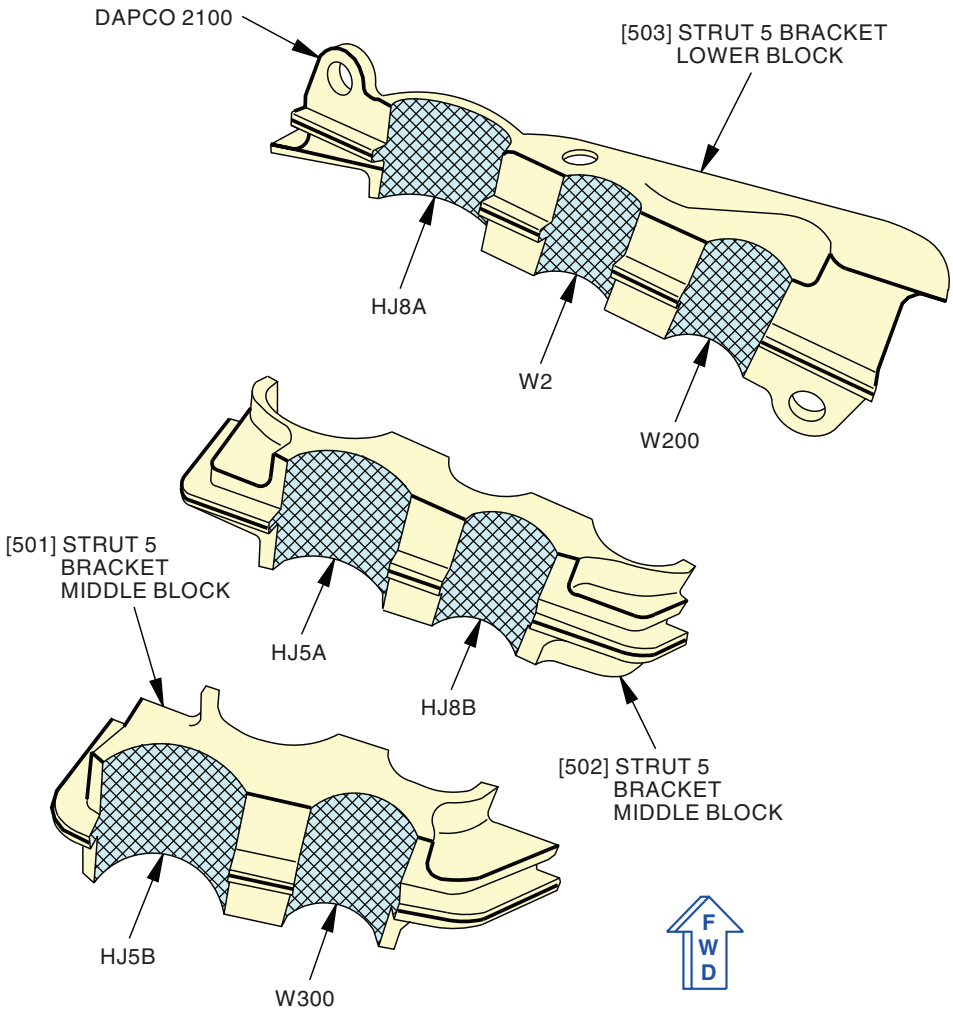
SEALANT PROVIDES FAY SEAL BETWEEN WIRE HARNESS AND BLOCKS AND SHOULD BE APPLIED TO THE BOTH SIDES OF BLOCKS.

2 SEALANT APPLIED TO ONE SIDE (TOP OR BOTTOM) OF THE BLOCKS.

2937799 S0000713388_V3

Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 8 of 9)

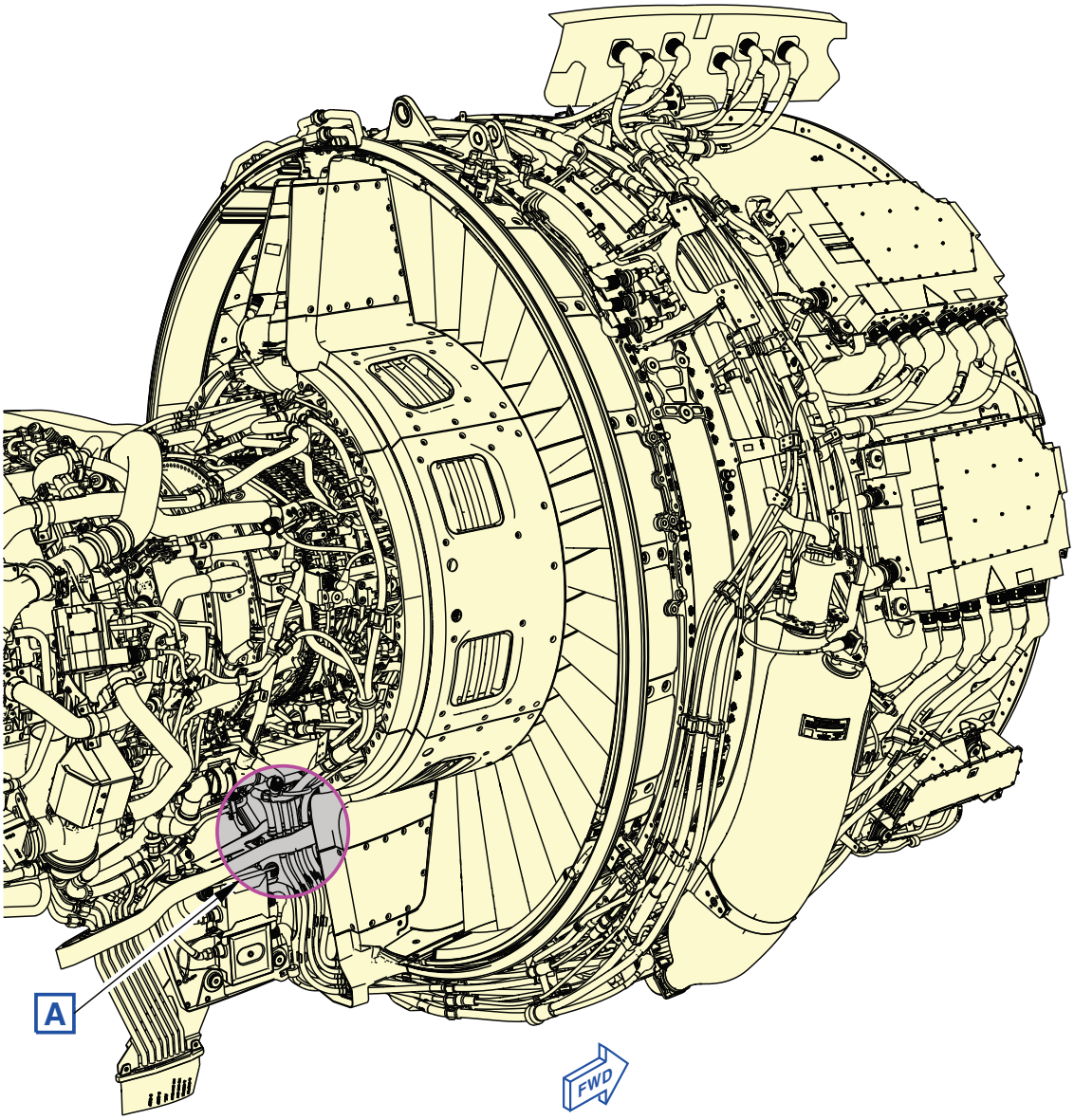
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2949941 S0000726813_V1

Strut Bracket Block Installation
Figure 404/73-22-01-990-870-G00 (Sheet 9 of 9)

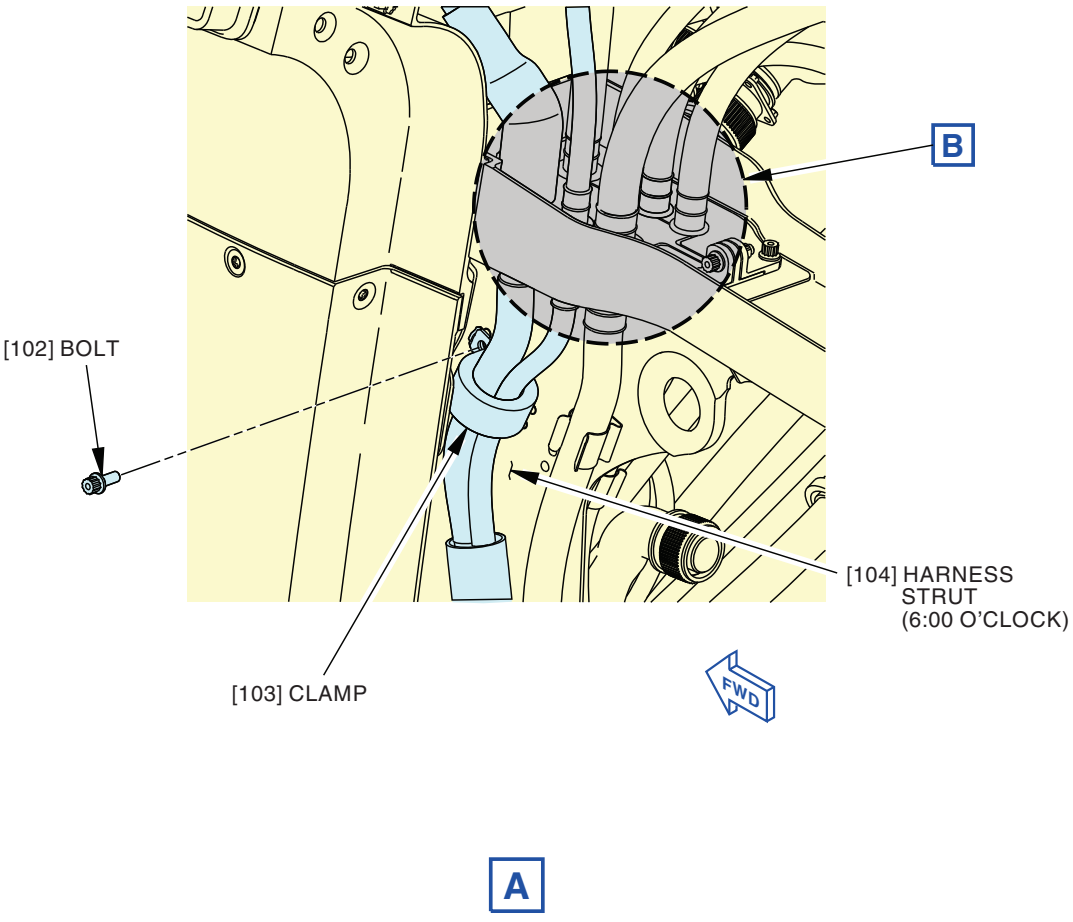
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
2938818 S0000714836_V3

Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 1 of 12)

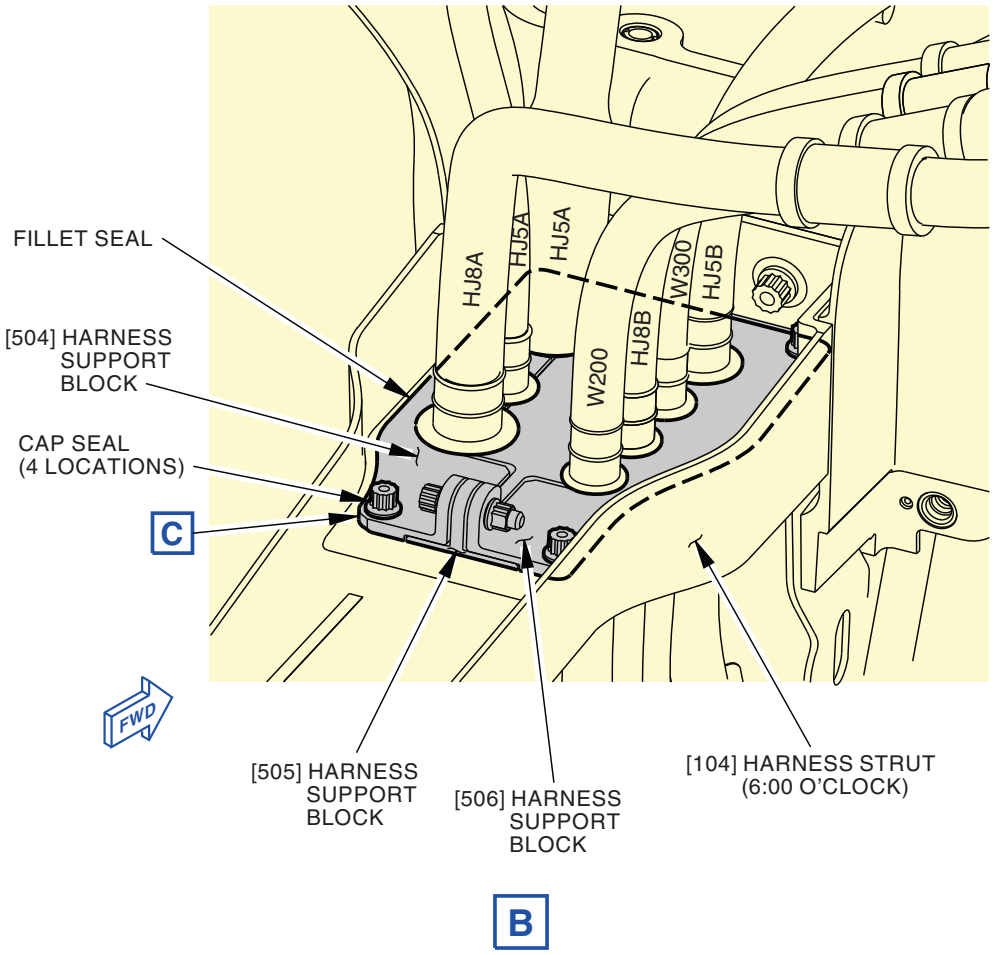
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937554 S0000713159_V2

Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 2 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

——— DAPCO 2100

NOTE:

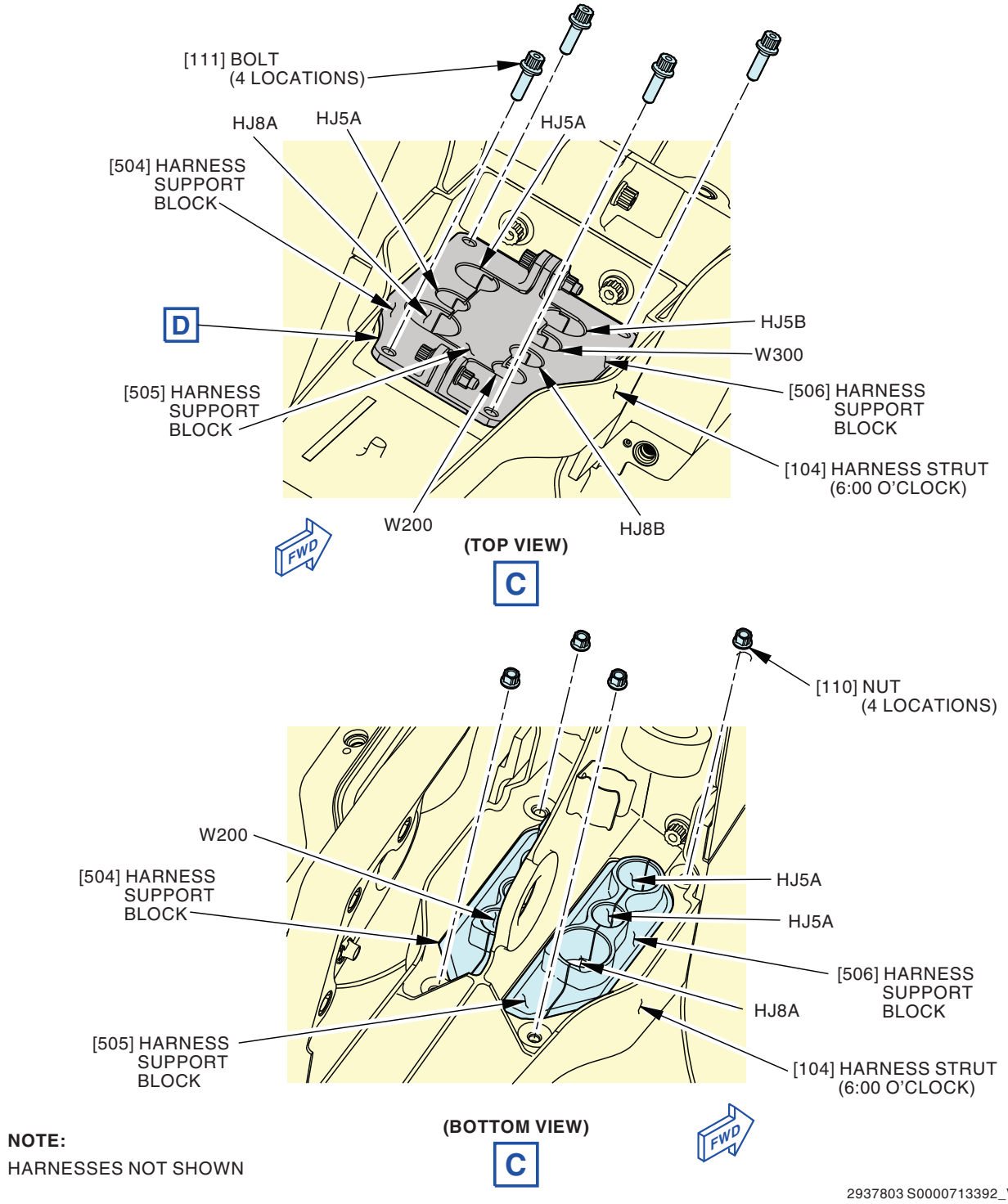
THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE HARNESS SUPPORT BLOCK. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937807 S0000713396_V4

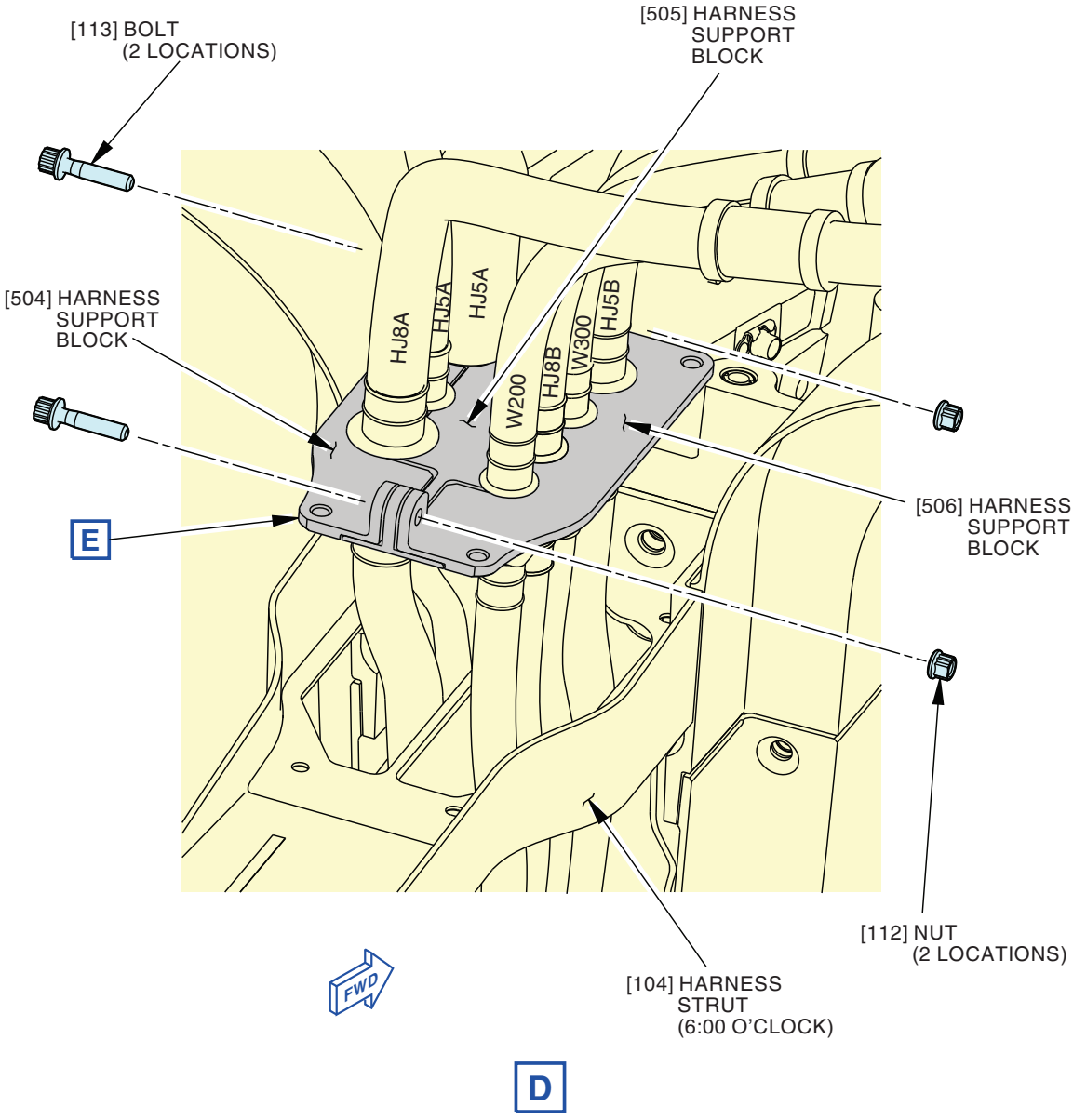
Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 3 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



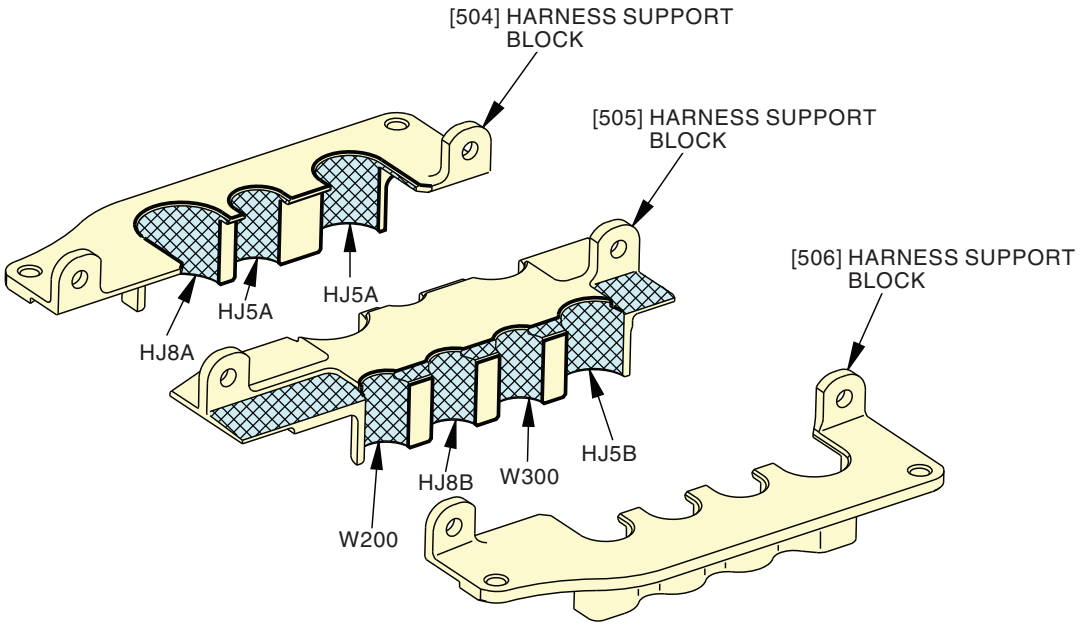
Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 4 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



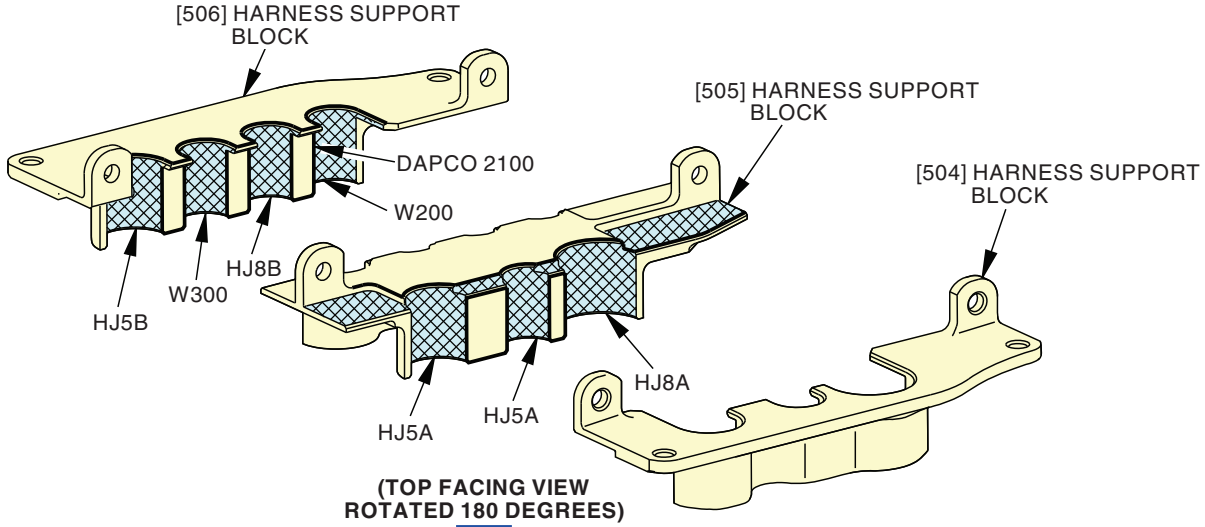
Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 5 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TOP FACING VIEW)

E



(TOP FACING VIEW ROTATED 180 DEGREES)

E

LEGEND:

DAPCO 2100

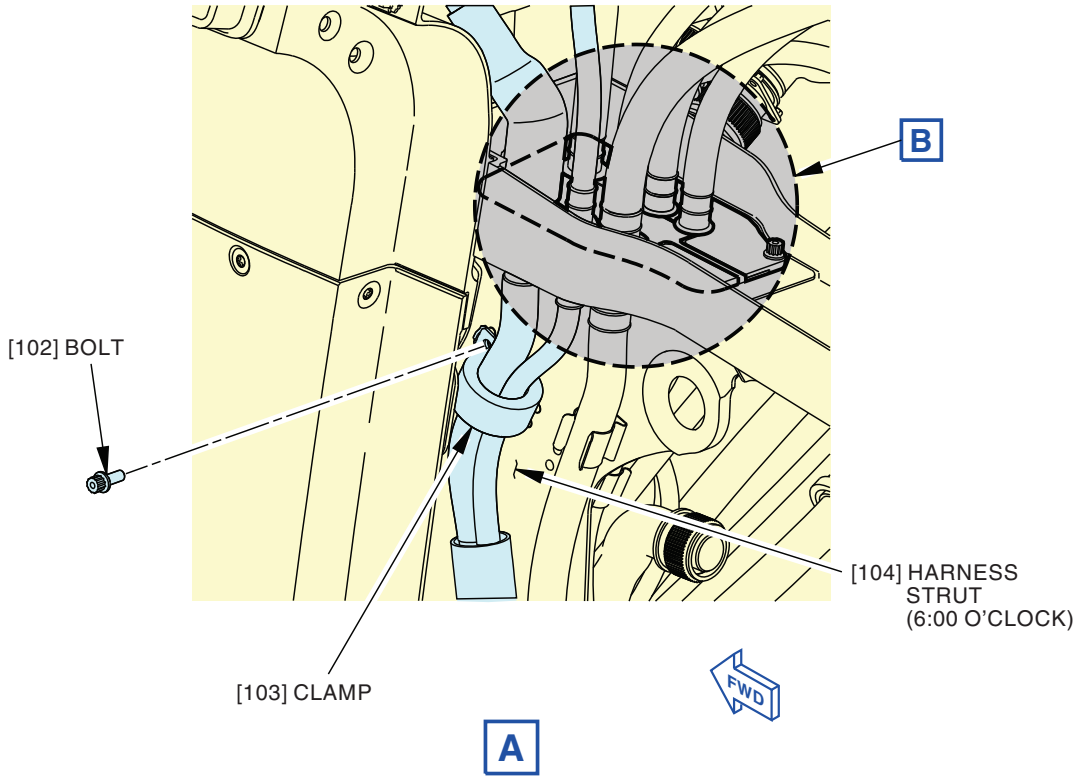
NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937805 S0000713394_V4

Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 6 of 12)

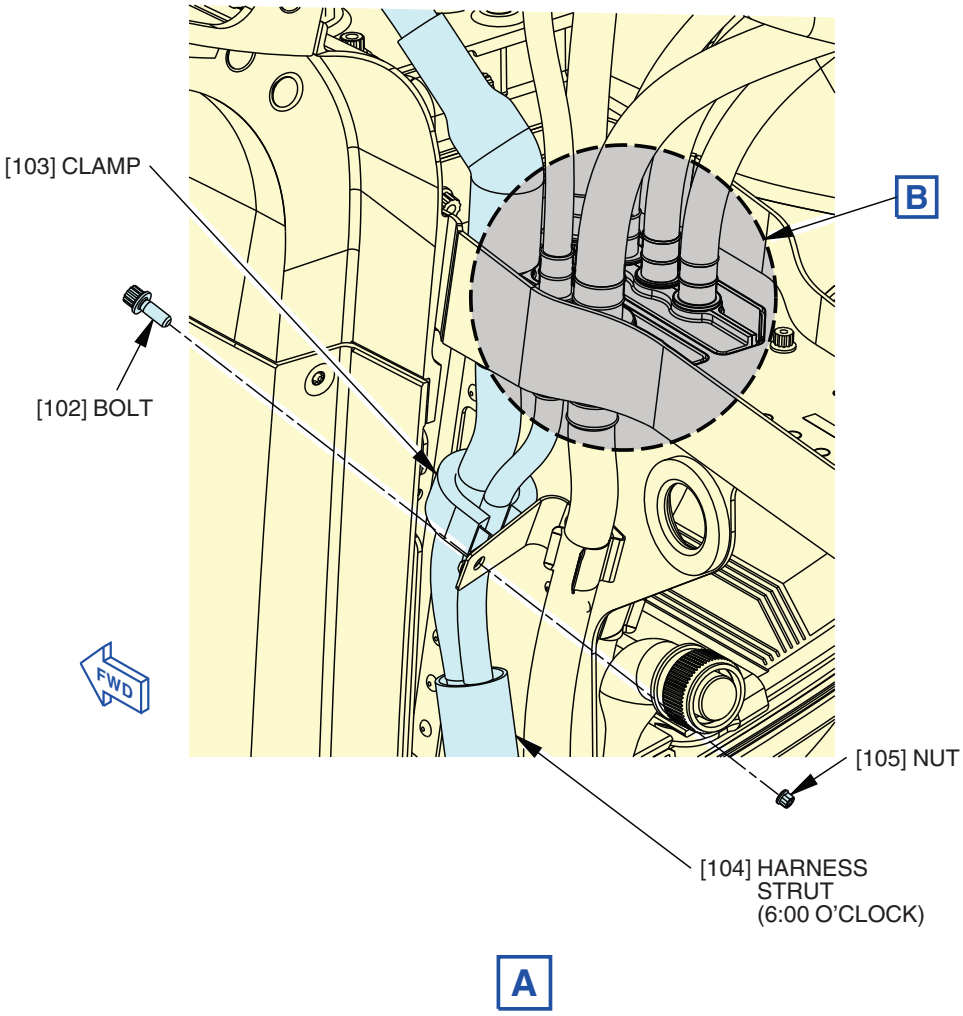
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2962027 S0000739197_V1

Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 7 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

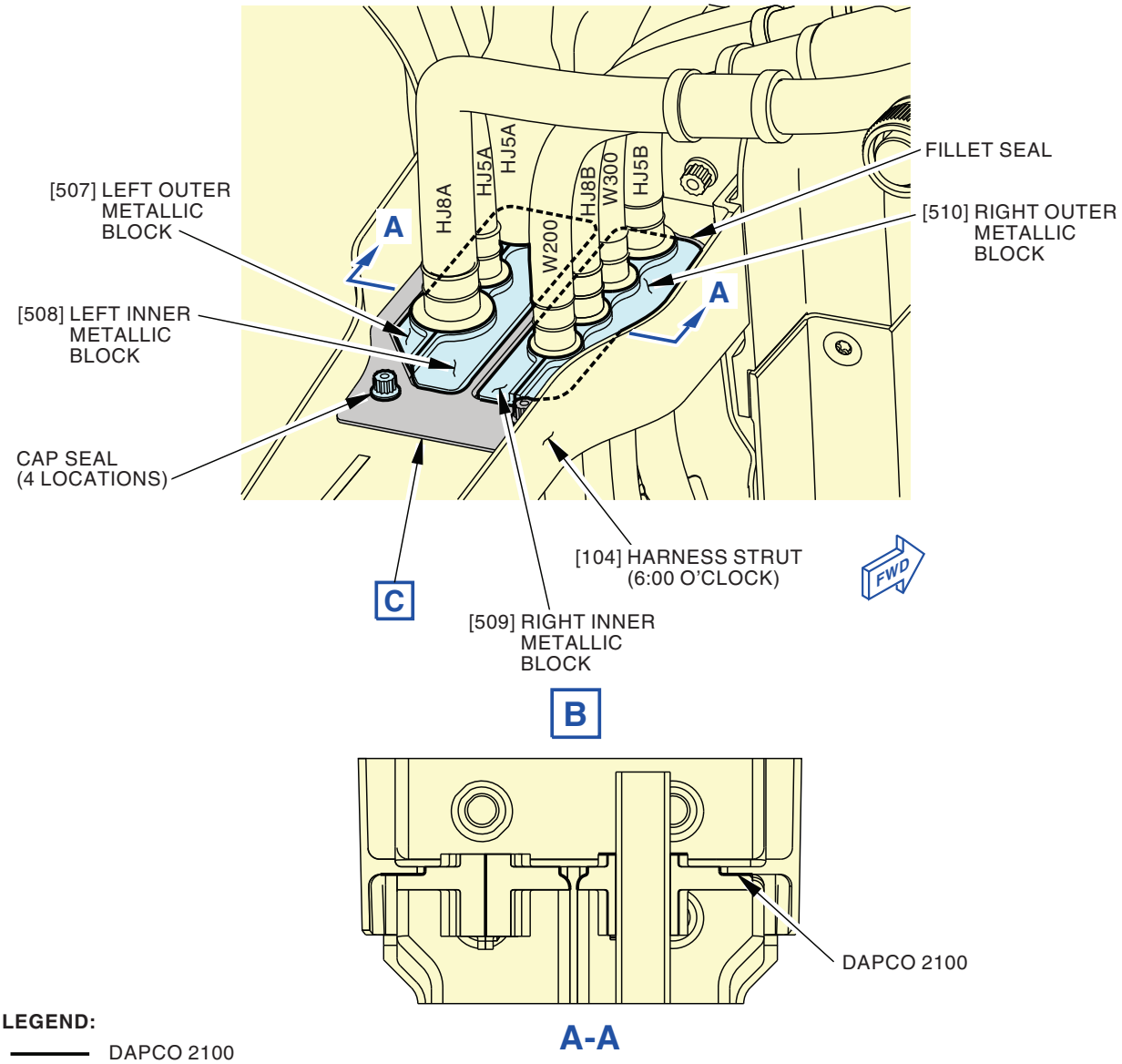


ICNLEAP1B732201D5882800167A00101

3120943 S0000855767_V1

Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 8 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

DAPCO 2100

NOTE:

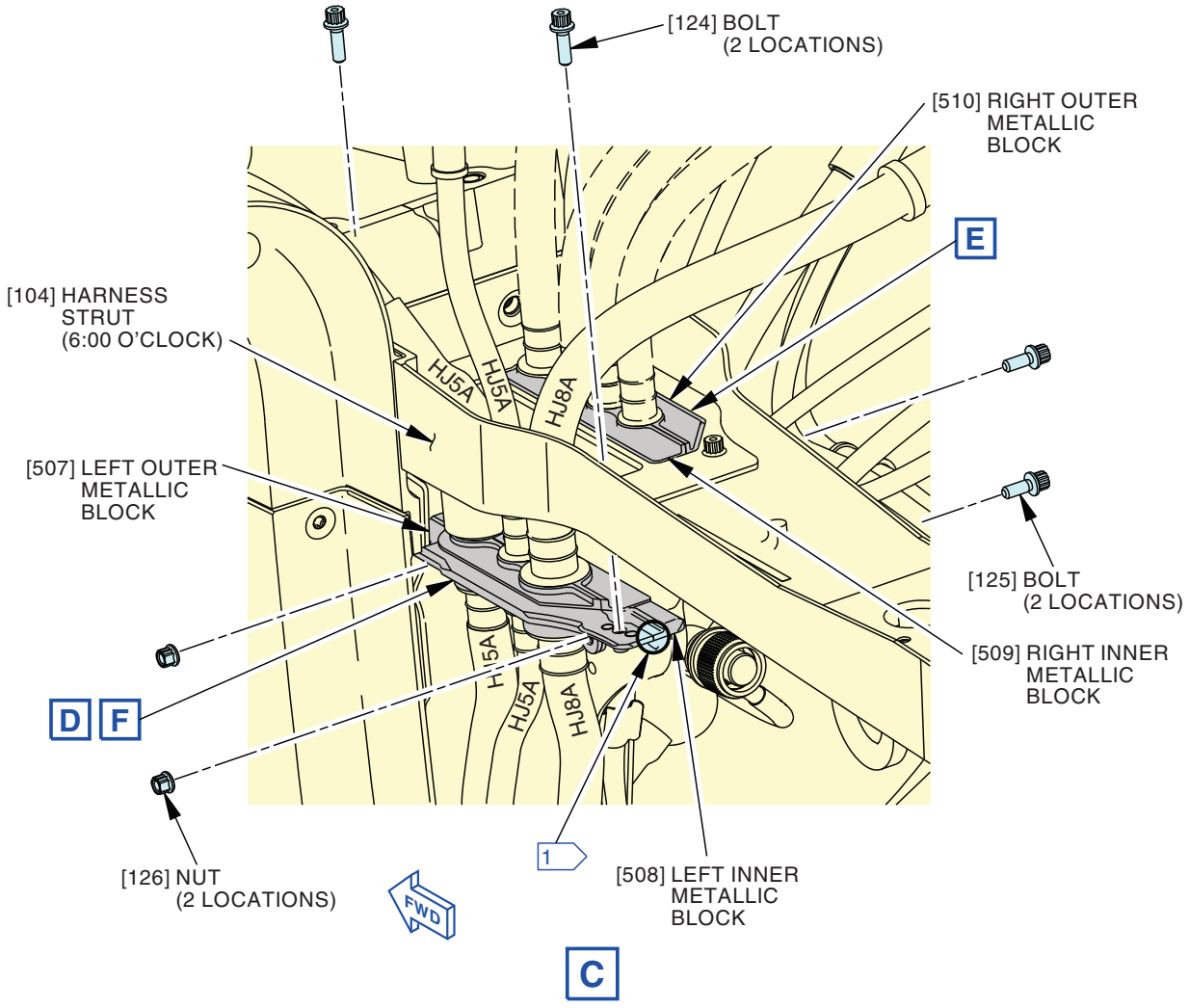
THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE METALLIC BLOCKS. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN BOTH RIGHT AND LEFT METALLIC BLOCKS HAVE BEEN REMOVED.

2962029 S0000739198_V2

Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 9 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

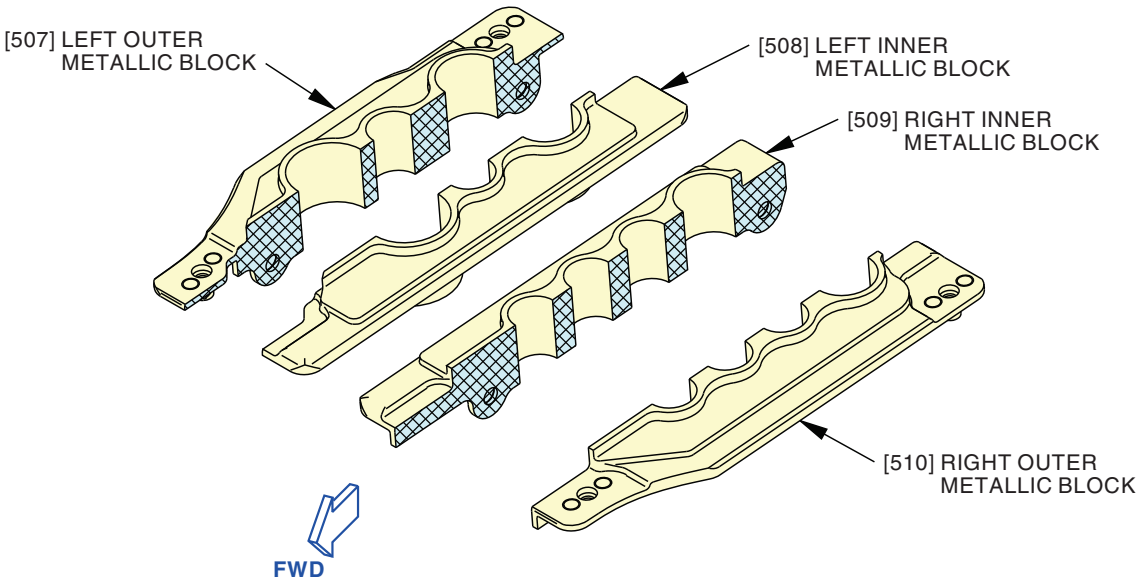
THIS VIEW IS SPECIFIC FOR LEFT INNER AND OUTER METALLIC BLOCKS. SAME LOGIC IS APPLIED FOR RIGHT INNER AND OUTER METALLIC BLOCKS.

1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

ICNLEAP1B732201D5882800060A00101
2962028 S0000739199_V2

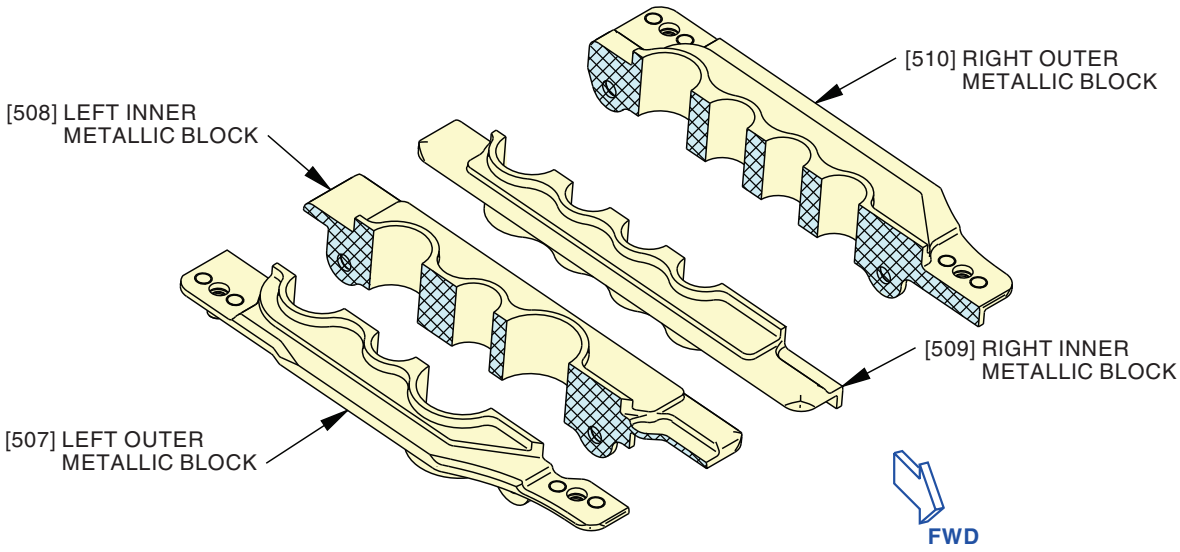
Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 10 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TOP FACING VIEW)

D



(TOP FACING VIEW ROTATED 90 DEGREES)

D

LEGEND:

DAPCO 2100

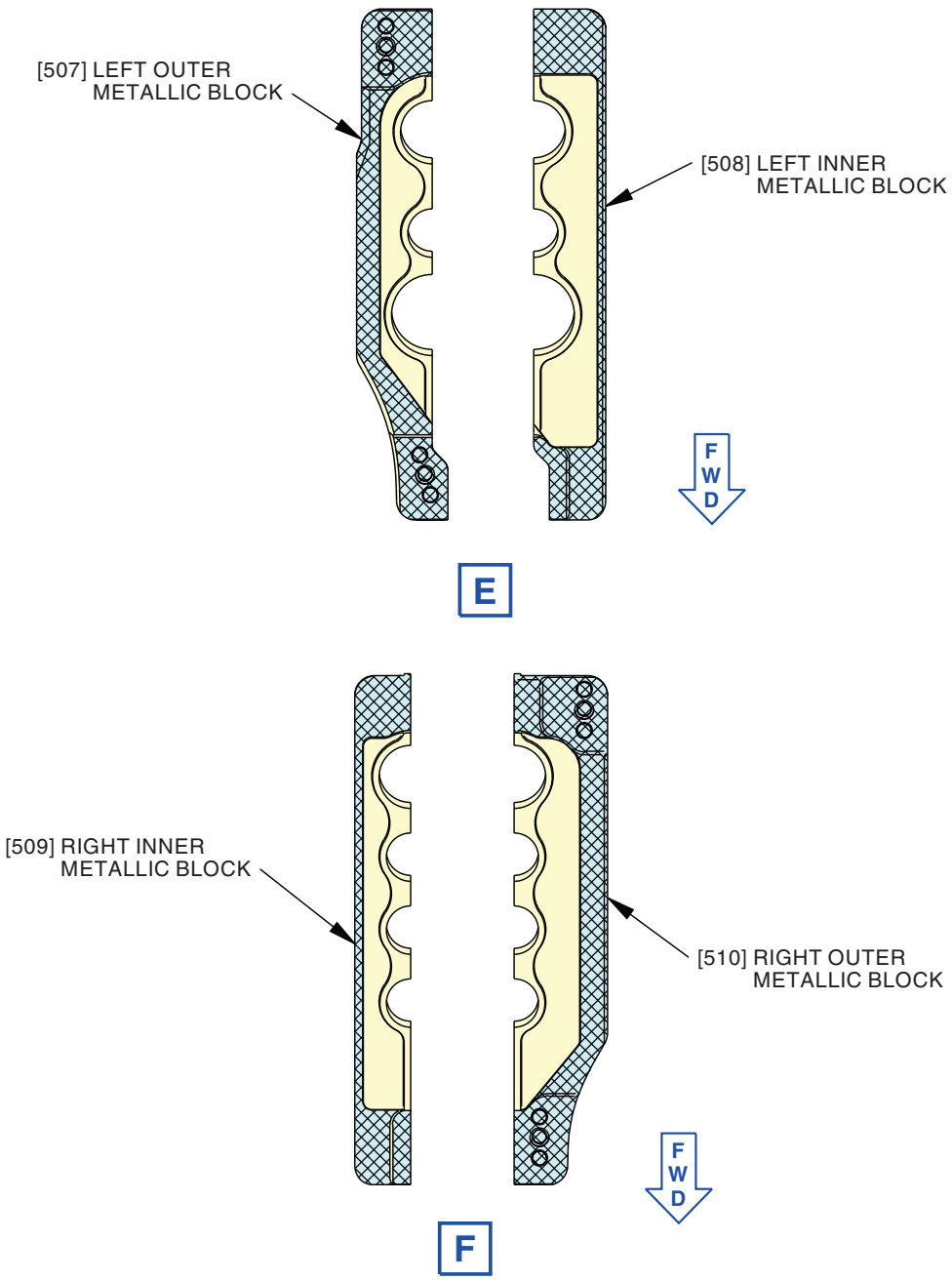
NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994662 S0000771309_V1

Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 11 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

DAPCO 2100

NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994663 S0000771310_V1

Harness Support Block Installation
Figure 405/73-22-01-990-871-G00 (Sheet 12 of 12)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-22-01-400-803-G00

7. Electronic Engine Control (EEC) Electrical Harness (HJ5A) Installation

(Figure 403, Figure 404, and Figure 405)

A. General

(1) This task gives the instructions to install the electronic engine control (EEC) electrical harness (HJ5A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-06-400-801-G00	Fan Frame Strut Panels Installation (P/B 401)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50396	Sealant - Dapco 2200 Primerless Silicone Firewall Sealant	BMS5-63 Type II Class B-1/2
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Electronic Engine Control (EEC) Electrical Harness (HJ5A) Installation

SUBTASK 73-22-01-210-001-G00

(1) During the installation, make sure that EEC electrical harness (HJ5A) [67] is not stressed or damage.

SUBTASK 73-22-01-210-002-G00

(2) Make sure that all components are clean and free of old sealant prior to adding new sealant.

SUBTASK 73-22-01-420-002-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(3) Install the EEC electrical harness (HJ5A) [67] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the EEC electrical harness (HJ5A) [67] in a correct position with the 24 clips [70] along its routing.

(d) Install the EEC electrical harness (HJ5A) [67] on the harness strut (6:00 o'clock) [104] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

1) Install the harness support block [504], harness support block [505], and harness support block [506] on the harness strut (6:00 o'clock) [104] as follows:

a) Lubricate the four bolts [111] with graphite mineral vaseline, D00601 [CP2101].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses (View E, Figure 405).

c) Make sure that the EEC electrical harness (HJ5A) [67] and harnesses are in the correct position on the harness support block [504], harness support block [505], and harness support block [506] (View D, Figure 405).

d) Install the two bolts [113] and two nuts [112] to assemble the harness support block [504], harness support block [505], and harness support block [506] in the correct position with the harnesses.

e) Tighten the nuts [112] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION

MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

f) Make sure that there is minimum amount of the squeeze-out on the lower side.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

g) Install the harness support block [504], harness support block [505], and harness support block [506] in the correct position on the harness strut (6:00 o'clock) [104].

h) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104] (View B, Figure 405).

i) Install the four bolts [111] and four nuts [110] that attach the assembly of the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

j) Tighten the nuts [110] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

k) Make sure that there is minimum amount of the squeeze-out on the lower side.

l) Apply more sealant to form a correct fillet seal around the contour of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

CAUTION MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

m) Clean the four nuts [110] and four bolts [111] that will have sealant applied to them.

n) Apply Dapco 2100 sealant, A50483 [CP2927], to the heads of the four bolts [111].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

2) Remove any remaining of old sealant residue left on the harnesses or harness strut (6:00 o'clock) [104].

3) Install the left outer metallic block [507] and left inner metallic block [508] on the harness strut (6:00 o'clock) [104] as follows:

a) Lubricate the two bolts [124] and two bolts [125] with graphite mineral vaseline, D00601 [CP2101].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507] and left inner metallic block [508].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

c) Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507], left inner metallic block [508], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the left outer metallic block [507] and left inner metallic block [508] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

f) Install the left outer metallic block [507] and left inner metallic block [508] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

g) Install the two bolts [124] that attach the left outer metallic block [507] and left inner metallic block [508] to the harness strut (6:00 o'clock) [104].

h) Tighten the two bolts [124] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

i) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

CAUTION

MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

j) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

<2> Make sure that the visible faces of the bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

k) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

l) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of left outer metallic block [507] and left inner metallic block [508].

4) Install the new right inner metallic block [509] and right outer metallic block [510] on the harness strut (6:00 o'clock) [104] as follows:

a) Lubricate the two bolts [124] and two bolts [125] with graphite mineral vaseline, D00601 [CP2101].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509] and right outer metallic block [510].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

d) Install the two bolts [125] and two nuts [126] to assemble the right inner metallic block [509] and right outer metallic block [510] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

f) Install the right inner metallic block [509] and right outer metallic block [510] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

g) Install the two bolts [124] that attach the right inner metallic block [509] and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

h) Tighten the two bolts [124] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

i) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

j) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

<2> Make sure that the visible faces of the two bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

k) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

l) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of right inner metallic block [509] and right outer metallic block [510].

PTF ALL

5) Install the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109] as follows:

a) Lubricate the five bolts [106] and two nuts [107] with graphite mineral vaseline, D00601 [CP2101].

b) Apply Dapco 2100 sealant, A50483 [CP2927], to the one side (either top or bottom) of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

NOTE: Fay seal applies to the two sides of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Make sure that the EEC electrical harness (HJ5A) [67] and harnesses are in the correct position on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] (View E, Figure 404).

d) Install the two bolts [108] and two nuts [107] to assemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

e) Tighten the nuts [107] to 42 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Make sure that the fillet seal of the squeeze-out is on the forward facing side.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

g) Make the sealant squeeze-out flush with the surface on the outer edges of the strut 5 bracket upper block [500] and strut 5 bracket lower block [503] (View F, Figure 404).

h) Install the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] in a correct position on the strut 5 outer fitting [109].

i) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 outer fitting [109] (View C, Figure 404).

j) Install the five bolts [106] that attach the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109].

k) Tighten the bolts [106] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

l) Make sure that there is a minimum amount of the squeeze-out on the aft side around the edges of the block assembly and that the sealant is flush with the block assembly (View C, Figure 404).

m) Make sure that the squeeze-out on the forward facing side is finished as a fillet seal.

CAUTION MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

n) Clean the nut plate side of the bolts [106] and nut plates.

o) Apply Dapco 2100 sealant, A50483 [CP2927], to the nut plate side of the bolts [106] and nut plates.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES WITH PRE LEAP-1B SERVICE BULLETIN 72-0462

(e) Lubricate the bolt [102] with graphite mineral vaseline, D00601 [CP2101].

(f) Install the bolt [102] on the clamp [103] that attaches the EEC electrical harness (HJ5A) [67] to the harness strut (6:00 o'clock) [104].

(g) Tighten the bolt [102] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

PTF ALL; AIRPLANES WITH POST LEAP-1B SERVICE BULLETIN 72-0462

(h) Lubricate the nut [105] with graphite mineral vaseline, D00601 [CP2101].

(i) Install the bolt [102] with the nut [105] on the clamp [103] that attaches the EEC electrical harness (HJ5A) [67] to the central bracket assembly [116].

(j) Tighten the nut [105] to 42 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

PTF ALL

(k) Install the Fan Frame Strut 5 Panel (left) (TASK 72-23-06-400-801-G00).

(l) Attach blanket [190] to the strut 5 outer fitting [109].

1) Apply Dapco 2100 sealant, A50483 [CP2927], between the blanket [190] and strut 5 outer fitting [109] (View B, Figure 404).

NOTE: Dapco 2200 firewall sealant, A50396, may use as optional. If this consumable meets the local regulatory requirements of the operator, it is acceptable as a chemical sealant for further usage.

(m) Lubricate the threads of the bolt [100], bolt [94], bolt [91], bolt [90], and bolt [88] with oil, D00599 [CP2442].

(n) Install the bolt [100] and clamp [101] that attach the EEC electrical harness (HJ5A) [67] to the fan case bracket.

(o) Tighten the bolt [100] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

(p) Install the bolt [94] and clamp [95] that attach the EEC electrical harness (HJ5A) [67] to the fan frame module bracket.

(q) Tighten the bolt [94] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

(r) Install the bolt [91] and clamp [92] that attach the EEC electrical harness (HJ5A) [67] on the fan frame module bracket.

(s) Tighten the bolt [91] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

(t) Install the bolt [90] and clamp [89] that attach the EEC electrical harness (HJ5A) [67] to the fan hub bracket.

(u) Tighten the bolt [90] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

(v) Install the bolt [88] and clamp [87] that attach the EEC electrical harness (HJ5A) [67] to the fan hub bracket.

(w) Tighten the bolt [88] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(x) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (HJ5A) [67] as follows (TASK 70-00-01-910-803-G00):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

1) Connect the EEC electrical harness (HJ5A) [67] to the lower overheat sensor [114].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0178 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

2) Connect the EEC electrical harness (HJ5A) [67] to the bracket assembly [115].

PTF ALL

3) Connect the EEC electrical harness (HJ5A) [67] connector to the left VSV actuator receptacle [99].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

4) Connect the EEC electrical harness (HJ5A) [67] connector to the left MTC actuator receptacle [98].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

5) Connect the EEC electrical harness (HJ5A) [67] connector to the electrical stow bracket [117].

PTF ALL

6) Connect the EEC electrical harness (HJ5A) [67] connector to the left VBV actuator receptacle [97].

7) Connect the EEC electrical harness (HJ5A) [67] connector to the SCU/SVA channel A receptacle [96].

8) Connect the EEC electrical harness (HJ5A) [67] connector to the fuel manifold transducer receptacle [93].

9) Connect the EEC electrical harness (HJ5A) [67] connector to the T25 sensor channel A receptacle [86].

10) Connect the EEC electrical harness (HJ5A) [67] connector to the N1 speed sensor Channel A receptacle [147A].

(y) Connect the connector socket [82] of the EEC electrical harness (HJ5A) [67] as follows:

1) Make sure that the mating faces are clean and free of unwanted materials.

2) Install the four screws [81] that attach the connector socket [82] to the bracket [83].

3) Tighten the four screws [81] to 10.2 in-lb (1.15 N·m) - 11.1 in-lb (1.25 N·m).

4) Connect the HJ5B engine wiring harness electrical connector [84] (at 5:00 o'clock) to the connector socket [82].

NOTE: If the HJ5B engine wiring harness has been installed, do not do this step.

5) Connect the EEC electrical harness (HJ6B) [80] to the EEC HJ6A harness receptacle [79].

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(z) Install the bolt [78] and clamp [77] that attach the EEC electrical harness (HJ5A) [67] to the fan case bracket.

(aa) Tighten the bolt [78] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(ab) Install the bolt [74] and clamp [73] that attach the EEC electrical harness (HJ5A) [67] to the fan case bracket.

(ac) Tighten the bolt [74] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(ad) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (HJ5A) [67] to the EEC CH A as follows:

1) Connect the EEC electrical harness (HJ5A) [67] connector to the FMU channel A receptacle [76].

2) Connect the EEC electrical harness (HJ5A) [67] connector to the fuel flow transmitter meter receptacle [75].

3) Connect the EEC electrical harness (HJ5A) [67] connector to the EEC channel A J5 receptacle [72].

(ae) Make sure that the work area is clean of all tools and other items.

SUBTASK 73-22-01-420-003-G00

(4) Install the segment shroud 4 [71].

SUBTASK 73-22-01-420-004-G00

(5) Install the segment shroud 3 [69].

SUBTASK 73-22-01-420-005-G00

(6) Install the segment shroud 2 [68].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-024-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-021-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-022-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-023-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-860-024-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

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(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

H. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ5A) Installation Test

SUBTASK 73-22-01-730-003-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-22-01-000-804-G00

8. Electronic Engine Control (EEC) Electrical Harness (HJ5B) Removal

(Figure 406, Figure 407, and Figure 408)

A. General

(1) This task gives the instructions to remove the electronic engine control (EEC) electrical harness (HJ5B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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F. Prepare for the Removal

SUBTASK 73-22-01-860-025-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-026-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-027-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-028-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

SUBTASK 73-22-01-010-026-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

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WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ5B) Removal

SUBTASK 73-22-01-020-006-G00

(1) Remove the segment shroud 2 [68].

SUBTASK 73-22-01-020-007-G00

(2) Remove the segment shroud 3 [69].

SUBTASK 73-22-01-020-008-G00

(3) Remove the segment shroud 4 [71].

SUBTASK 73-22-01-020-009-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(4) Use the teflon-jawed pliers, STD-664, to disconnect the EEC electrical harness (HJ5B) [135] as follows (TASK 70-00-01-910-803-G00):

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(a) Disconnect the EEC electrical harness (HJ5B) [135] connector from the EEC channel B J5 receptacle [136].

(b) Disconnect the EEC electrical harness (HJ5B) [135] connector from the FMU channel B receptacle [139].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

(c) Disconnect the EEC electrical harness (HJ5B) [135] from the lower overheat sensor [114].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0178 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

(d) Disconnect the EEC electrical harness (HJ5B) [135] from the bracket assembly [115].

PTF ALL

(e) Disconnect the HJ6B engine wiring harness electrical connector [142] from the EEC HJ6A harness receptacle [143].

(f) Disconnect the HJ8B engine wiring harness electrical connector [146] from the EEC HJ8A harness receptacle [145].

(g) Disconnect the EEC electrical harness (HJ5B) [135] from the EEC HJ5A harness receptacle [144].

(h) Remove the EEC electrical harness (HJ5B) [135] as follows:

1) Remove the bolt [137] from the clamp [138] that attaches the EEC electrical harness (HJ5B) [135] to the fan case bracket.

2) Remove the bolt [140] from the clamp [141] that attaches the EEC electrical harness (HJ5B) [135] to the fan case bracket.

(i) Disconnect the EEC electrical harness (HJ5B) [135] connector from the T25 sensor channel B receptacle [155].

(j) Disconnect the EEC electrical harness (HJ5B) [135] connector from the N1 speed sensor channel B receptacle [147].

(k) Disconnect the EEC electrical harness (HJ5B) [135] connector from the SCU/SVA channel B receptacle [160].

(l) Disconnect the EEC electrical harness (HJ5B) [135] connector from the right VBV actuator receptacle [154].

(m) Disconnect the EEC electrical harness (HJ5B) [135] connector from the right VSV actuator (top) receptacle [158].

(n) Disconnect the EEC electrical harness (HJ5B) [135] connector from the right VSV actuator (bottom) receptacle [159].

(o) Remove the bolt [148] from the clamp [149] that attaches the EEC electrical harness (HJ5B) [135] to the forward stator case bracket.

(p) Remove the bolt [150] from the clamp [151] that attaches the EEC electrical harness (HJ5B) [135] to the forward stator case bracket.

(q) Remove the bolt [153] from the clamp [152] that attaches the EEC electrical harness (HJ5B) [135] to the forward stator case bracket.

(r) Remove the bolt [157] from the clamp [156] that attaches the EEC electrical harness (HJ5B) [135] to the forward stator case bracket.

(s) Remove the EEC electrical harness (HJ5B) [135] from the 24 clips [136A] along its routing.

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(t) Remove blanket [190] from the strut 5 outer fitting [109] (View B, Figure 407).

(u) Remove the EEC electrical harness (HJ5B) [135] from the harness strut (6:00 o'clock) [104] as follows:

1) Remove the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] from the strut 5 outer fitting [109] as follows:

a) Remove the five bolts [106] that attach the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] to the strut 5 outer fitting [109].

b) Cut and remove the sealant between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502],strut 5 bracket lower block [503], and strut 5 outer fitting [109].

c) Move the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] rearward from the electrical harnesses.

d) Remove the two nuts [107] and two bolts [108] to disassemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

e) Cut and remove the sealant on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], strut 5 bracket lower block [503], and around the harnesses.

f) Remove harnesses from the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

g) Remove any remaining sealant residue.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

2) Remove the harness support block [504], harness support block [505], and harness support block [506] from the harness strut (6:00 o'clock) [104] as follows:

a) Remove the four bolts [111] and four nuts [110] that attach the harness support block [504], harness support block [505], and harness support block [506] to the harness strut (6:00 o'clock) [104].

b) Cut and remove the sealant between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

c) Move the assembly of the harness support block [504], harness support block [505], and harness support block [506] upward and away from the harness strut (6:00 o'clock) [104].

d) Remove the two nuts [112] and two bolts [113] to disassemble the harness support block [504], harness support block [505], and harness support block [506].

e) Cut and remove the sealant on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses.

f) Remove the harnesses from the harness support block [504], harness support block [505], and harness support block [506].

g) Remove any remaining sealant residue.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

3) Remove the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] from the harness strut (6:00 o'clock) [104] as follows:

a) Remove the four bolts [124] that attach the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

b) Cut and remove the sealant between the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

c) Move the assembly of the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510] upward and away from the harness strut (6:00 o'clock) [104].

d) Remove the four nuts [126] and four bolts [125] to disassemble the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

e) Cut and remove the sealant on the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], right outer metallic block [510], and around the harnesses.

f) Remove the harnesses from the left outer metallic block [507], left inner metallic block [508], right inner metallic block [509], and right outer metallic block [510].

g) Remove any remaining sealant residue.

PTF ALL

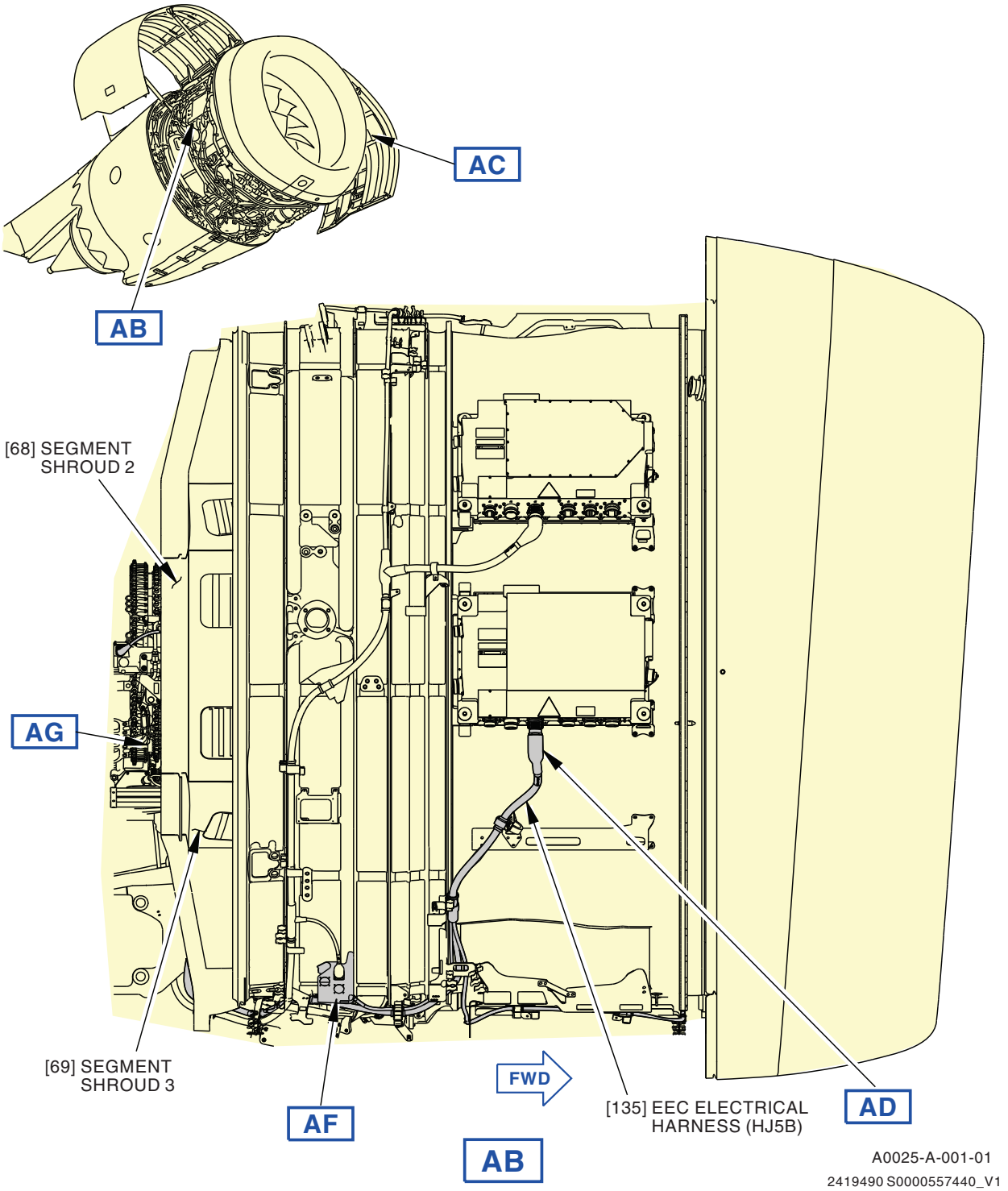
4) Remove the EEC electrical harness (HJ5B) [135] from the harness strut (6:00 o'clock) [104].

(v) Carefully remove the EEC electrical harness (HJ5B) [135] from the engine.

(w) Install the protective covers on all of the harness connectors and the component receptacles.

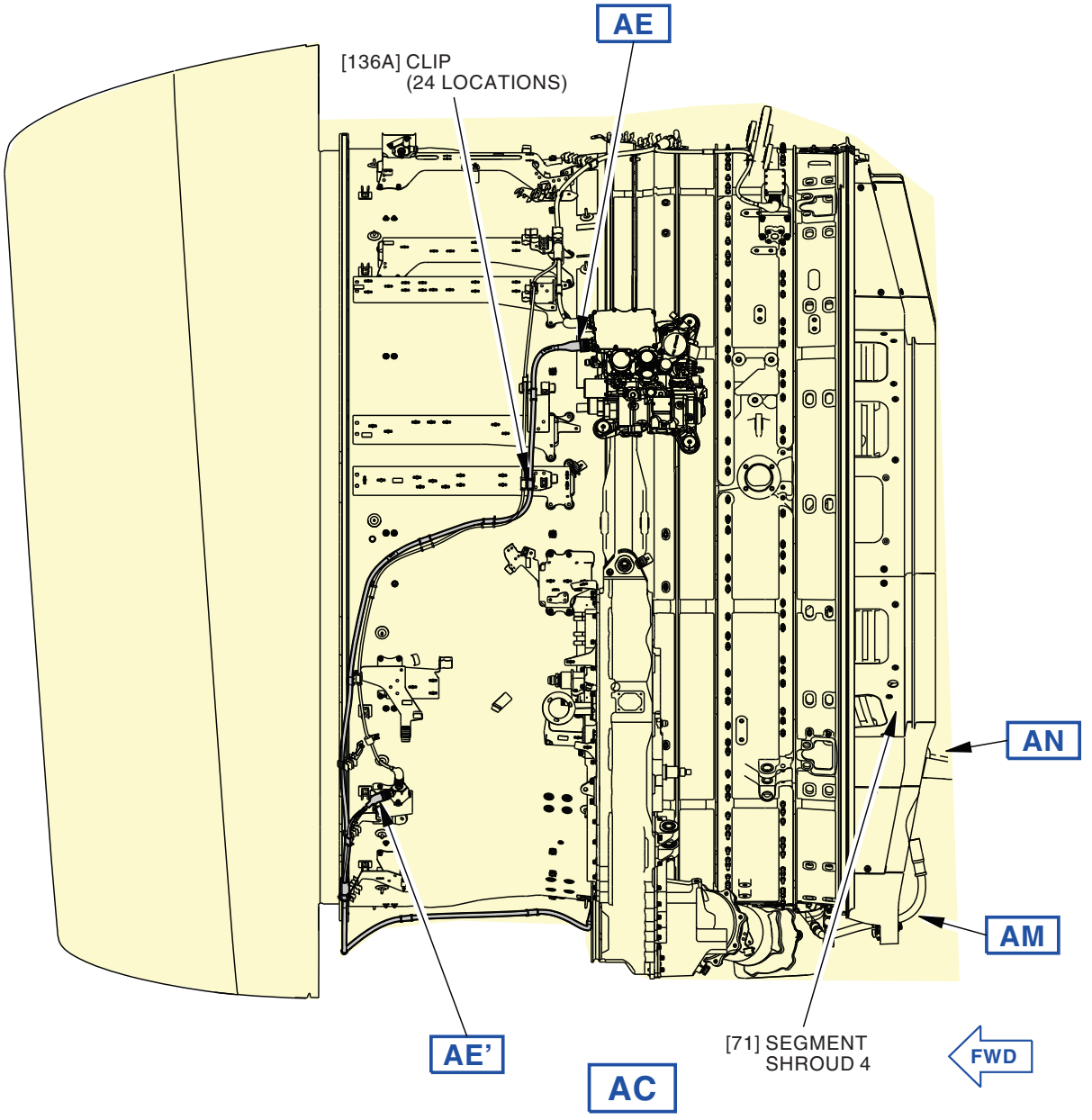
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 1 of 9)

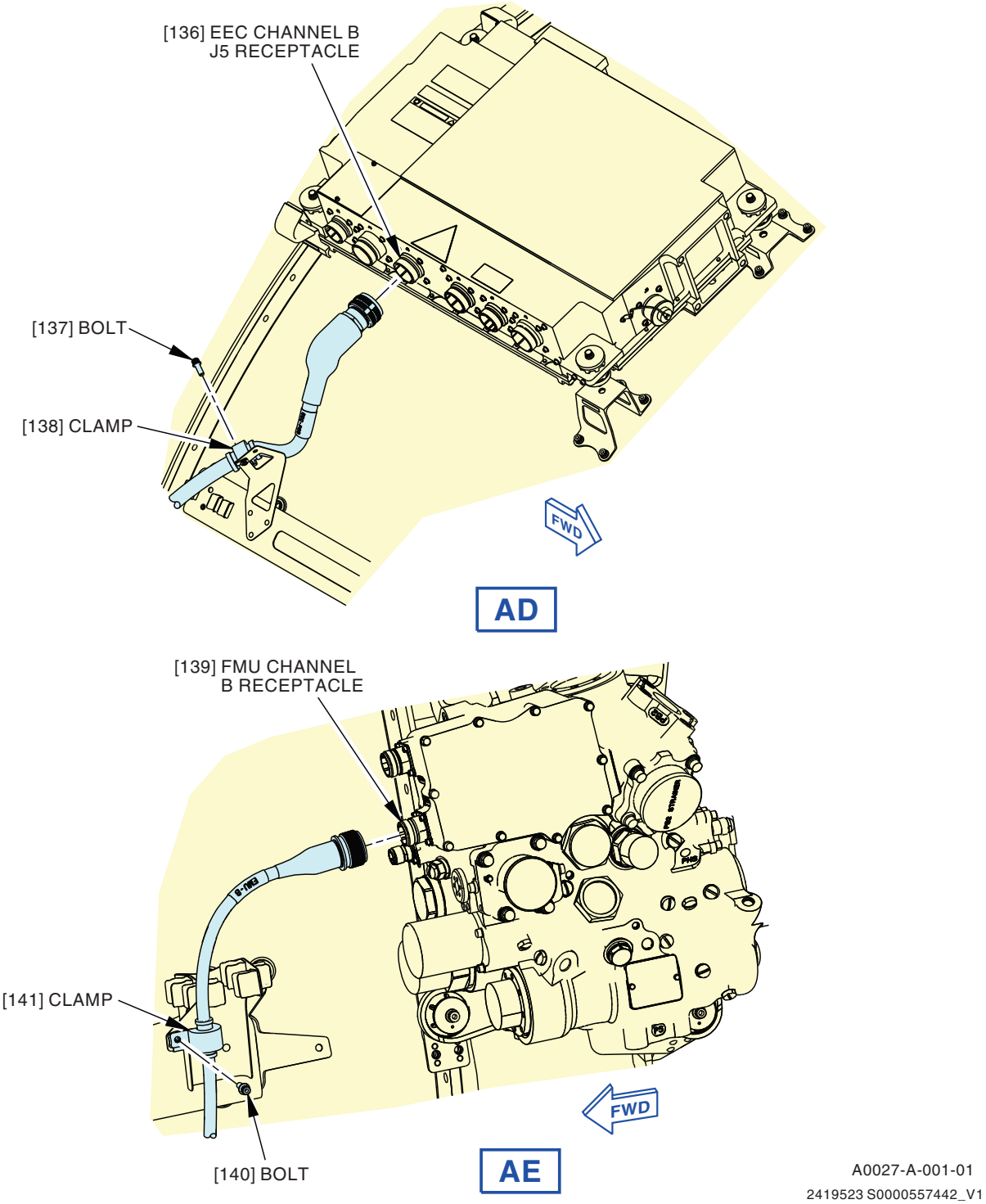
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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2419504 S0000557441_V2

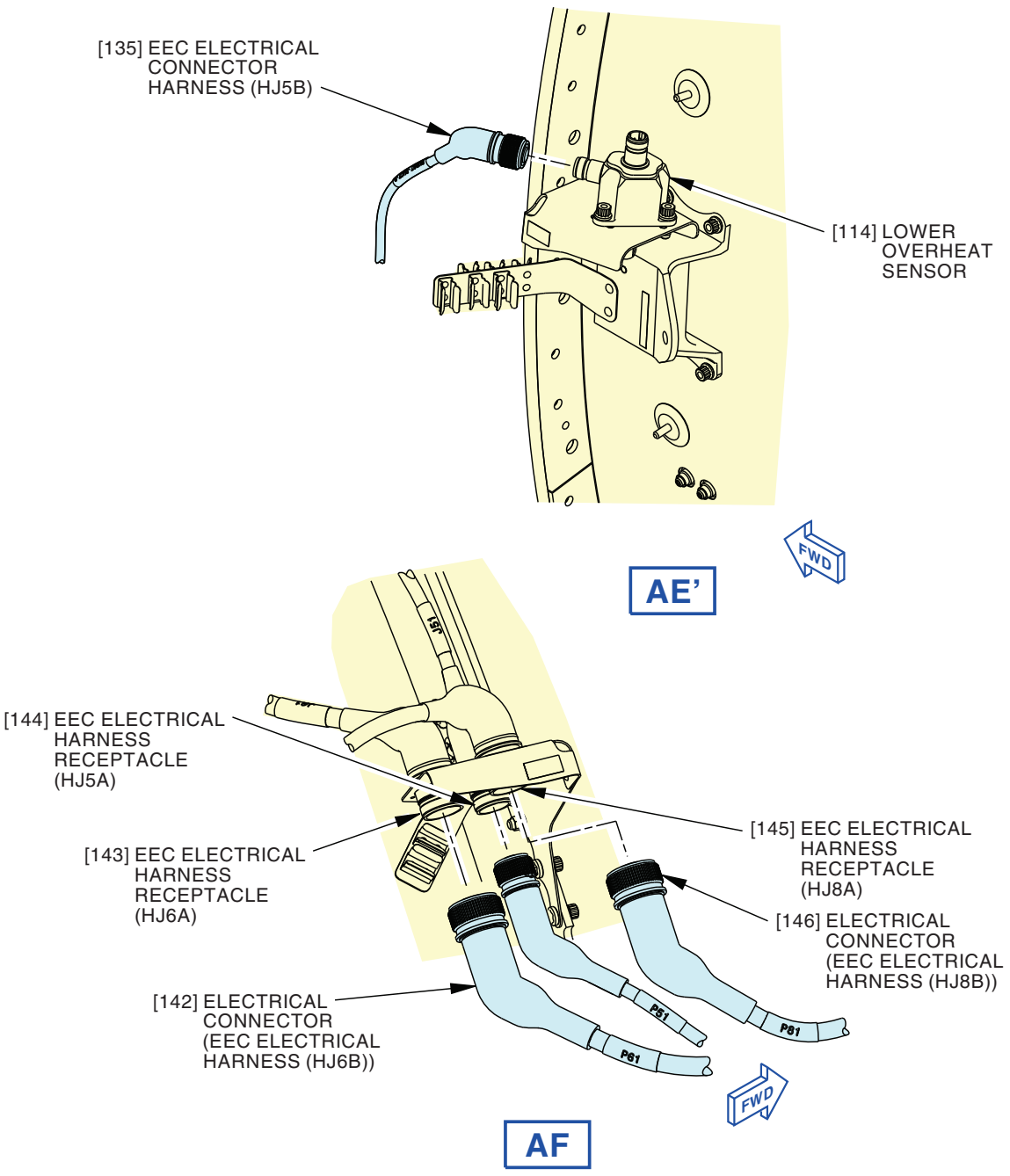
EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



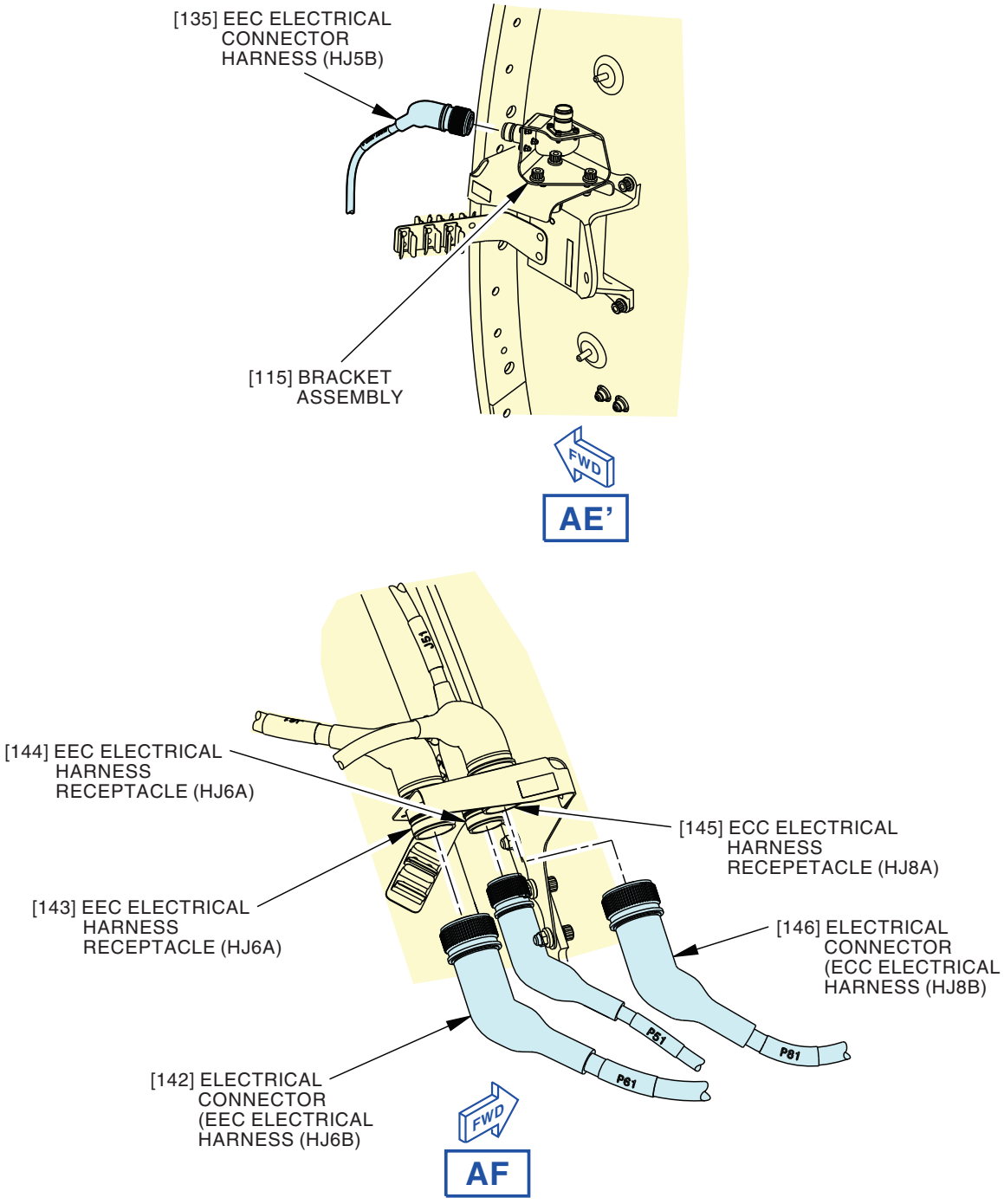
EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 4 of 9)

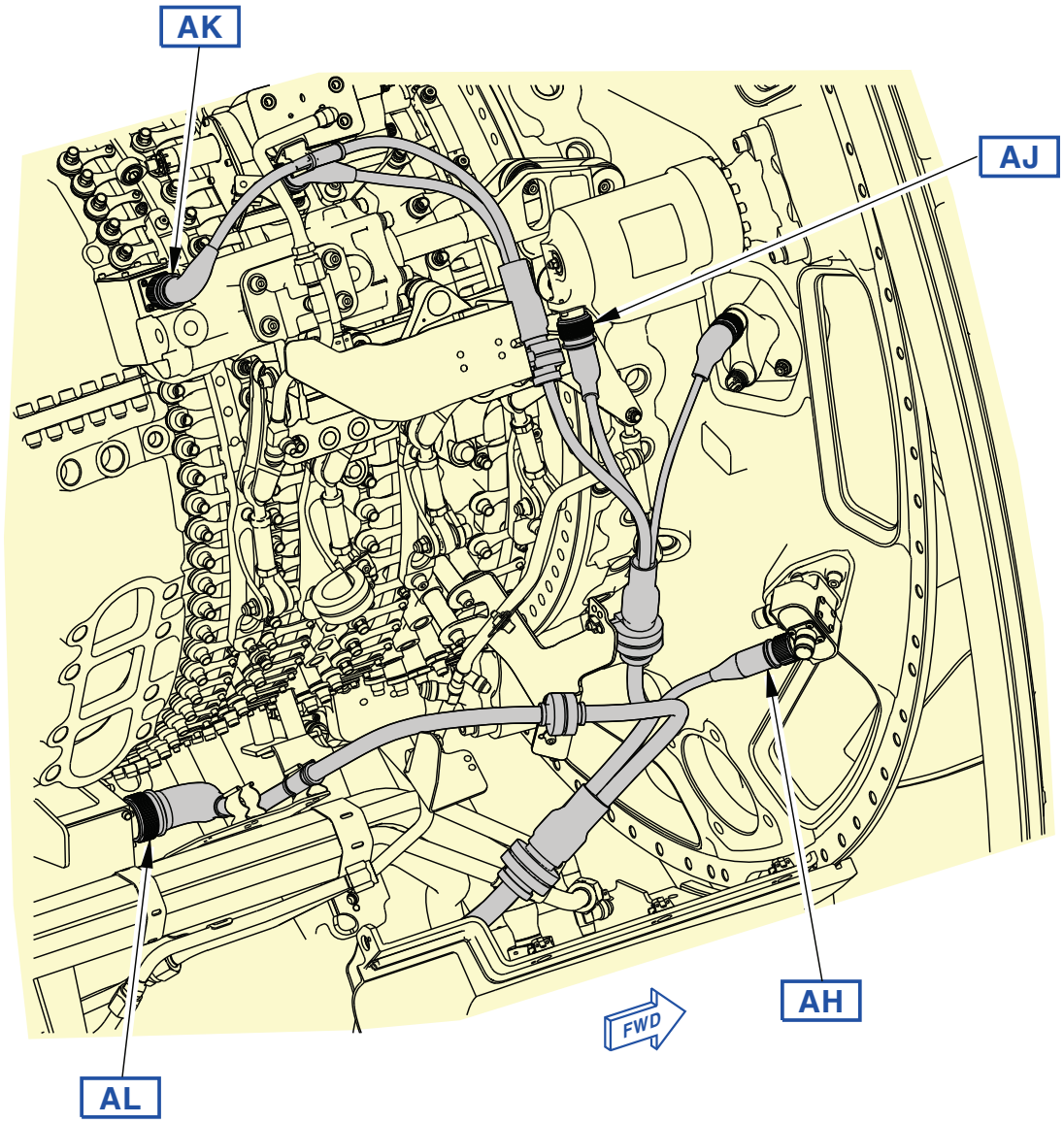
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EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 5 of 9)

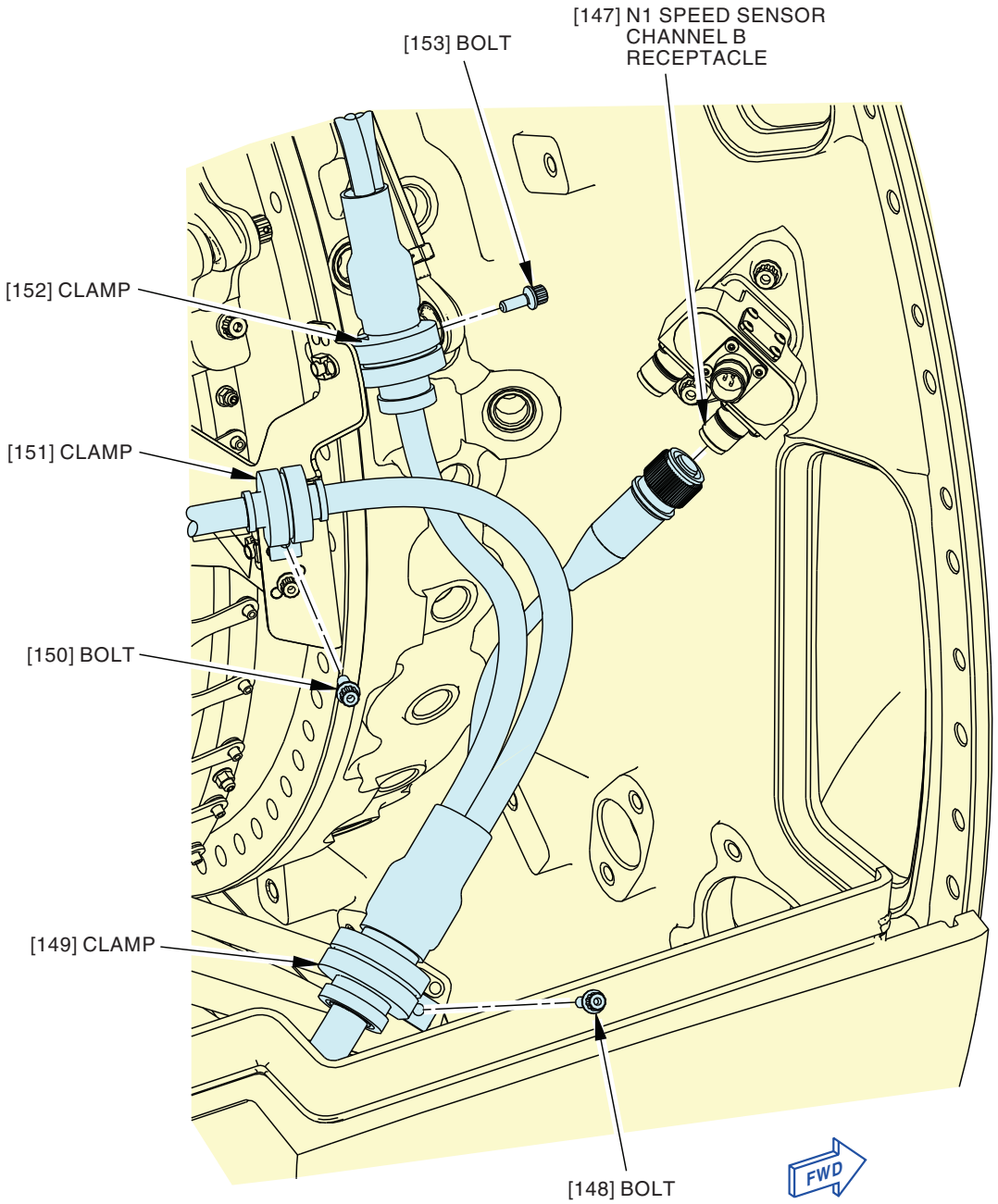
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A0029-A-001-01
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EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

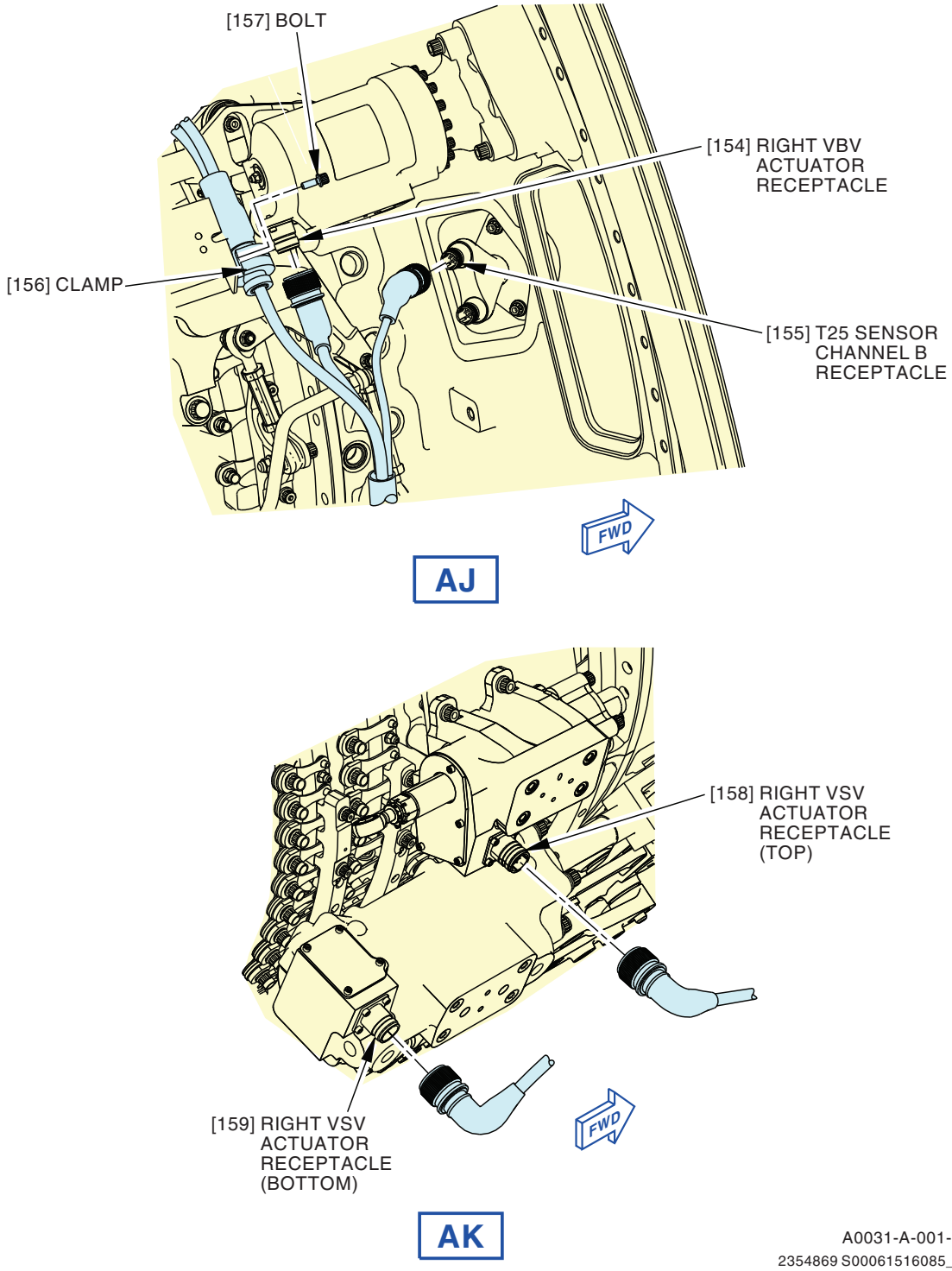


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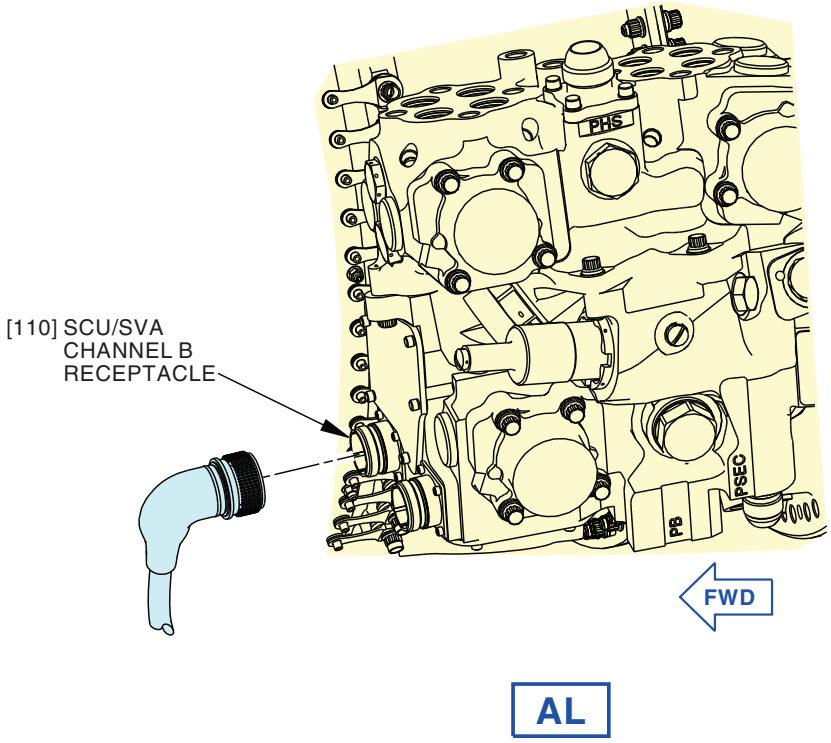
EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 8 of 9)

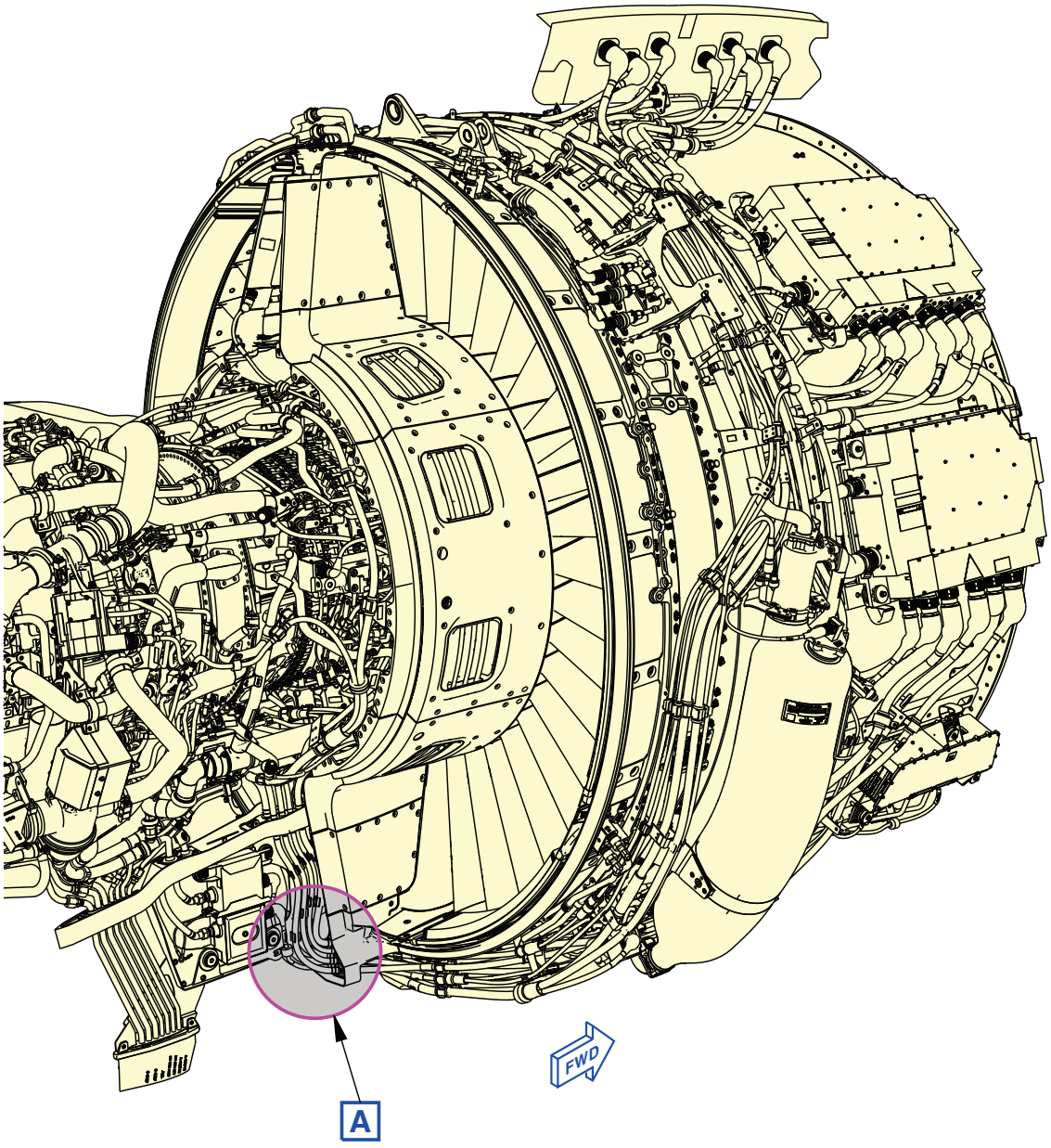
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0032-A-001-01
2354870 S00061516086_V2

EEC Electrical Harness (HJ5B) Installation
Figure 406/73-22-01-990-854-G00 (Sheet 9 of 9)

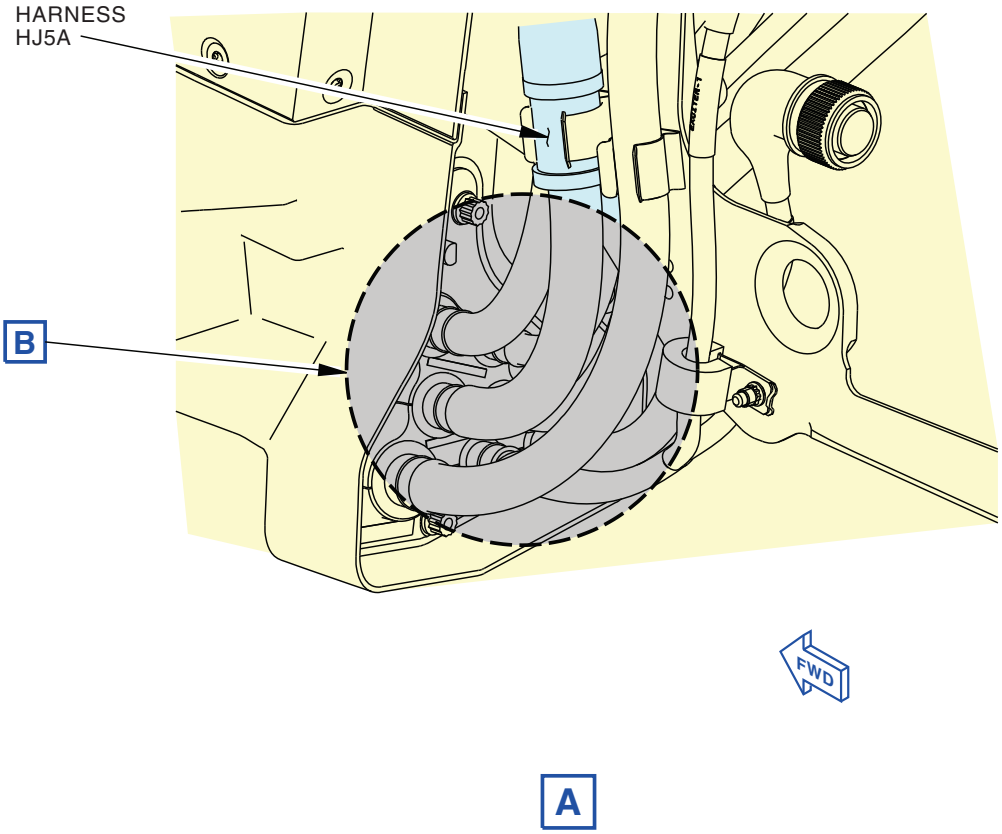
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937794 S0000713383_V3

Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 1 of 9)

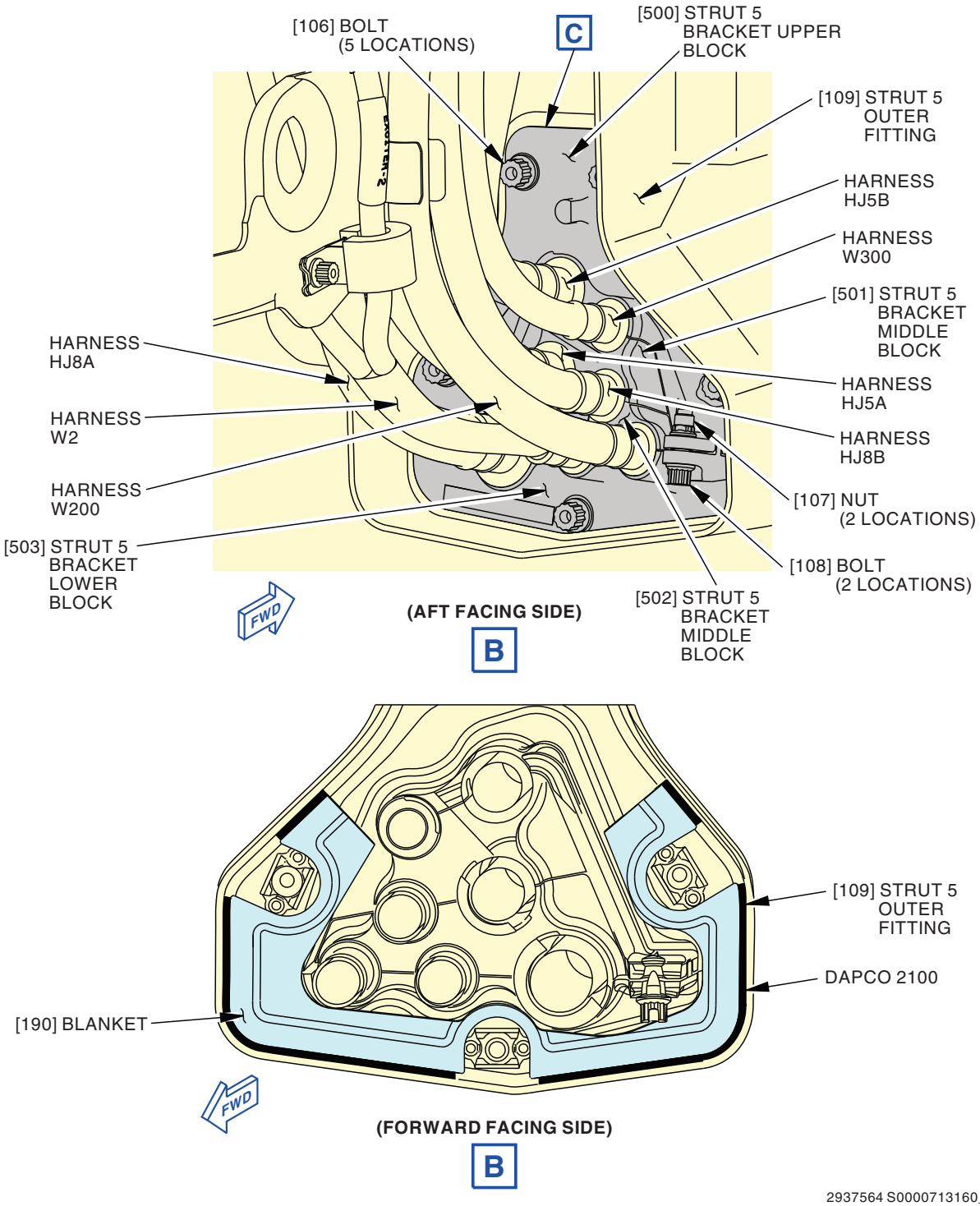
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2937795 S0000713384_V2

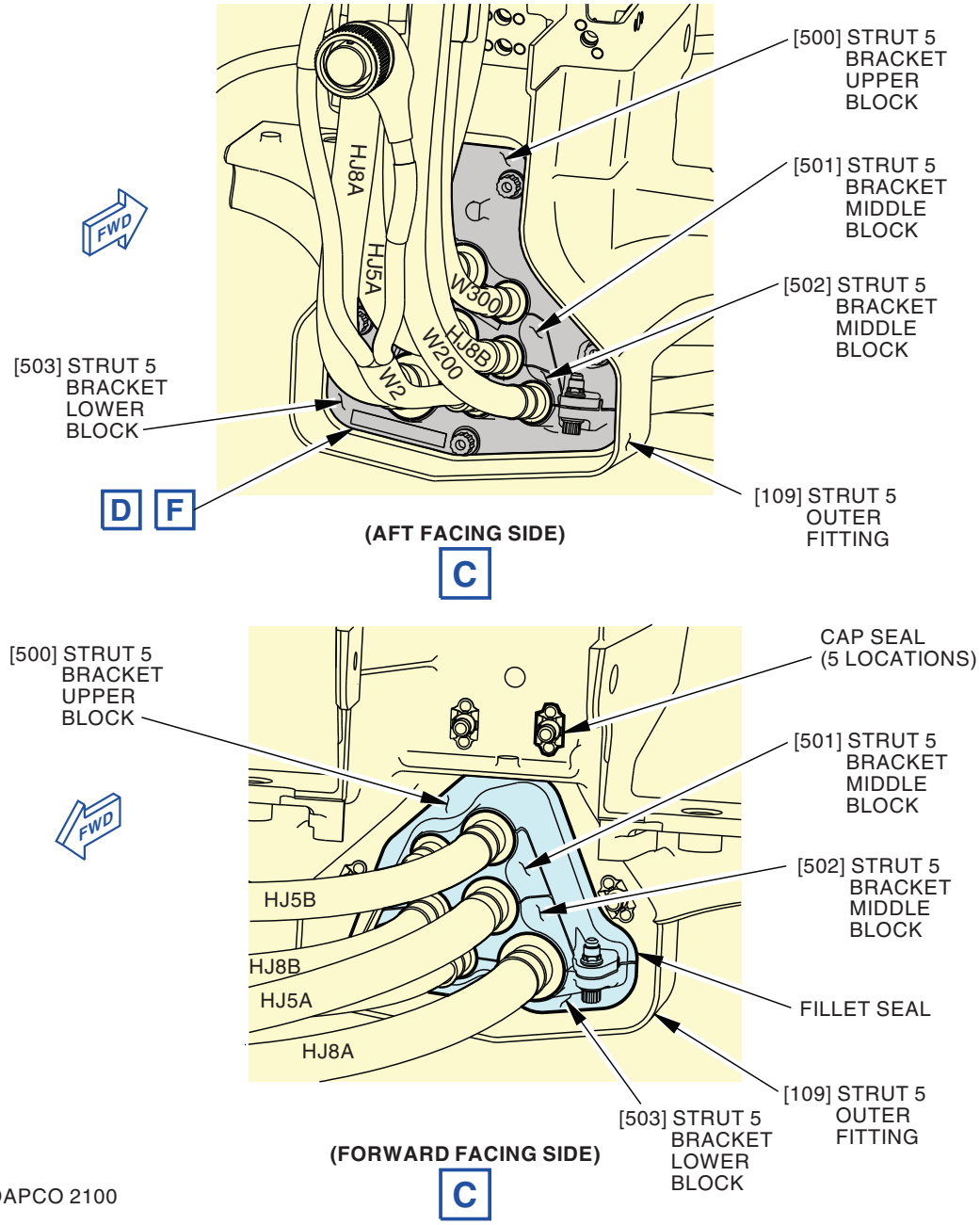
Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 2 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 3 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

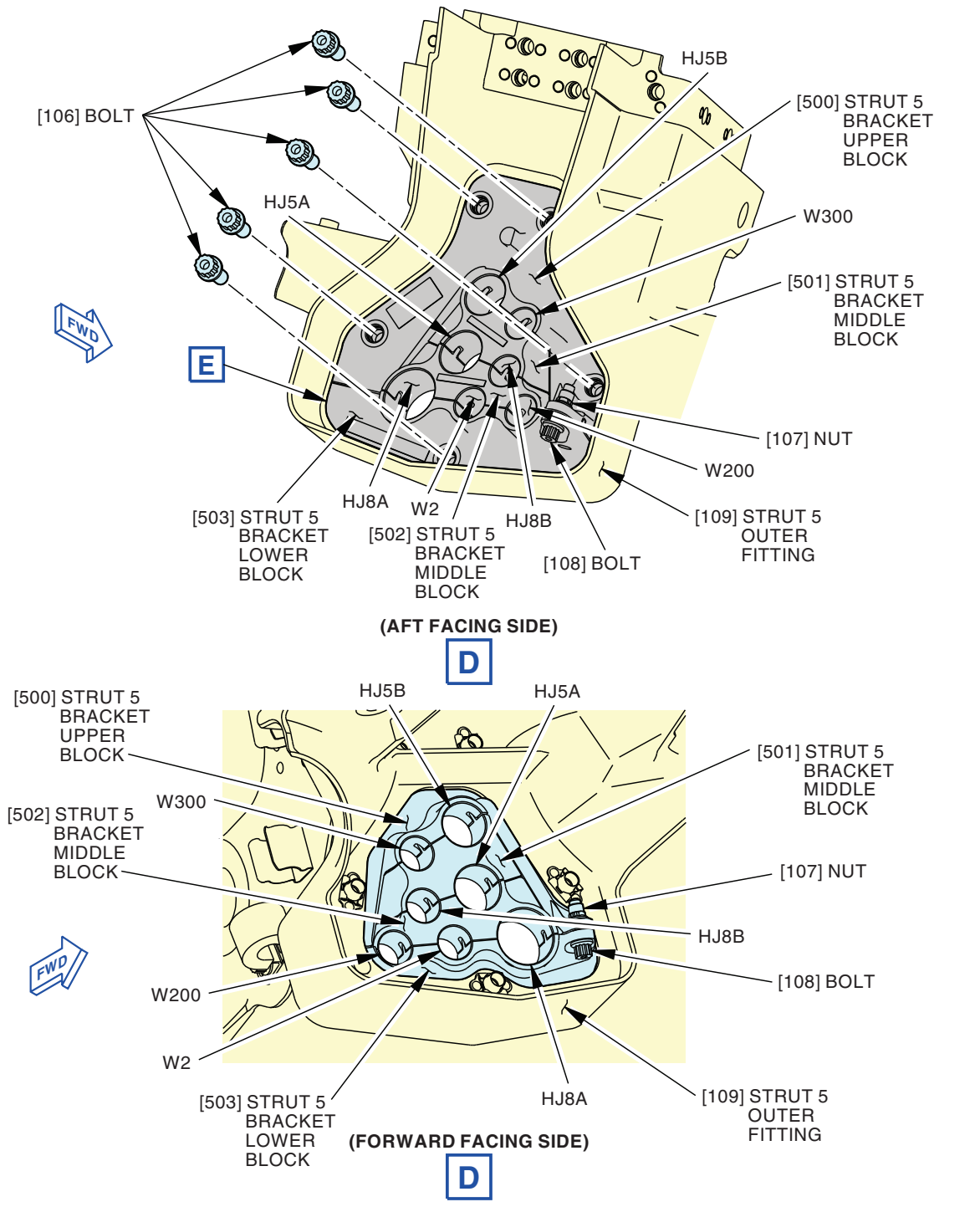
FAY SEAL PER AMM 51-22-00 ON INNER CONTOUR OF THE FITTING (AFT SIDE). SEALANT SQUEEZE-OUT SHOULD BE CONTINUOUS AND EVEN.

IT IS ACCEPTABLE TO APPLY THE FILLET SEAL ALONG INNER CONTOUR OF THE FITTING ON THE FORWARD SIDE.

2937801 S0000713390_V3

Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 4 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



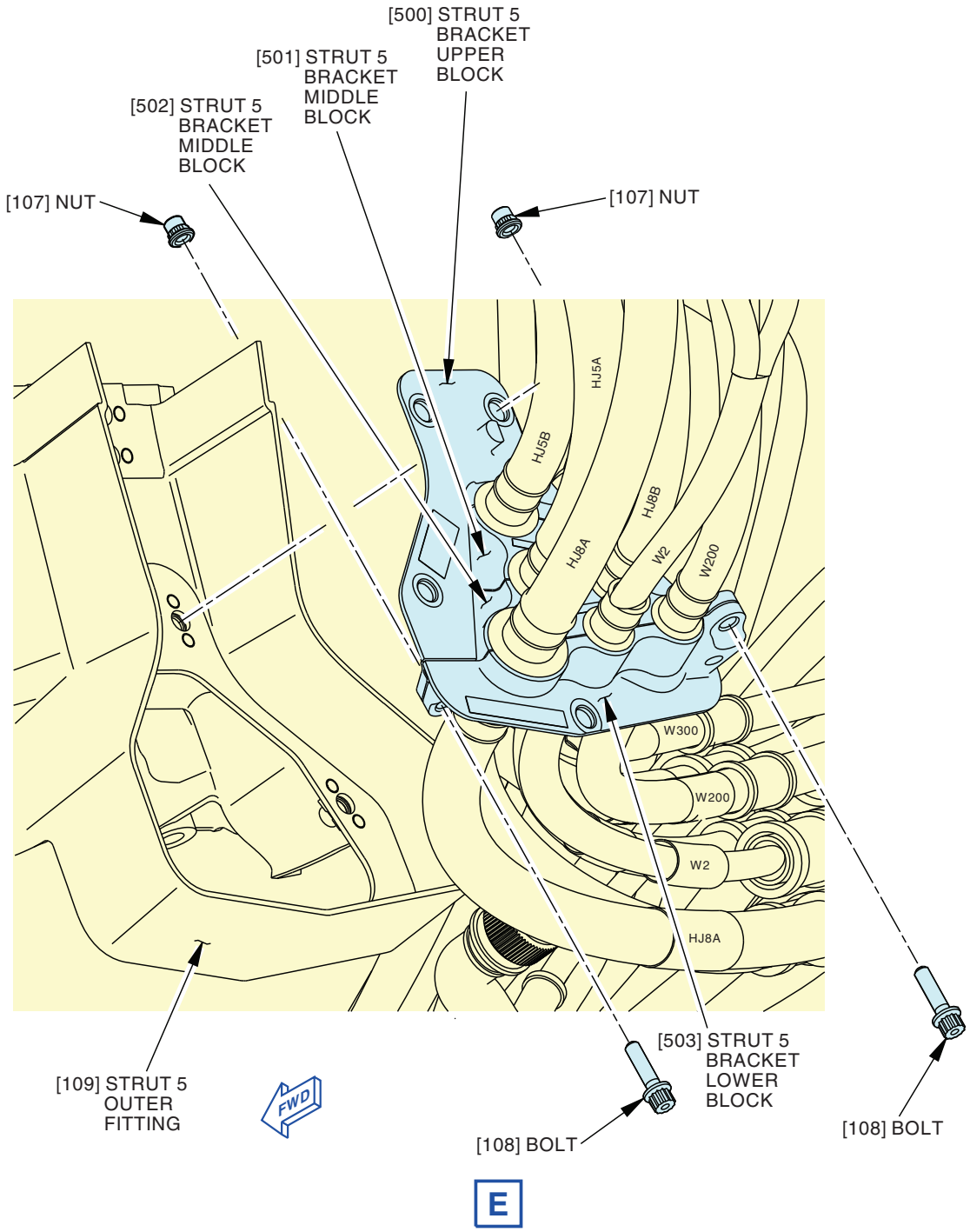
NOTE:

HARNESSES NOT SHOWN

2943980 S0000720369_V2

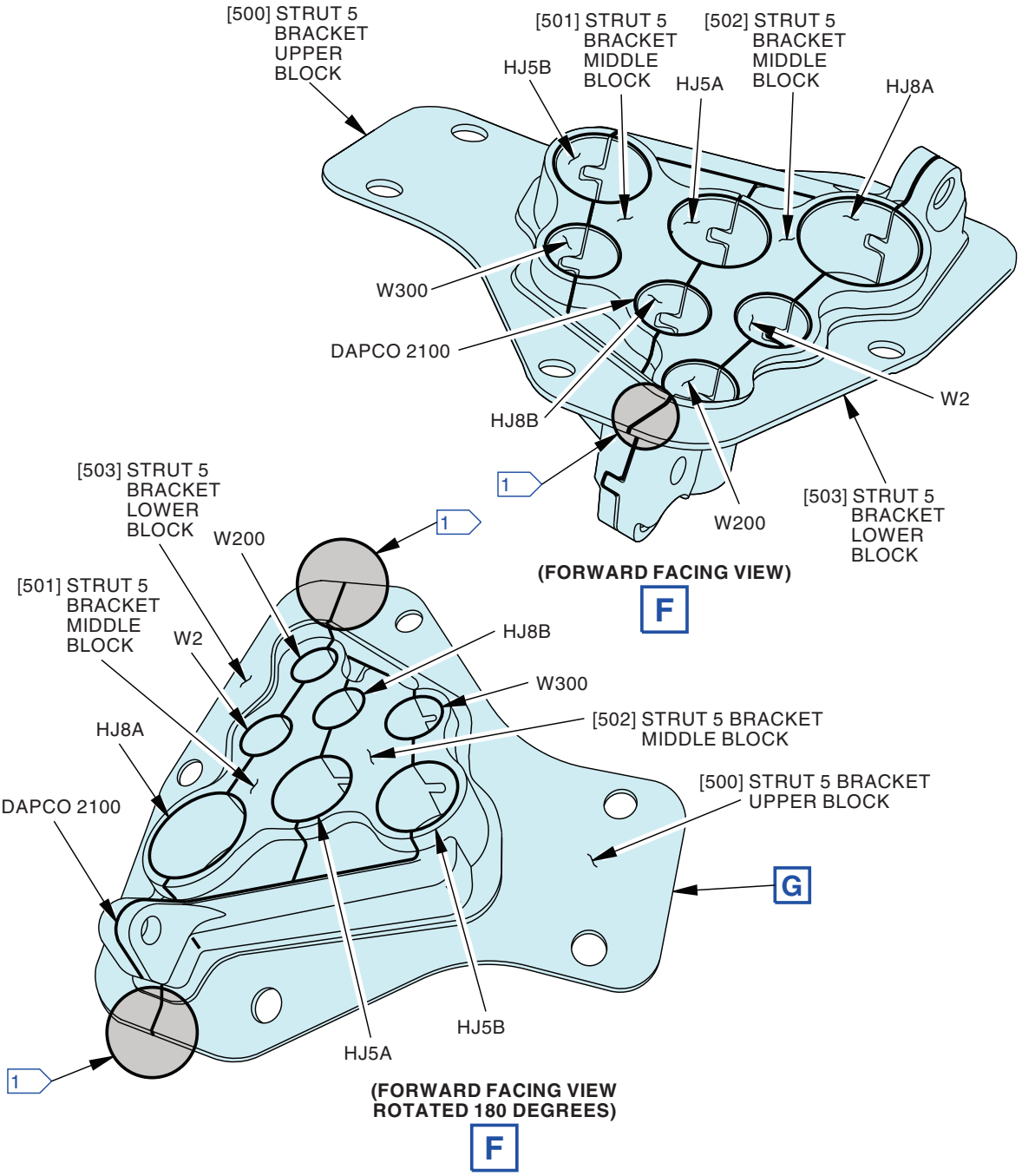
Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 5 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 6 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

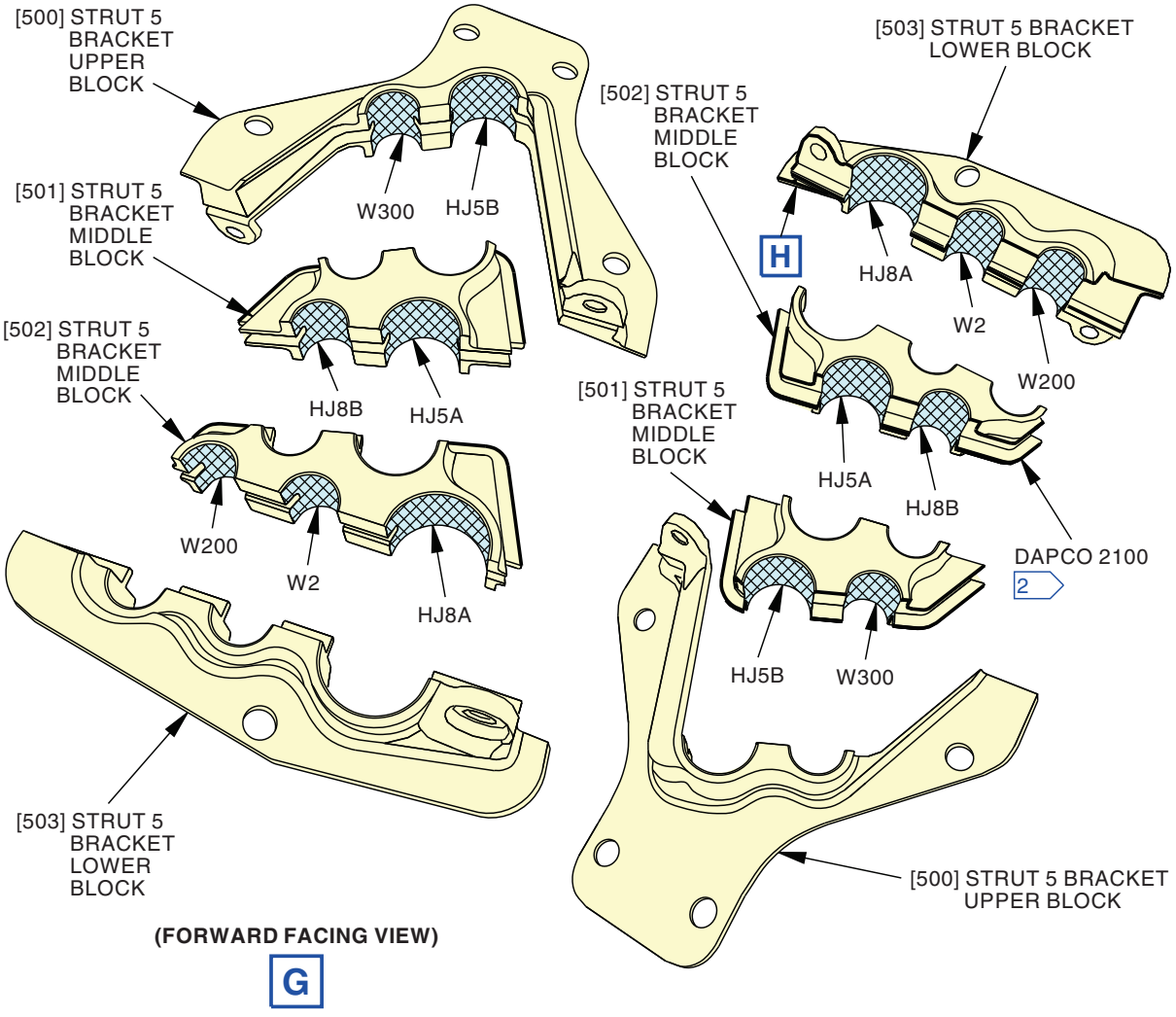
HARNESSES NOT SHOWN.

1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

2937800 S0000713389_V3

Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 7 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

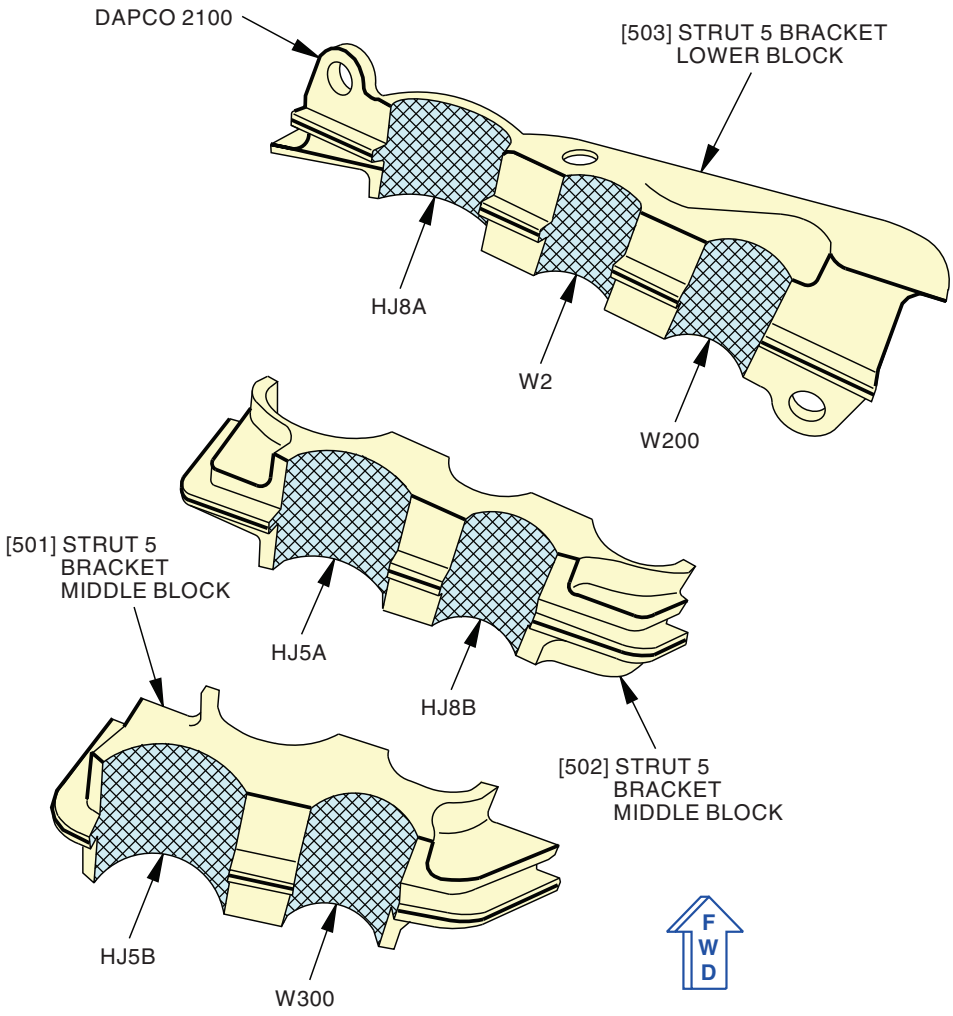
SEALANT PROVIDES FAY SEAL BETWEEN WIRE HARNESS AND BLOCKS AND SHOULD BE APPLIED TO THE BOTH SIDES OF BLOCKS.

2 SEALANT APPLIED TO ONE SIDE (TOP OR BOTTOM) OF THE BLOCKS.

2937799 S0000713388_V3

Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 8 of 9)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

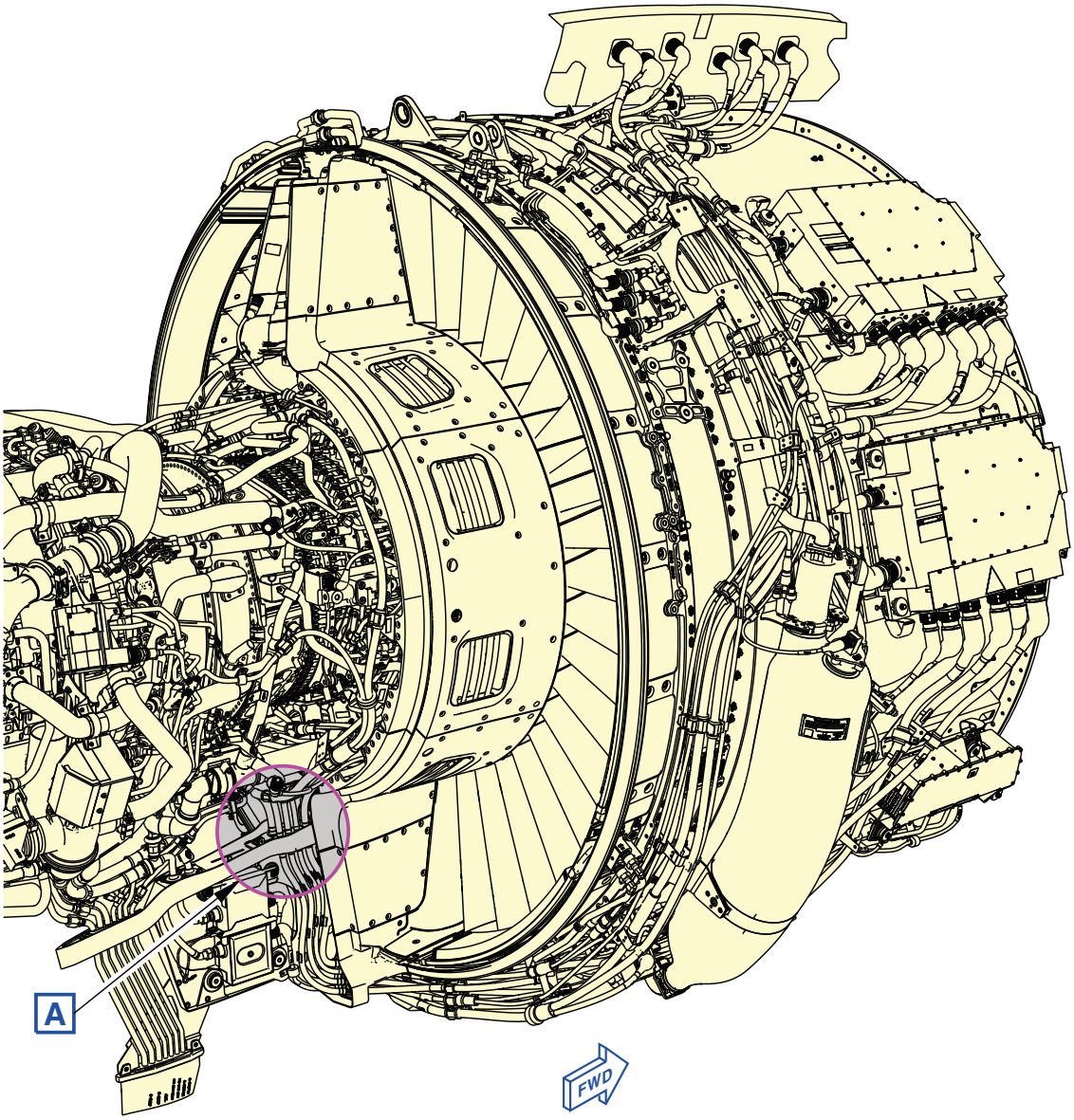


H

2949941 S0000726813_V1

Strut Bracket Block Installation
Figure 407/73-22-01-990-872-G00 (Sheet 9 of 9)

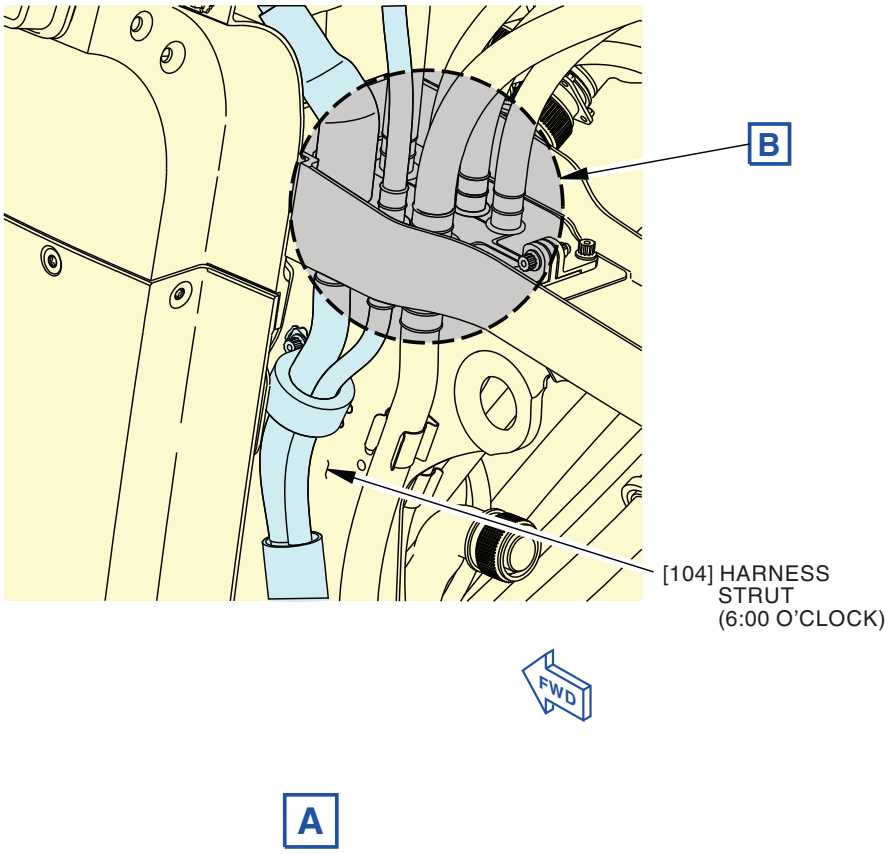
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-732201-D-58828-00045-A-001-01
2938818 S0000714836_V3

Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 1 of 11)

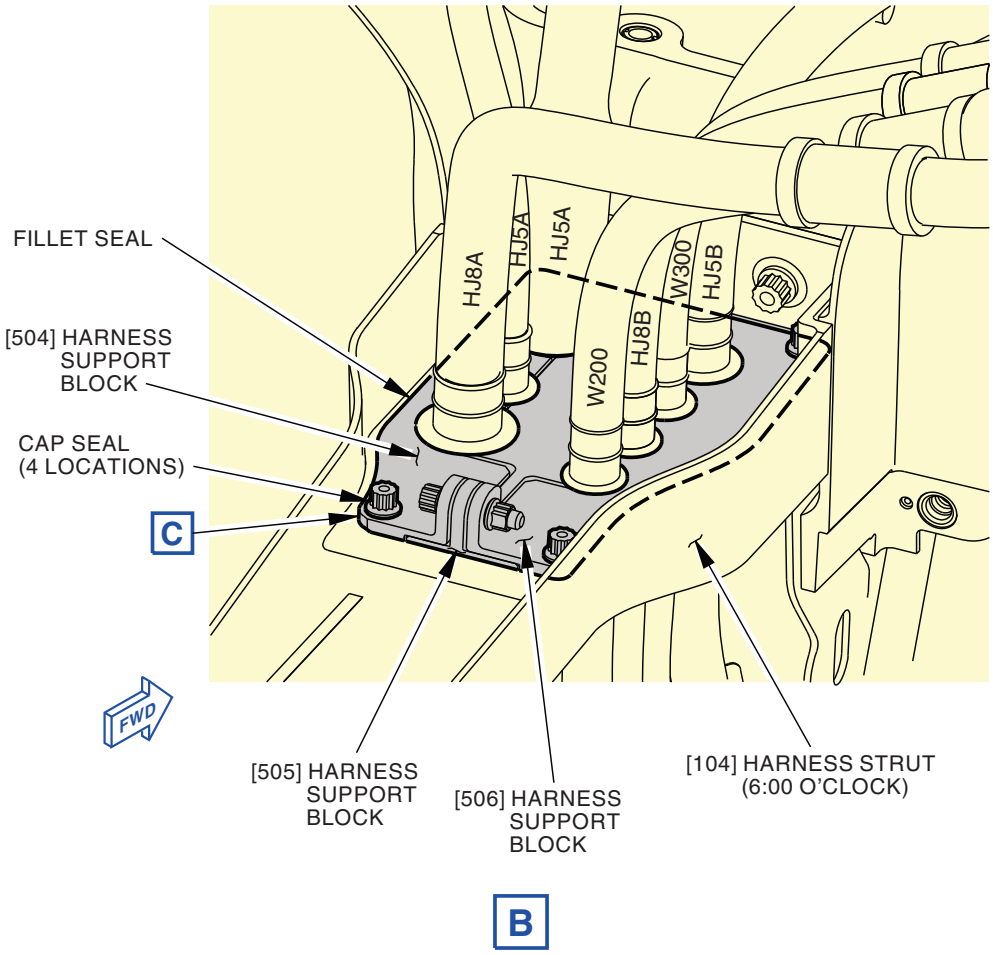
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2952925 S0000730238_V1

Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 2 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

——— DAPCO 2100

NOTE:

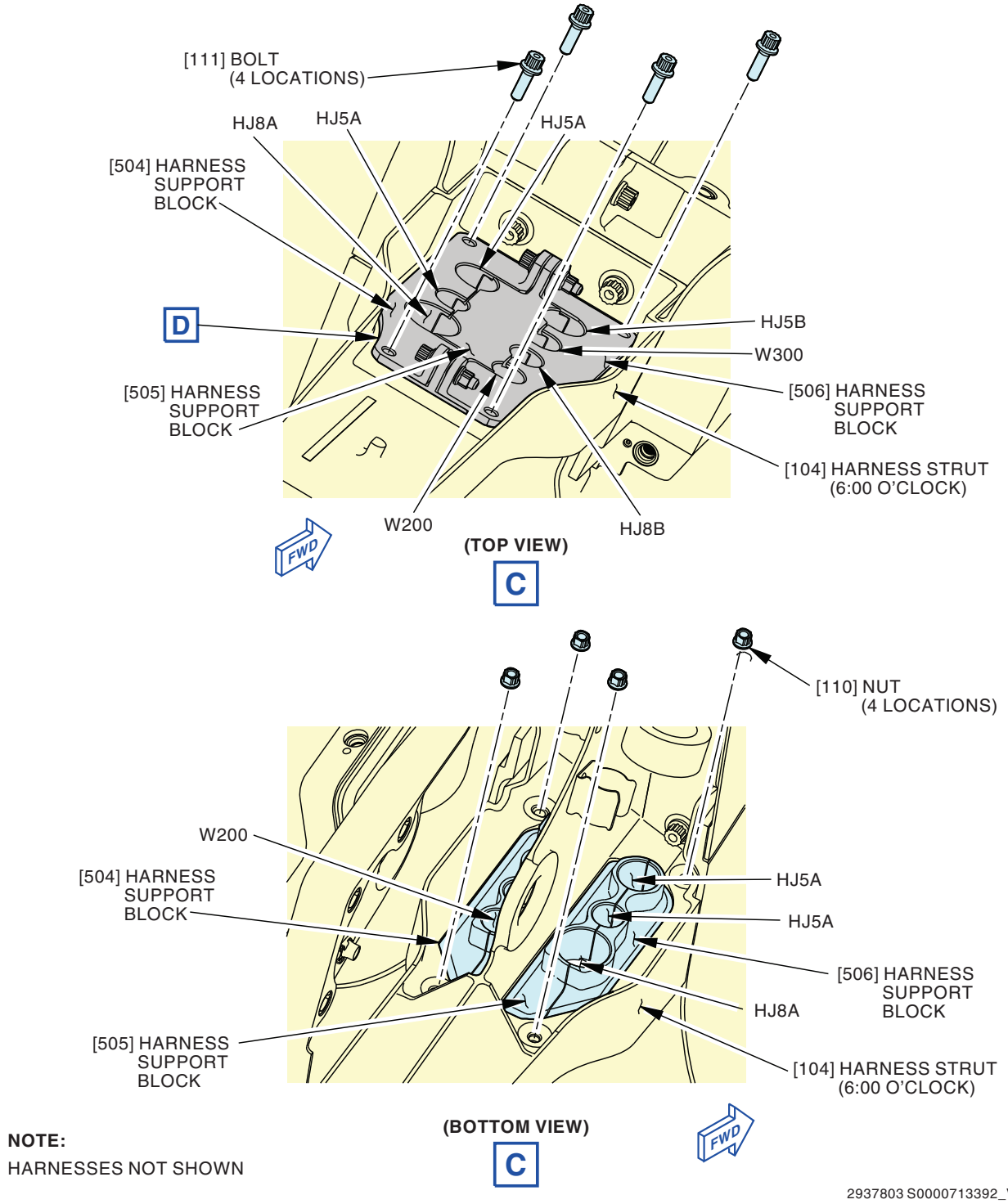
THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE HARNESS SUPPORT BLOCK. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937807 S0000713396_V4

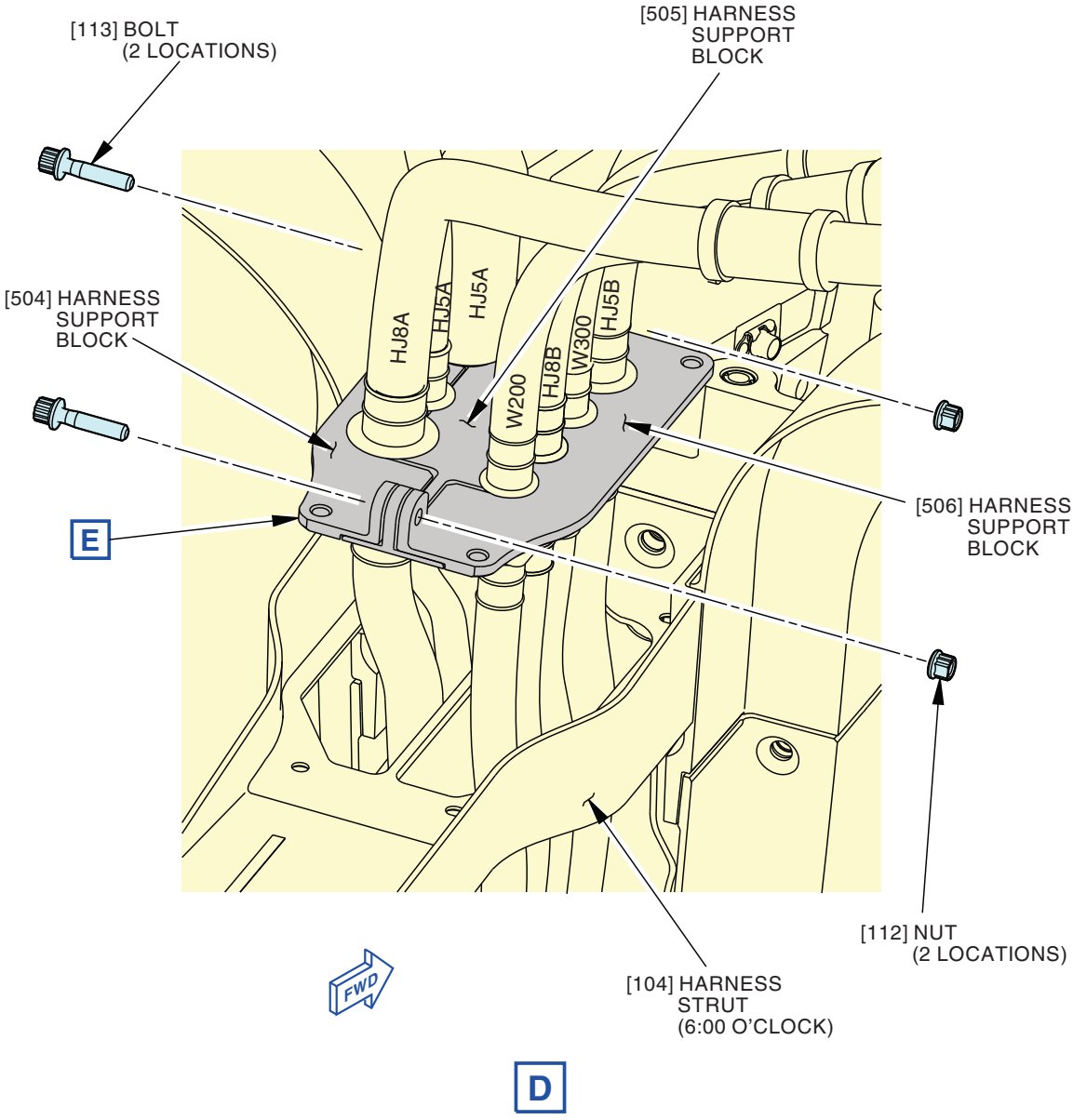
Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 3 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



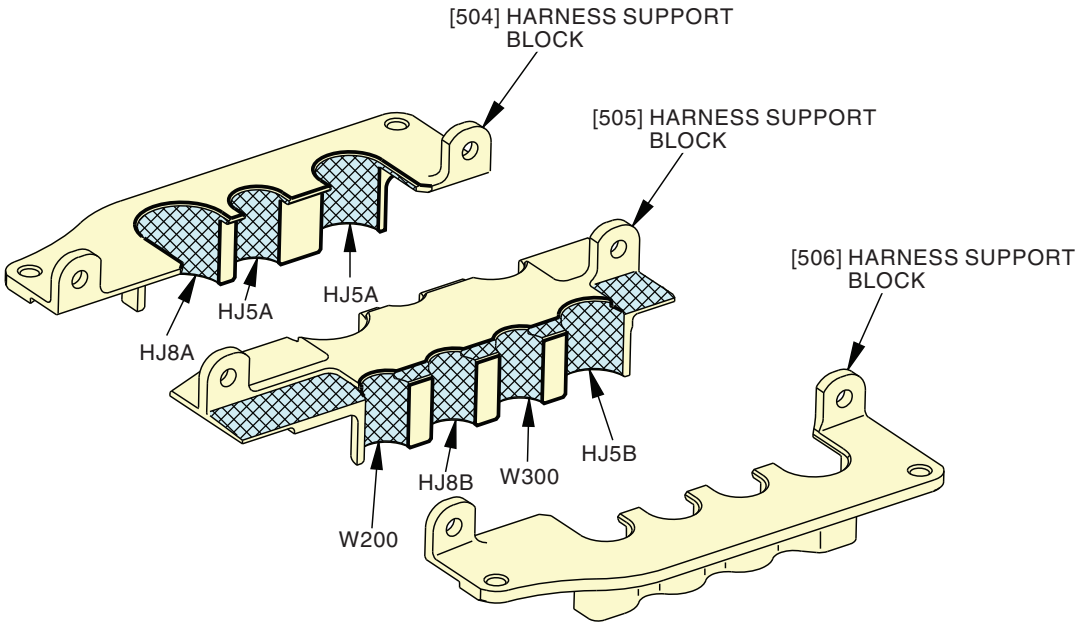
Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 4 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



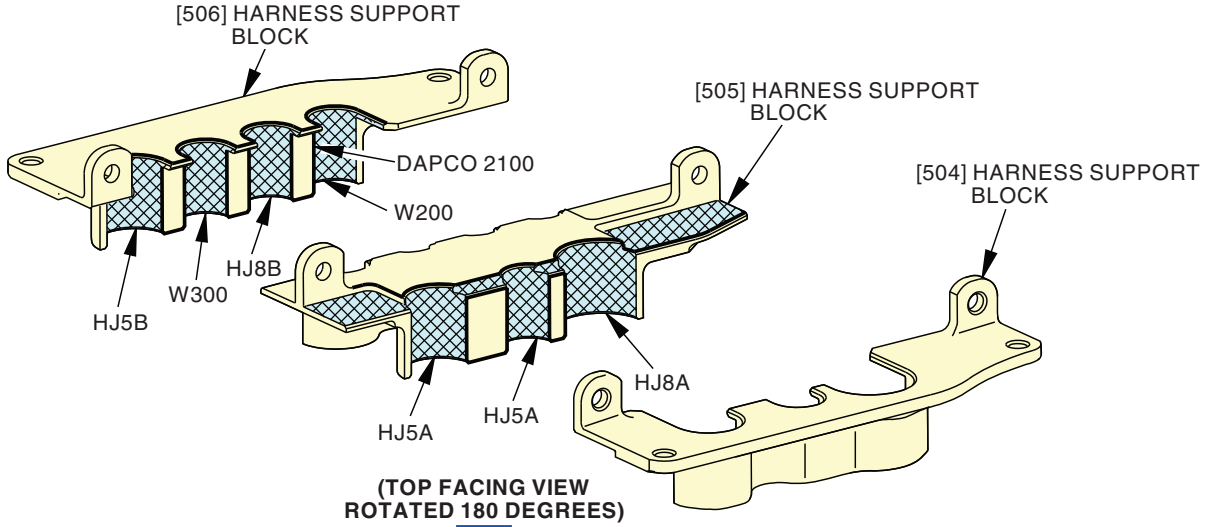
Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 5 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TOP FACING VIEW)

E



(TOP FACING VIEW ROTATED 180 DEGREES)

E

LEGEND:

DAPCO 2100

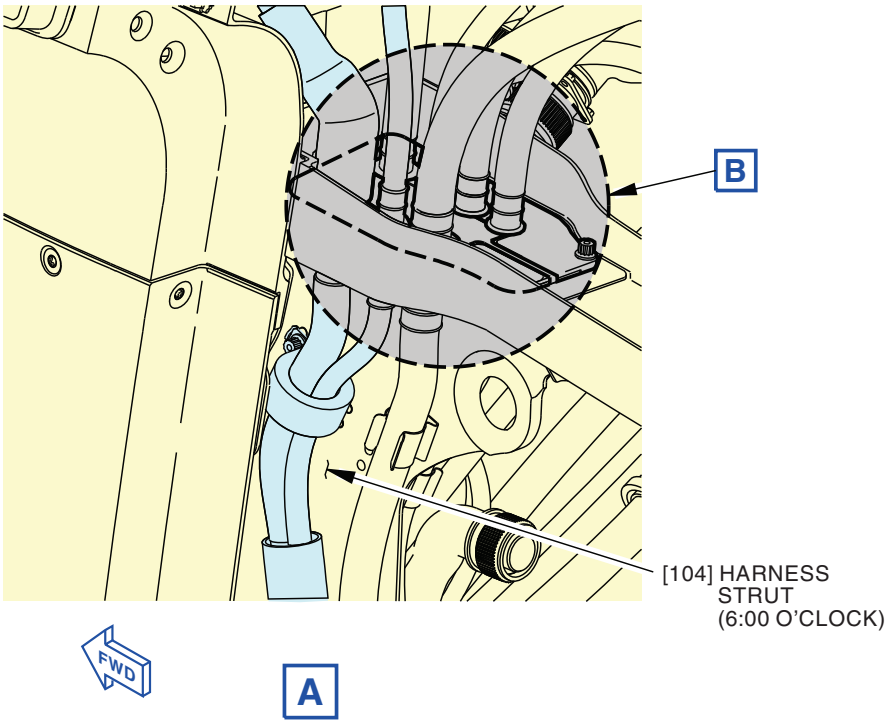
NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2937805 S0000713394_V4

Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 6 of 11)

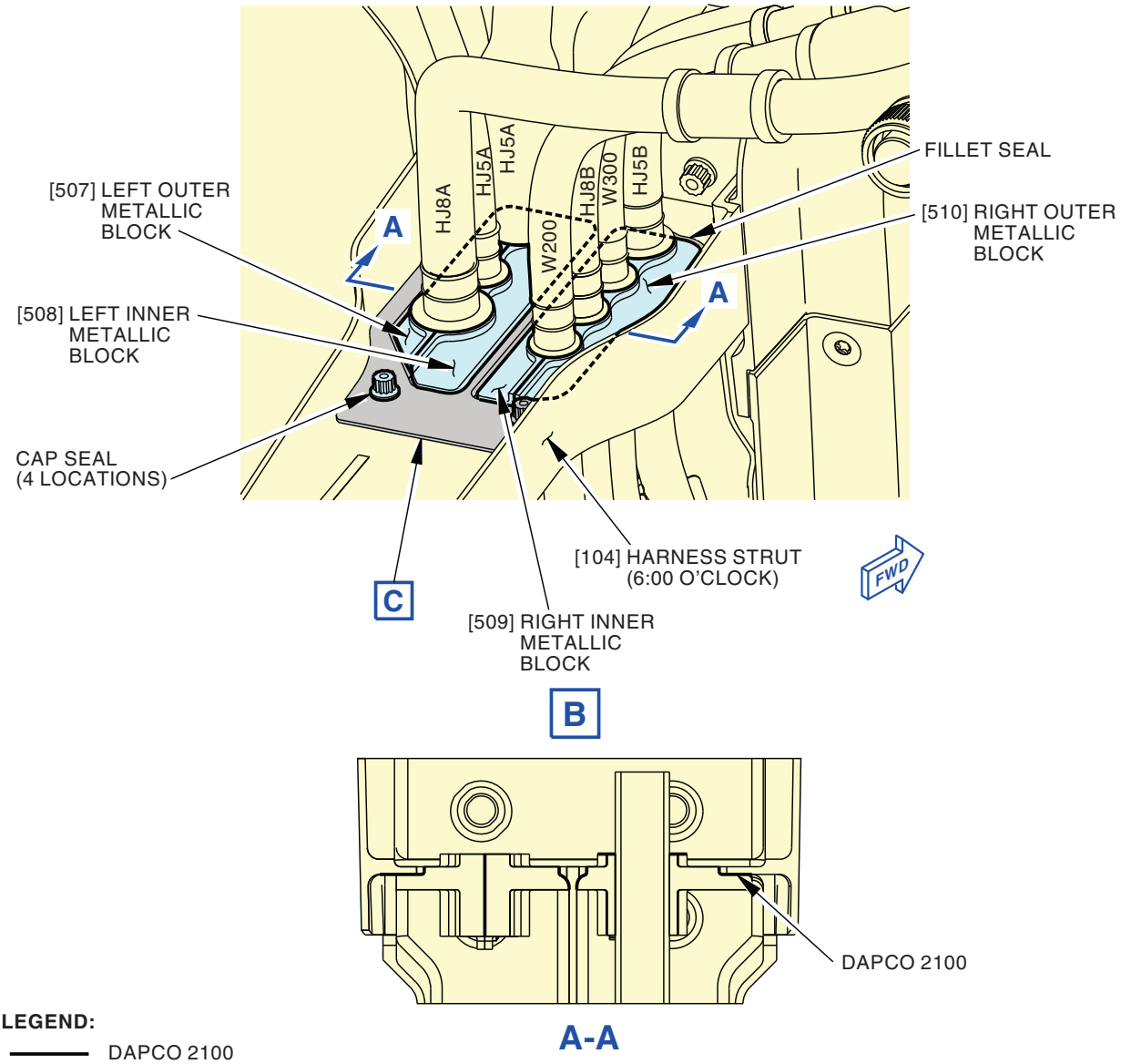
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2962032 S0000739201_V1

Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 7 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

DAPCO 2100

NOTE:

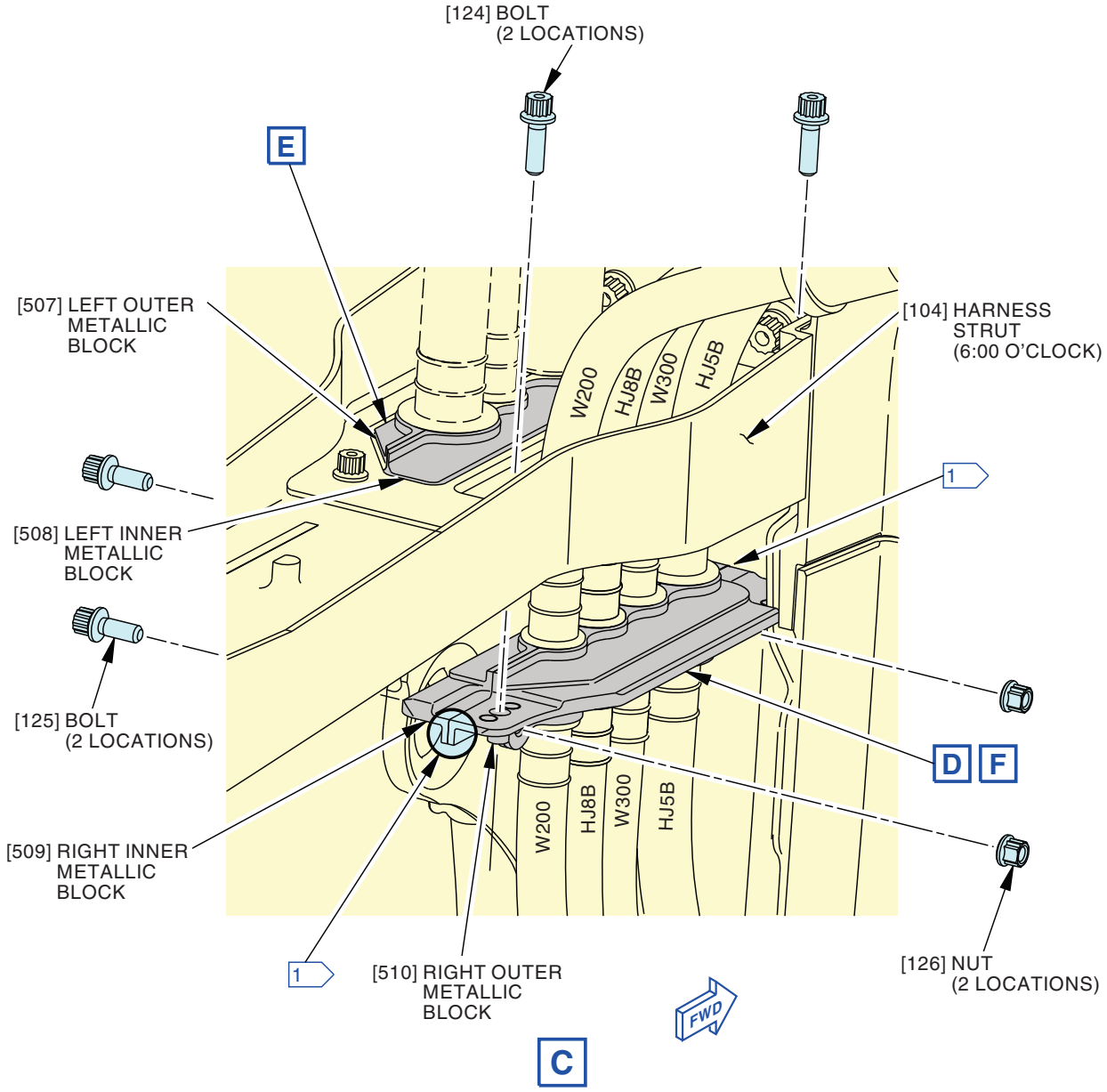
THE FILLET SEAL IS FINISH OF THE SQUEEZE OUT AROUND THE CONTOUR OF THE METALLIC BLOCKS. IT IS ACCEPTABLE TO APPLY A FILLET SEAL AROUND THE CONTOUR OF THE BLOCK AFTER IT IS INSTALLED.

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN BOTH RIGHT AND LEFT METALLIC BLOCKS HAVE BEEN REMOVED.

ICNLEAP1B732201D5882800059A00101
2962033 S0000739202_V2

Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 8 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

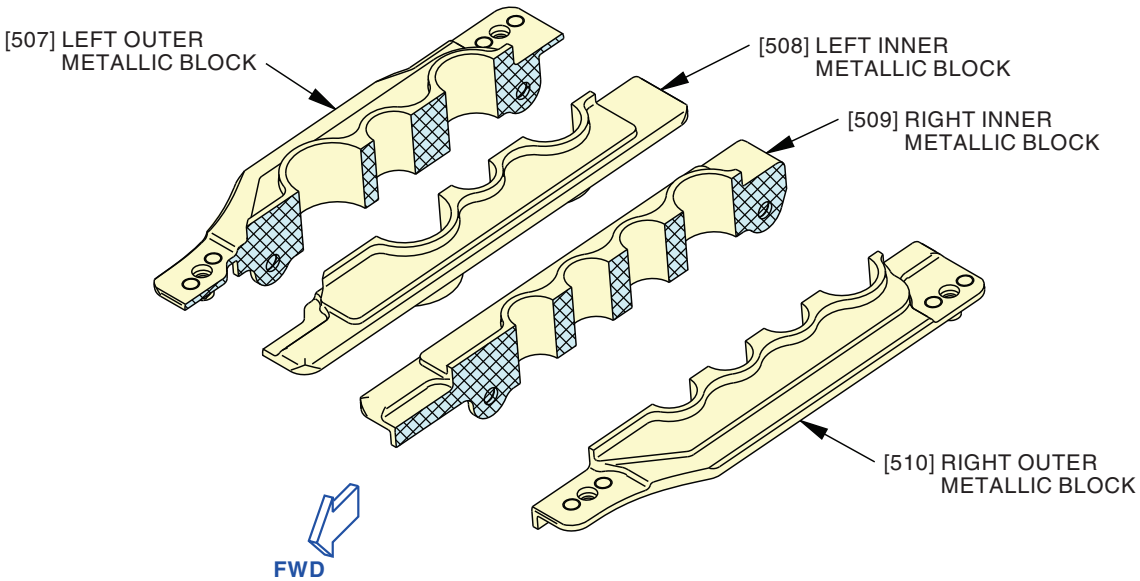
THIS VIEW IS SPECIFIC FOR RIGHT INNER AND OUTER METALLIC BLOCKS. SAME LOGIC IS APPLIED FOR LEFT INNER AND OUTER METALLIC BLOCKS.

1 SEALANT SQUEEZE OUT IS FLUSH WITH SURFACE.

ICNLEAP1B732201D5882800063A00101
2962034 S0000739203_V2

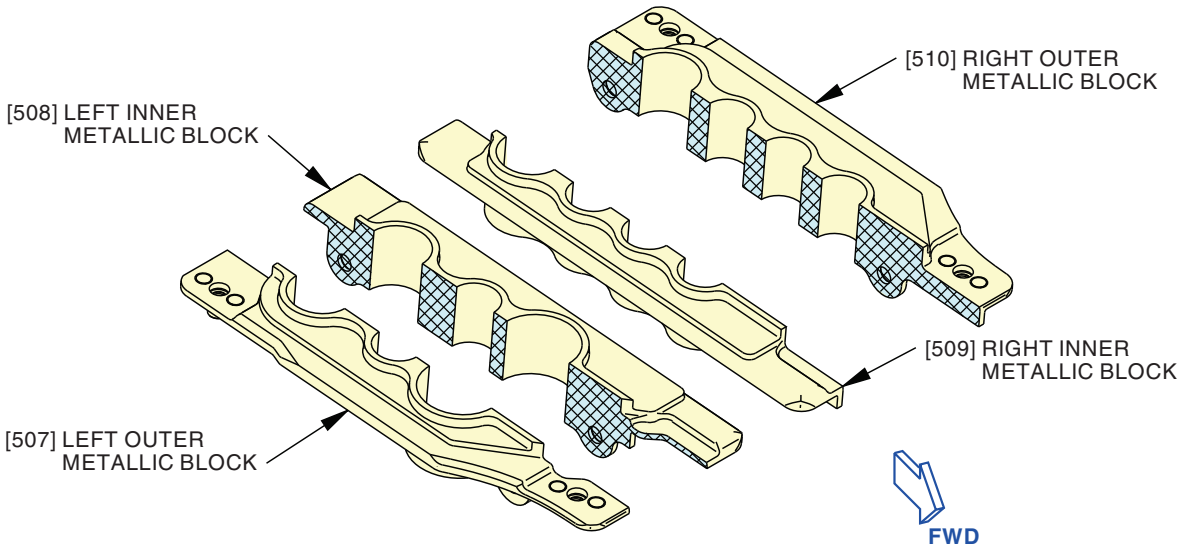
Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 9 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TOP FACING VIEW)

D



(TOP FACING VIEW ROTATED 90 DEGREES)

D

LEGEND:

DAPCO 2100

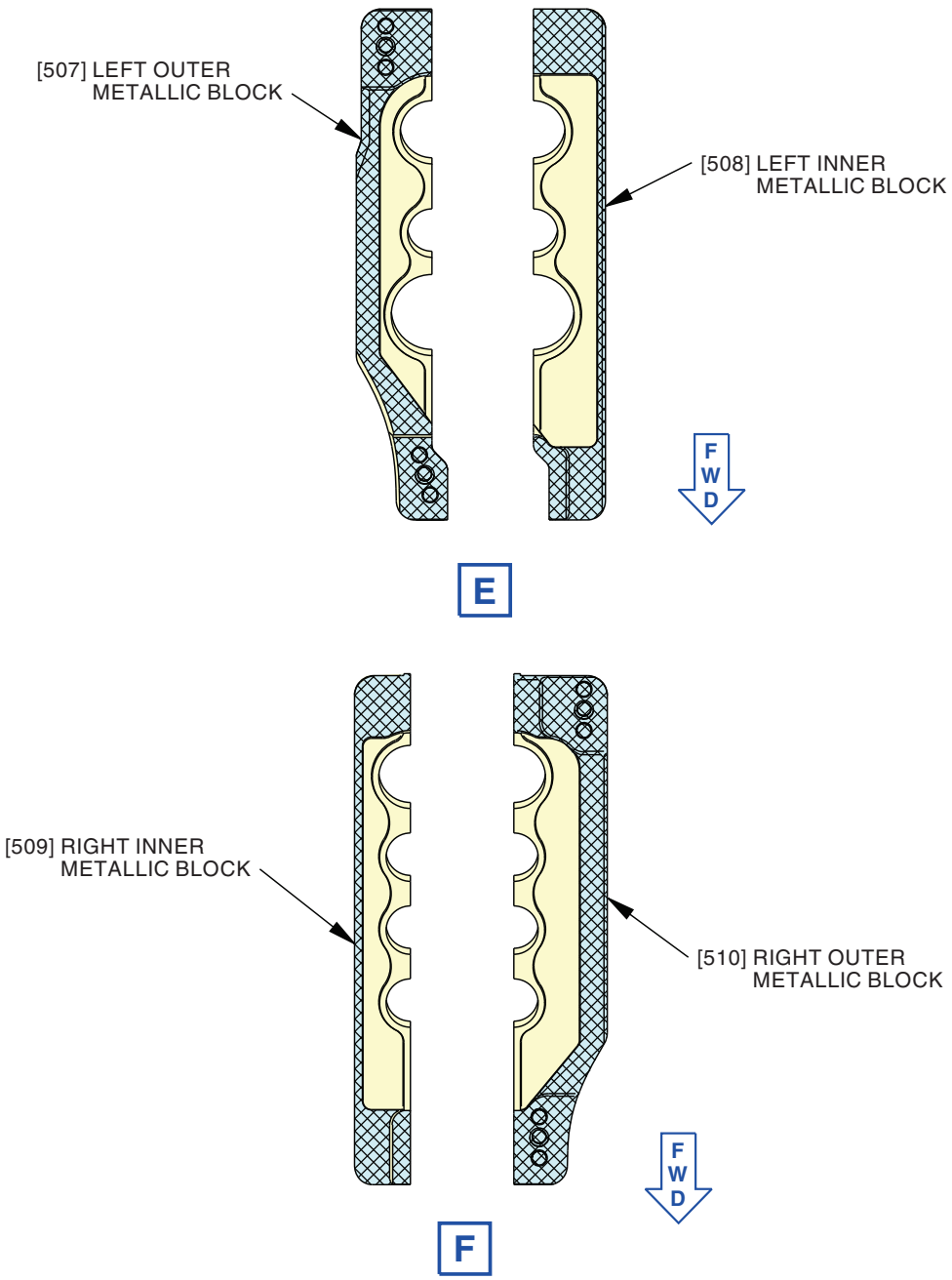
NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994662 S0000771309_V1

Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 10 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

DAPCO 2100

NOTE:

THIS VIEW SHOWS DAPCO 2100 APPLICATION WHEN ALL HARNESS SUPPORT BLOCKS HAVE BEEN REMOVED.

2994663 S0000771310_V1

Harness Support Block Installation
Figure 408/73-22-01-990-873-G00 (Sheet 11 of 11)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-22-01-400-804-G00

9. Electronic Engine Control (EEC) Electrical Harness (HJ5B) Installation

(Figure 406, Figure 407, and Figure 408)

A. General

(1) This task gives the instructions to install the electronic engine control (EEC) electrical harness (HJ5B).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

(3) You must do the tests that are given in the power plant test reference table after you install the applicable engine control electrical harnesses.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50396	Sealant - Dapco 2200 Primerless Silicone Firewall Sealant	BMS5-63 Type II Class B-1/2
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Electronic Engine Control (EEC) Electrical Harness (HJ5B) Installation

SUBTASK 73-22-01-210-003-G00

(1) During the installation, make sure that the EEC electrical harness (HJ5B) [135] is not stressed or damaged.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-210-004-G00

(2) Make sure that all components are clean and free of old sealant prior to adding new sealant.

SUBTASK 73-22-01-420-006-G00

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(3) Install the EEC electrical harness (HJ5B) [135] as follows:

(a) Make sure that the mating faces are clean and free of unwanted materials.

(b) Remove the protective covers on the connectors and receptacles.

(c) Install the EEC electrical harness (HJ5B) [135] in the correct position with the 24 clips [136A] along its routing.

(d) Install the EEC electrical harness (HJ5B) [135] on the harness strut (6:00 o'clock) [104] as follows:

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288

1) Install the harness support block [504], harness support block [505], and harness support block [506] on the harness strut (6:00 o'clock) [104] as follows:

a) Lubricate the four bolts [111] with graphite mineral vaseline, D00601 [CP2101].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the harness support block [504], harness support block [505], harness support block [506], and around the harnesses (View E, Figure 408).

c) Make sure that the EEC electrical harness (HJ5B) [135] and harnesses are in the correct position on the harness support block [504], harness support block [505], and harness support block [506] (View D, Figure 408).

d) Install the two bolts [113] and two nuts [112] to assemble the harness support block [504], harness support block [505], and harness support block [506] in the correct position with the harnesses.

e) Tighten the nuts [112] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION

MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

f) Make sure that there is minimum amount of the squeeze-out on the lower side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

g) Install the harness support block [504], harness support block [505], and harness support block [506] in the correct position on the harness strut (6:00 o'clock) [104].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

h) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104] (View B, Figure 408).

i) Install the four bolts [111] and four nuts [110] that attach the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

j) Tighten the nuts [110] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

k) Make sure that the minimum amount of the sealant squeeze out is on the lower side.

l) Apply more sealant to form a correct fillet seal around the contour of the harness support block [504], harness support block [505], harness support block [506], and harness strut (6:00 o'clock) [104].

CAUTION MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

m) Clean the four nuts [110] and four bolts [111] that will have sealant applied to them.

n) Apply Dapco 2100 sealant, A50483 [CP2927], to the heads of the four bolts [111].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288

2) Remove any remaining of old sealant residue left on the harnesses or harness strut (6:00 o'clock) [104].

3) Install the left outer metallic block [507] and left inner metallic block [508] on the harness strut (6:00 o'clock) [104] as follows:

a) Lubricate the two bolts [124] and two bolts [125] with graphite mineral vaseline, D00601 [CP2101].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507] and left inner metallic block [508].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the left outer metallic block [507], left inner metallic block [508], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the left outer metallic block [507] and left inner metallic block [508] in the correct position with the harnesses.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

e) Tighten the two nuts [126] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

f) Install the left outer metallic block [507] and left inner metallic block [508] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

g) Install the two bolts [124] that attach the left outer metallic block [507] and left inner metallic block [508] to the harness strut (6:00 o'clock) [104].

h) Tighten the two bolts [124] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

i) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

j) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

<2> Make sure that the visible faces of the bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

k) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

l) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of left outer metallic block [507] and left inner metallic block [508].

4) Install the new right inner metallic block [509] and right outer metallic block [510] on the harness strut (6:00 o'clock) [104] as follows:

a) Lubricate the two bolts [124] and two bolts [125] with graphite mineral vaseline, D00601 [CP2101].

b) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509] and right outer metallic block [510].

NOTE: Dapco 2100 sealant, A50483 [CP2927], must be evenly applied in large quantity on each hatching area to make sure that there is enough Dapco 2100 sealant, A50483 [CP2927], between parts after the squeeze out when assembling and tightening the bolts and nuts.

c) Apply Dapco 2100 sealant, A50483 [CP2927], on the mating faces between the right inner metallic block [509], right outer metallic block [510], and harness strut (6:00 o'clock) [104].

d) Install the two bolts [125] and two nuts [126] to assemble the right inner metallic block [509] and right outer metallic block [510] in the correct position with the harnesses.

e) Tighten the two nuts [126] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0288 (Continued)

f) Install the right inner metallic block [509] and right outer metallic block [510] in the correct position on the harness strut (6:00 o'clock) [104] by the bottom side.

g) Install the two bolts [124] that attach the right inner metallic block [509] and right outer metallic block [510] to the harness strut (6:00 o'clock) [104].

h) Tighten the two bolts [124] to 41.8 in-lb (4.72 N·m) - 46.2 in-lb (5.22 N·m).

i) Remove the squeezed out excedent and remainings of Dapco 2100 sealant, A50483 [CP2927], using a profiler tool, a swipe or with fingers using protective gloves.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE BOTTOM SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

j) Make sure that there is minimum amount of the squeeze-out on the bottom side.

<1> Make sure that the sealant squeeze-out is finished flush with the surface.

<2> Make sure that the visible faces of the two bolts [124] and harness strut (6:00 o'clock) [104] are clean and clear of graphite mineral vaseline, D00601 [CP2101].

k) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around the head of the two bolts [124], in contact with the harness strut (6:00 o'clock) [104].

l) Apply Dapco 2100 sealant, A50483 [CP2927], fillet around each electrical harness, between the metal bobbin of each wire harness and top side of right inner metallic block [509] and right outer metallic block [510].

PTF ALL

5) Install the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109] as follows:

a) Lubricate the five bolts [106] and two nuts [107] with the graphite mineral vaseline, D00601 [CP2101].

b) Apply the Dapco 2100 sealant, A50483 [CP2927], to the one side (either top or bottom) of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

NOTE: Fay seal applies to the two sides of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

c) Make sure that the EEC electrical harness (HJ5B) [135] and harnesses are in the correct position on the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503](View E, Figure 407).

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d) Install the two bolts [108] and two nuts [107] to assemble the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503].

e) Tighten the nuts [107] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

f) Make sure that the fillet seal of the squeeze out is on the forward facing side.

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

g) Make the sealant squeeze-out flush on the outer edges of the strut 5 bracket upper block [500] and strut 5 bracket lower block [503] (View F, Figure 407).

h) Install the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] in a correct position on the strut 5 outer fitting [109].

i) Apply Dapco 2100 sealant, A50483 [CP2927], between the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 outer fitting [109] (View C, Figure 408).

j) Install the five bolts [106] that attach the assembly of the strut 5 bracket upper block [500], strut 5 bracket middle block [501], strut 5 bracket middle block [502], and strut 5 bracket lower block [503] on the strut 5 outer fitting [109].

k) Tighten the bolts [106] to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

CAUTION MAKE SURE THAT YOU REMOVE UNWANTED SEALANT FROM THE AFT SIDE OF THE ASSEMBLY. THIS CAN DECREASE THE RISK OF IGNITION DURING A FIRE. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

l) Make sure that there is a minimum amount of the squeeze-out on the aft side around the edges of the block assembly and that the sealant is flush with the block assembly (View C, Figure 407).

m) Make sure that the squeeze-out on the forward facing side is finished as a fillet seal.

CAUTION MAKE SURE THAT THE BOLTS ARE CLEAN BEFORE YOU APPLY SEALANT TO THEM. IF YOU DO NOT REMOVE UNWANTED MATERIALS, THE SEALANT WILL NOT BOND. THIS CAN CAUSE DAMAGE TO EQUIPMENT.

n) Clean the nut plate side of the bolts [106] and nut plates.

o) Apply Dapco 2100 sealant, A50483 [CP2927], to the nut plate side of the bolts [106] and nut plates.

(e) Attach blanket [190] to the strut 5 outer fitting [109].

1) Apply Dapco 2100 sealant, A50483 [CP2927], between the blanket [190] and strut 5 outer fitting [109] (View B, Figure 407).

NOTE: Dapco 2200 firewall sealant, A50396, may use as optional. If this consumable meets the local regulatory requirements of the operator, it is acceptable as a chemical sealant for further usage.

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(f) Lubricate the threads of the bolt [148], bolt [150], bolt [153], and bolt [157] with the oil, D00599 [CP2442].

(g) Install the bolt [157] and clamp [156] that attach the EEC electrical harness (HJ5B) [135] to the forward stator case bracket.

(h) Tighten the bolt [157] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

(i) Install the bolt [153] and clamp [152] that attach the EEC electrical harness (HJ5B) [135] to the forward stator case bracket.

(j) Tighten the bolt [153] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

(k) Install the bolt [150] and clamp [151] that attach the EEC electrical harness (HJ5B) [135] to the forward stator case bracket.

(l) Tighten the bolt [150] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

(m) Install the bolt [148] and clamp [149] that attach the EEC electrical harness (HJ5B) [135] to the forward stator case bracket.

(n) Tighten the bolt [148] to 33.25 in-lb (3.8 N·m) - 36.75 in-lb (4.2 N·m).

CAUTION

MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION

USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(o) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (HJ5B) [135] as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (HJ5B) [135] connector to the SCU/SVA channel B receptacle [160].

2) Connect the EEC electrical harness (HJ5B) [135] connector to the right VSV actuator (bottom) receptacle [159].

3) Connect the EEC electrical harness (HJ5B) [135] connector to the right VSV actuator (top) receptacle [158].

4) Connect the EEC electrical harness (HJ5B) [135] connector to the right VBV actuator receptacle [154].

5) Connect the EEC electrical harness (HJ5B) [135] connector to the T25 sensor channel B receptacle [155].

6) Connect the EEC electrical harness (HJ5B) [135] connector to the N1 speed sensor channel B receptacle [147].

(p) Install the bolt [140] and clamp [141] that attach the EEC electrical harness (HJ5B) [135] to the fan case bracket.

(q) Tighten the bolt [140] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

(r) Install the bolt [137] and clamp [138] that attach the EEC electrical harness (HJ5B) [135] to the fan case bracket.

(s) Tighten the bolt [137] to 42.0 in-lb (4.75 N·m) - 46.5 in-lb (5.25 N·m).

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CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO TIGHTEN THE ELECTRICAL CONNECTORS. DO NOT USE PLIERS THAT HAVE METAL JAWS. METAL JAWS CAN CAUSE DAMAGE TO THE ELECTRICAL CONNECTORS.

(t) Use the teflon-jawed pliers, STD-664, to connect the EEC electrical harness (HJ5B) [135] as follows (TASK 70-00-01-910-803-G00):

1) Connect the EEC electrical harness (HJ5B) [135] connector to the EEC HJ5A harness receptacle [144].

2) Connect the HJ8B engine wiring harness electrical connector [146] connector to the EEC HJ8A harness receptacle [145].

3) Connect the HJ6B engine wiring harness electrical connector [142] connector to the EEC HJ6A harness receptacle [143].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

4) Connect the EEC electrical harness (HJ5B) [135] to the lower overheat sensor [114].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0178 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

5) Connect the EEC electrical harness (HJ5B) [135] to the bracket assembly [115].

PTF ALL

6) Connect the EEC electrical harness (HJ5B) [135] connector to the FMU channel B receptacle [139].

7) Connect the EEC electrical harness (HJ5B) [135] connector to the EEC channel B J5 receptacle [136].

(u) Make sure that the work area is clean of all tools and other items.

SUBTASK 73-22-01-420-007-G00

(4) Install the segment shroud 4 [71].

SUBTASK 73-22-01-420-008-G00

(5) Install the segment shroud 3 [69].

SUBTASK 73-22-01-420-009-G00

(6) Install the segment shroud 2 [68].

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-22-01-410-025-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

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1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-22-01-860-029-G00

(2) Remove the safety tags and close these circuit breakers:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-030-G00

(3) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-031-G00

(4) For the left engine, remove the safety tag and close this circuit breaker:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-860-032-G00

(5) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

H. Electronic Engine Control Electrical (EEC) Electrical Harness (HJ5B) Installation Test

SUBTASK 73-22-01-730-004-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

TASK 73-22-01-000-805-G00

10. Electronic Engine Control (EEC) Electrical Harness (HJ6A) Removal

(Figure 409)

A. General

(1) This task gives the instructions to remove the electronic engine control (EEC) electrical harness (HJ6A).

(2) The electronic engine control (EEC) electrical harness is referred to as the EEC electrical harness.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-664	Pliers - Teflon-jawed (or Equivalent Soft-Jawed)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Removal

SUBTASK 73-22-01-860-033-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-22-01-860-034-G00

(2) Open these applicable circuit breakers and install safety tags:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
A	4	C04008	ENGINE 1 ALTN PWR CHAN B
A	5	C04007	ENGINE 1 ALTN PWR CHAN A
B	3	C01312	ENGINE 1 RUN/PWR

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
B	5	C01313	ENGINE 2 RUN/PWR
D	7	C04010	ENGINE 2 ALTN PWR CHAN B
D	8	C04009	ENGINE 2 ALTN PWR CHAN A

F/O Electrical System Panel, P6-5

Row	Col	Number	Name
E	3	C04018	ENG 1 STBY ALTN PWR RELAY CHAN A
E	4	C04019	ENG 1 STBY ALTN PWR CHAN A

SUBTASK 73-22-01-860-035-G00

(3) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-2

Row	Col	Number	Name
C	4	C00154	ENGINE 2 START VALVE

SUBTASK 73-22-01-860-036-G00

(4) For the left engine, open this circuit breaker and install safety tag:

CAPT Electrical System Panel, P18-2

Row	Col	Number	Name
B	8	C01103	ENGINE 1 START VALVE

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-22-01-010-027-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Electronic Engine Control (EEC) Electrical Harness (HJ6A) Removal

(Figure 409)

SUBTASK 73-22-01-020-011-G00

CAUTION MAKE SURE THAT THE ELECTRICAL CONNECTORS AND RECEPTACLES ARE CLEAN AND CLEAR OF UNWANTED MATERIALS BEFORE YOU DISCONNECT, OR CONNECT THEM. CONTAMINATION OF THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO EQUIPMENT.

CAUTION USE TEFLON-JAWED PLIERS TO LOOSEN ELECTRICAL CONNECTORS. DO NOT USE METAL-JAWED PLIERS. DAMAGE TO THE ELECTRICAL CONNECTORS COULD OCCUR.

(1) Use the teflon-jawed pliers, STD-664 to disconnect the EEC electrical harness (HJ6A) [192] as follows (TASK 70-00-01-910-803-G00):

(a) Disconnect the EEC electrical harness (HJ6A) [192] connector from the EEC channel A J6 receptacle [219].

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(b) Disconnect the EEC electrical harness (HJ6A) [192] connector from the PSS channel A receptacle [199].

(c) Disconnect the EEC electrical harness (HJ6A) [192] connector from the main f/o heat exchanger DP sensor receptacle [202].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0178

(d) Disconnect the EEC electrical harness (HJ6A) [192] from the upper overheat sensor [220].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0178 AND PRE LEAP-1B SERVICE BULLETIN 73-0019

(e) Disconnect the EEC electrical harness (HJ6A) [192] from the bracket assembly [116].

PTF ALL

(f) Disconnect the EEC electrical harness (HJ6A) [192] connector from the T12 Channel A receptacle [203].

(g) Disconnect the EEC electrical harness (HJ6A) [192] connector from the FMU fuel strainer DP transducer receptacle [204].

(h) Disconnect the EEC electrical harness (HJ6A) [192] connector from the fuel filter DP transducer receptacle [207].

(i) Disconnect the EEC electrical harness (HJ6A) [192] connector from the oil filter DP sensor receptacle [208].

(j) Disconnect the EEC electrical harness (HJ6A) [192] connector from the SAV receptacle [209].

(k) Disconnect the EEC electrical harness (HJ6A) [192] connector from the N2 speed sensor channel A receptacle [210].

(l) Disconnect the EEC electrical harness (HJ6A) [192] connector from the oil P/T sensor channel A receptacle [214].

(m) Disconnect the connector socket [216] of the EEC electrical harness (HJ6A) [192] as follows:

1) Disconnect the HJ6B engine wiring harness electrical connector [215] (at 5:00 o'clock) from the connector socket [216].

NOTE: If the HJ6B engine wiring harness has been removed, do not do this step.

2) Remove the four nuts [217] and four bolts [218] that attach the connector socket [216] on the bracket [83].

3) Disconnect the connector socket [216] from the bracket [83].

(n) Remove the bolt [195] from the clamp [196] that attaches the EEC electrical harness (HJ6A) [192] to the fan case bracket.

(o) Remove the bolt [197] from the clamp [198] that attaches the EEC electrical harness (HJ6A) [192] to the fan case bracket.

(p) Remove the bolt [200] from the clamp [201] that attaches the EEC electrical harness (HJ6A) [192] to the fan case bracket.

(q) Remove the bolt [205] from the clamp [206] that attaches the EEC electrical harness (HJ6A) [192] to the fan case bracket.

(r) Remove the nut [211] and bolt [212] from the clamp [213] that attaches the EEC electrical harness (HJ6A) [192] to the fan case bracket.

(s) Remove the EEC electrical harness (HJ6A) [192] from the 44 clips [194] along its routing.

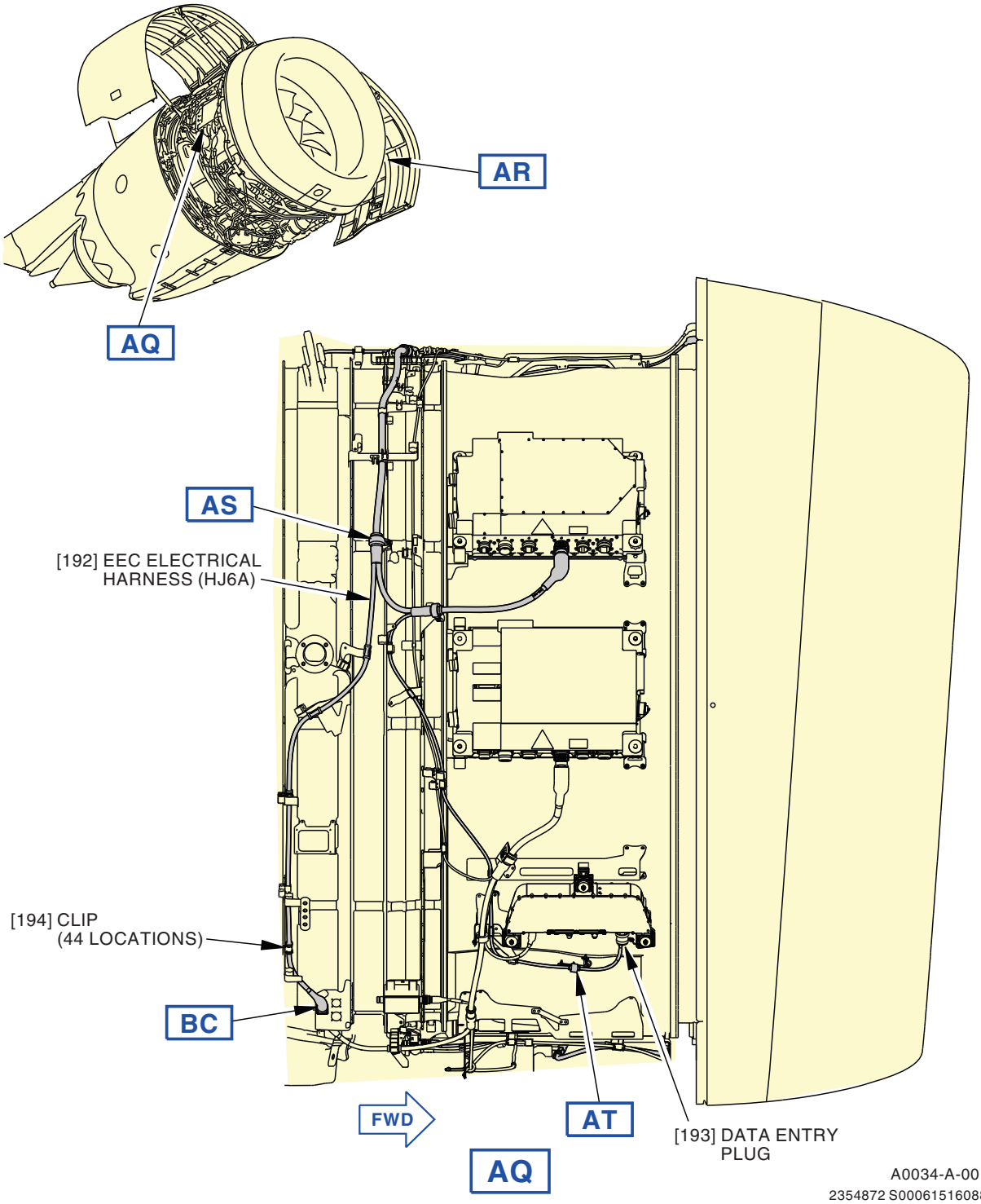
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(t) Carefully remove the EEC electrical harness (HJ6A) [192] from the engine.

(u) Install the protective covers on all of the harness connectors and the component receptacles.

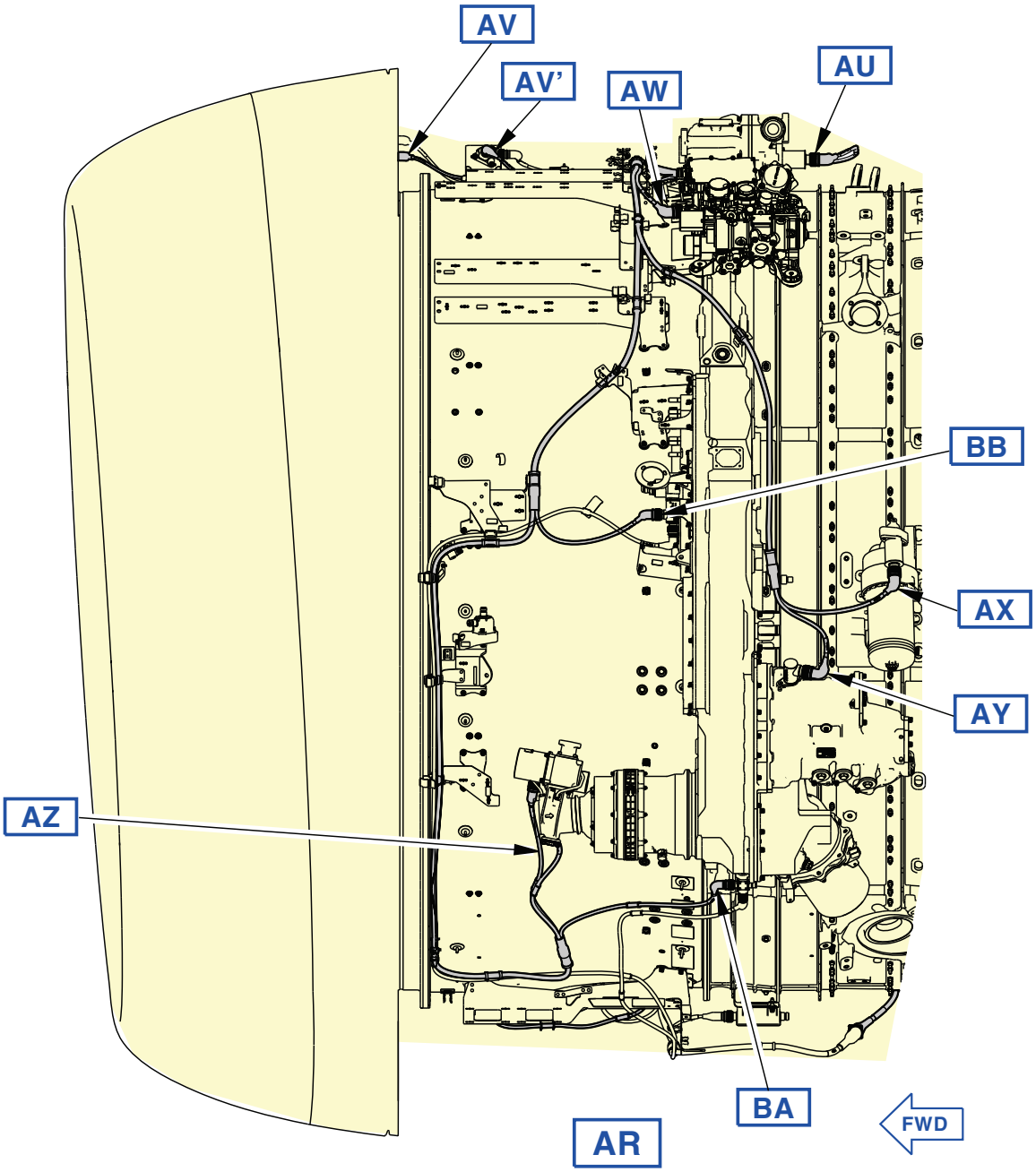
——— END OF TASK ———

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EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 1 of 9)

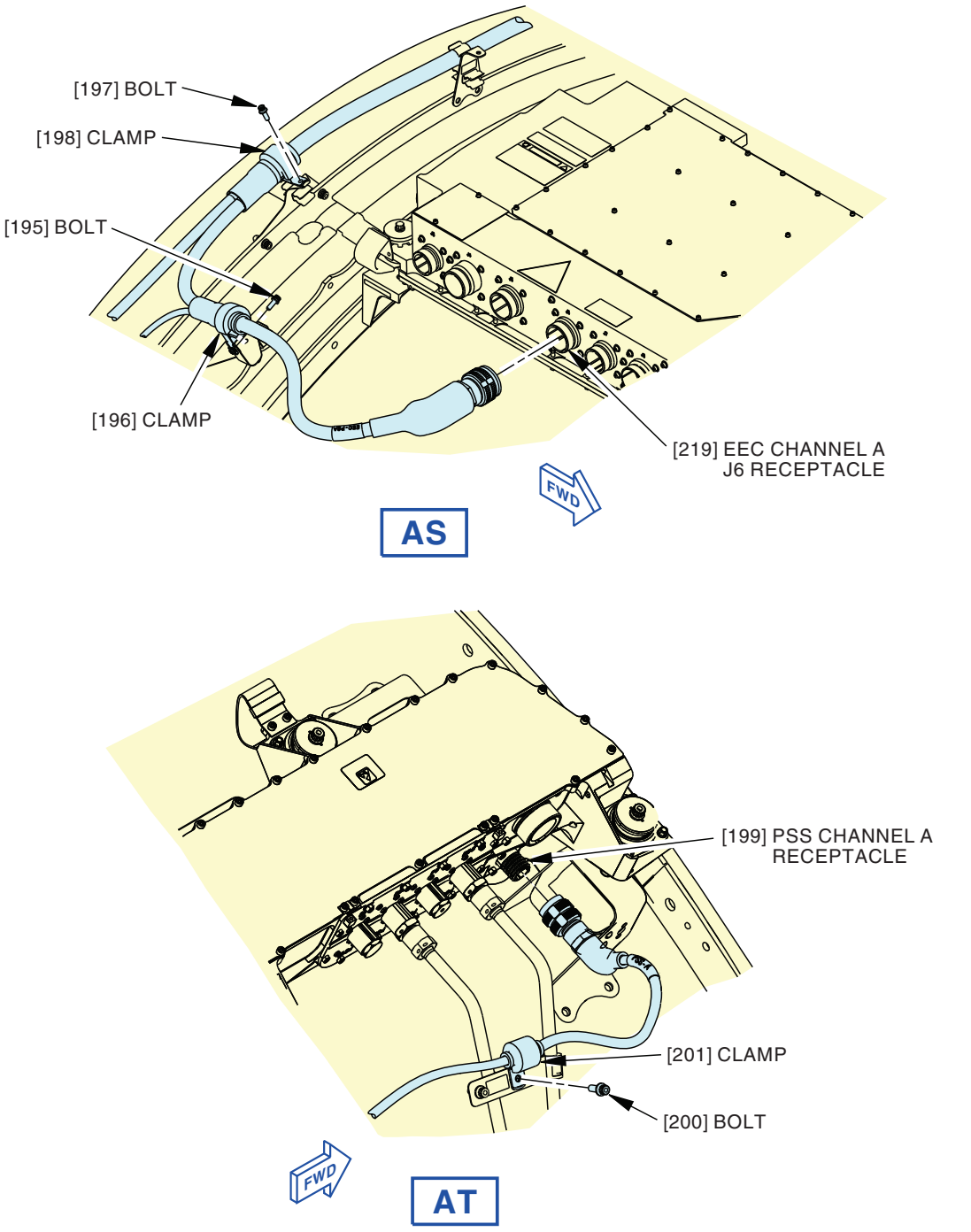
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EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 2 of 9)

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EEC Electrical Harness (HJ6A) Installation
Figure 409/73-22-01-990-855-G00 (Sheet 3 of 9)

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